Algebraic Geometry—Mumford's Red Book

Summer 2026

1 Varieties

1.1 Some algebra

Theorem 1 (Noether's Normalization Lemma)**.** *Let R be an integral domain, finitely generated over a field k. If R has transcendence degree n over k, then there exist elements $x_1, \ldots, x_n \in R$, algebraically independent over k, such that R is integrally dependent on the subring $k[x_1, \ldots, x_n]$ generated by the x_i's.*

Theorem 2 (Going-up theorem of Cohen-Seidenberg)**.** *Let R be a ring (commutative) and $S \subseteq R$ a subring such that R is integrally dependent on S. For all prime ideals $P \subseteq S$, there exist prime ideals $P' \subseteq R$ such that $P' \cap S = P$.*

Lemma 3. *Let R be a field and $S \subseteq R$ a subring such that R is integrally dependent on S. Then S is a field.*

Theorem 4 (Weak Nullstellensatz)**.** *Let k be an algebraically closed field. Then the maximal ideals in the ring $k[X_1, \ldots, X_n]$ are the ideals*

$$(X_1 - a_1, \ldots, X_n - a_n)$$

where $a_1, \ldots, a_n \in k$.

Example 5. Let $A = k[x]$. Then:

- $B_1 = A[y]/(y^2 - x)$ is integral over A
- $B_2 = A[x^{-1}]$ is not integral over A (but is algebraic over A)
- $B_3 = A[y]$ is not integral over A (not even algebraic)

Geometrically, A is the affine coordinate ring of the affine line ($= x - axis$).

- B_1 is a parabola
- B_2 is a hyperbola; $A[x^{-1}] \cong k[x, y]/(xy - 1)$ via $y \mapsto x^{-1}$
  You can either visualize B_2 as the affine line minus the origin, or a hyperbola
- B_3 is a plane

Given $A \subseteq B_i$ (say A is the affine coordinate ring of X, and B_i is the affine coordinate ring of Y_i), there exists some map $Y_i \to X$ via precomposition. To see this, think of an element of A as a map $X \to k$. Then given $\varphi : A \to B_i$, we get a map $Y_i \to X \to k$ i.e. an element of B_i.

Remark 6. If k is an infinite field, different polynomial functions in $k[x]$ are different maps:

Let $f(x) = a_n x^n + \cdots + a_1 x + a_0 \in k[x]$. If f is not the zero polynomial, then at least one $a_i \neq 0$; let n be the highest index such that $a_n \neq 0$.

If $f(a) = 0$ for all $a \in k$, then every element of k is a root of $f(x)$. Since k is an infinite field, $f(x)$ has infinitely many roots. But a nonzero polynomial of degree n over a field has at most n roots, so f must have at most n roots in k. So f must be the zero polynomial.

Then if $f(a) - g(a) = 0$ for all $a \in k$, then $f(a) - g(a)$ must be the zero polynomial, meaning $f(x) = g(x)$.

Example 7. Again let $A = k[x]$ and

- $B_1 = A[y]/(y^2 - x)$
- $B_2 = A[x^{-1}]$
- $B_3 = A[y]$

Say A is the affine coordinate ring of X, and B_i is the affine coordinate ring of Y_i.
Given $A \subseteq B_i$ there exists some map $Y_i \to X$ via precomposition. To see this, think of an element of A as a map $X \to k$. Then given $\varphi : A \to B_i$, we get a map $Y_i \to X \to k$ i.e. an element of B_i.

- The map $Y_1 \to X$ is surjective and quasi-finite (i.e. only finitely many points map to a given point)
- The map $Y_2 \to X$ is not surjective but is quasi-finite
- The map $Y_3 \to X$ is surjective but is not quasi-finite

Recall: integral and finite type (= finitely generated as an algebra) $\implies$ quasi-finite.
Finitely generated as an algebra is weaker than finitely generated as a module (e.g. B_3/A is finitely generated as an algebra but not as a module, since you need all powers of y. B_1/A is, however, finitely generated as a module.

Example 8. Geometrically, the "Going up" (=Lying over) theorem says:
the map $V(B_1) = Y_1 \to V(A) = X$ (parabola to affine line) is surjective, since B_1 is integral over A.

Remark 9. If $A \subseteq B$ are domains and B/A is integral, then (say $A = k[Y]$, $B = k[X]$) the induced map $X \to Y$ is surjective.

Recall: $k[x]$ and $\mathbb{Z}$ are the key examples of Euclidean domains, and $\mathbb{Z} \subseteq \mathbb{Q}$ is the key example where an ideal in $\mathbb{Q}$ is maximal but its contraction is not.

1.2 Irreducible algebraic sets

Definition 1. A **closed algebraic subset of** k^n is a set consisting of all roots of a finite collection of polynomial equations, i.e.

$$\{(x_1, \ldots, x_n) \mid f_1(x_1, \ldots, x_n) = \cdots = f_m(x_1, \ldots, x_n) = 0\}.$$

The above set depends only on the ideal $A = (f_1, \ldots, f_m)$ in $k[X_1, \ldots, X_n]$. So for any ideal $A \trianglelefteq k[X_1, \ldots, X_n]$, define

$$V(A) := \{x \in k^n \mid f(x) = 0 \text{ for all } f \in A\}.$$

The subsets of k^n of the form $V(A)$ are exactly the closed algebraic sets.

If Σ is a closed algebraic set, we define

$$I(\Sigma) := \{f \in k[X_1, \ldots, X_n] \mid f(x) = 0 \text{ for all } x \in \Sigma\}.$$

$I(\Sigma)$ is an ideal, and

$$\Sigma = V(I(\Sigma)).$$

Theorem 2 (Hilbert's Nullstellensatz)**.**

$$I(V(A)) = \sqrt{A}.$$

Corollary 3. *V and I set up a bijection between the set of closed algebraic subsets of k^n and the set of ideals $A \subseteq k[X_1, \ldots, X_n]$ such that $A = \sqrt{A}$.*

Remark 4. This really is a generalization of Weak Nullstellensatz, since maximal ideals are radical, and points are minimal closed subsets.

Example 5. An example of the correspondence: in $k[x, y]$, the chain

$$(0) \subseteq (x) \subseteq (x, y)$$

corresponds to the chain in k^2 looking like

$$\text{Plane} \supseteq \text{(Vertical) line} \supseteq \text{Point}$$

.

Example 6. In $k[x, y]$, the ideal (xy) corresponds to the two coordinate axes. This ideal is not prime but is still radical, hence the geometric image is not irreducible but is still closed, since the correspondence is

$$\text{Closed subsets} \leftrightarrow \text{Radical ideals}$$

and

$$\text{Irreducible closed subsets} \leftrightarrow \text{Prime ideals}$$

and

$$\text{Minimal irreducible closed subsets (=points)} \leftrightarrow \text{Maximal ideals}$$

Definition 7. A closed algebraic set is **irreducible** if it is not the union of two strictly smaller closed algebraic sets. We will omit "closed" when referring to these sets.

Primary Decomposition Theorem says any radical ideal can be written in exactly one way as an intersection of a finite set of prime ideals, none of which contain any other. And a prime ideal is not the intersection of any two strictly bigger ideals. Then:

Proposition 8. *In the bijection in our corollary, the irreducible algebraic sets correspond exactly to the prime ideals of* $k[X_1, \ldots, X_n]$.
Every closed algebraic set Σ *can be written in exactly one way as*

$$\Sigma = \Sigma_1 \cup \cdots \cup \Sigma_k$$

where the Σ_i *are irreducible sets and* $\Sigma_i \not\subseteq \Sigma_j$ *if* $i \neq j$.

Definition 9. The Σ_i of the proposition are called the **components of** Σ.

Now that we have affine algebraic sets, we can also define projective algebraic sets:

$\mathbb{P}_n(k)$ is the set of $(n+1)$-tuples $(x_0, \ldots, x_n) \in k^{n+1}$ such that some $x_i \neq 0$, modulo the equivalence relation

$$(x_0, \ldots, x_n) \sim (\alpha x_0, \ldots, \alpha x_n), \quad \alpha \in k^\times.$$

An $(n+1)$-tuple $(x_0 : \cdots : x_n)$ is called a set of **homogeneous coordinates** for the point associated to it.

$\mathbb{P}_n(k)$ can be covered by $n+1$ subsets $U_0, \ldots, U_n$ where

$$U_i = \{\text{points represented by homogeneous coordinates } (x_0 : \cdots : x_n) \text{ with } x_i \neq 0\}.$$

And each U_i is naturally isomorphic to k^n under the map

$$U_i \to k^n, \quad (x_0 : \cdots : x_n) \mapsto \left(\frac{x_0}{x_i}, \ldots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \ldots, \frac{x_n}{x_i}\right).$$

$\mathbb{P}_n(k)$ adds to the affine space $\mathbb{A}^n(k) \cong U_0$ the extra points at infinity $\mathbb{P}_n(k) \setminus U_0$.

- The **linear subspaces** L of $\mathbb{P}_n(k)$ are the sets of points $P \in \mathbb{P}_n(k)$ with homogeneous coordinates in W for some vector subspace $W \subseteq k^{n+1}$.
- If W has codimension 1, we get a hyperplane.
- The "points at infinity" with respect to the affine piece U_i form the hyperplane associated to the vector subspace $x_i = 0$.
- And the linear subspace L associated to W is isomorphic to $\mathbb{P}_r(k)$ where $r = \dim W - 1$.

Linear subspaces are examples of projective algebraic sets.

Definition 10. A **closed algebraic set** in $\mathbb{P}_n(k)$ is a set consisting of all roots of a finite collection of homogeneous polynomials $f_i \in k[X_0, \ldots, X_n], 1 \leq i \leq m$.

This makes sense because if f is homogeneous (i.e. all terms have the same degree) and $(x_0 : \cdots : x_n) = (\alpha x_0 : \cdots : \alpha x_n)$, then $f(x_0 : \cdots : x_n) = 0 \iff f(\alpha x_0 : \cdots : \alpha x_n) = 0$.

Homogeneous ideals $A \subseteq k[X_0, \ldots, X_n]$ refer to ideals that, when they contain a polynomial f, also contain the homogeneous components of f. Equivalently, homogeneous ideals are the ideals generated by a finite set of homogeneous polynomials.

If A is a homogeneous ideal, define

$$V(A) := \{P \in \mathbb{P}_n(k) \mid \text{ If } (x_0 : \cdots : x_n) \text{ are coordinates of } P, \text{ then } f(x) = 0 \forall f \in A\}.$$

If $\Sigma \subseteq \mathbb{P}_n(k)$ is a closed algebraic set, define

$$I(\Sigma) := \langle F \in k[X_0, \ldots, X_n] \text{ homogeneous} \mid F|_\Sigma = 0 \rangle.$$

Theorem 11. *V and I set up a bijection between the closed algebraic subsets of $\mathbb{P}_n(k)$ and the homogeneous radical ideals $A \subseteq k[X_0, \ldots, X_n]$ satisfying*

$$A : (X_0, \ldots, X_n) = A.$$

Equivalently, one must identify homogeneous ideals that define the same projective zero set; the condition above chooses the largest such ideal.

Define irreducible algebraic sets analogously to the affine case.

Proposition 12. *In the bijection of the theorem, the nonempty irreducible algebraic sets correspond exactly to the homogeneous prime ideals other than $(X_0, \ldots, X_n)$. And every closed algebraic set Σ in $\mathbb{P}_n(k)$ can be written in exactly one way as*

$$\Sigma = \Sigma_1 \cup \cdots \cup \Sigma_k$$

where the Σ_i are irreducible algebraic sets and $\Sigma_i \not\subseteq \Sigma_j$ if $i \neq j$.

Problem 1.1. Let $\Sigma \subseteq \mathbb{P}_n(k)$ be a closed algebraic set, and let H be the hyperplane $X_0 = 0$. Identify $\mathbb{P}_n \setminus H$ with $\mathbb{A}^n(k)$ in the usual way.
Prove that $\Sigma \cap (\mathbb{P}_n \setminus H)$ is a closed algebraic subset of k^n and show that the ideal of $\Sigma \cap (\mathbb{P}_n \setminus H)$ is derived from the ideal of Σ in a very natural way.

Identify $\mathbb{P}_n \setminus H$ with k^n via

$$\varphi : \mathbb{P}_n \setminus H \to k^n, \quad (x_0 : \cdots : x_n) = (1 : x_1/x_0 : \cdots : x_n/x_0) \mapsto (x_1/x_0, \ldots, x_n/x_0)$$

Observe: $(1 : y_1 : \cdots : y_n) \in \Sigma \iff F(1, y_1, \ldots, y_n) = 0$ for all homogeneous $F \in I(\Sigma)$. Thus,

$$\varphi(\Sigma \cap (\mathbb{P}_n \setminus H)) =$$

$$\{(x_1/x_0, \ldots, x_n/x_0) \in k^n \mid F(1, x_1/x_0, \ldots, x_n/x_0) = 0 \text{ for all homogeneous } F \in I(\Sigma)\}$$

$$= \{x \in k^n \mid F(1, x) = 0 \text{ for all homogeneous } F \in I(\Sigma)\}$$

Define $f(x_1, \ldots, x_n) := F(1, x_1, \ldots, x_n)$ for each homogeneous F. Then $f \in k[x_1, \ldots, x_n]$, so

$$\varphi(\Sigma \cap (\mathbb{P}_n \setminus H)) = \{x \in k^n \mid f(x) = F(1, x) = 0 \text{ for all homogeneous } F \in I(\Sigma)\}$$

is closed.

For the ideal $I(\varphi(\Sigma \cap (\mathbb{P}_n \setminus H)))$, we should consider

$$J := \{f(x) = F(1, x) \mid F \in I(\Sigma) \text{ homogeneous}\}.$$

This is an ideal in $k[x_1, \ldots, x_n]$:

- If $F(1, x) \in J$ and $h \in k[x_1, \ldots, x_n]$, homogenize h to a homogeneous polynomial H of sufficiently large degree. Then $HF \in I(\Sigma)$, so $h(x)F(1, x) = (HF)(1, x) \in J$.
- If $F(1, x), G(1, x) \in J$, first multiply F and G by suitable powers of X_0 so they have the same degree. Their sum is then homogeneous and lies in $I(\Sigma)$, so $F(1, x) + G(1, x) \in J$.

We know (by construction) that $V(J) = \varphi(\Sigma \cap (\mathbb{P}_n \setminus H))$.
By Nullstellensatz, $I(\varphi(\Sigma \cap (\mathbb{P}_n \setminus H))) = \sqrt{J}$.
And

$$\sqrt{J} = \left\{f \in k[x_1, \ldots, x_n] \mid f^\ell \in J \text{ for some } \ell \in \mathbb{Z}_{>0}\right\}$$

The natural construction $F \mapsto f$ is called **dehomogenization:**
Given $F(x_0, \ldots, x_n)$ homogeneous polynomial, its dehomogenization with respect to x_0 is

$$F^\#(x_1, \ldots, x_n) := F(1, x_1, \ldots, x_n).$$

So the affine ideal is

$$I(\Sigma)^\# = \{F^\# \mid F \in I(\Sigma)\}$$

Thus, the ideal of the affine chart is obtained by setting $X_0 = 1$.

To summarize:

1. Let $I(\Sigma) \subseteq k[x_0, \ldots, x_n]$ be the homogeneous ideal defining Σ. Under the identification $\mathbb{P}^n \setminus H \cong k^n$, $\quad (1 : x_1 : \cdots : x_n) \leftrightarrow (x_1, \ldots, x_n)$ we have

$$\Sigma \cap (\mathbb{P}^n \setminus H) = \{x \in k^n \mid F(1, x) = 0 \forall F \in I(\Sigma)\} \implies$$

$$\Sigma \cap (\mathbb{P}^n \setminus H) = V(\{F(1, x_1, \ldots, x_n) \mid F \in I(\Sigma)\})$$

   so it's a closed algebraic subset of k^n.

2. The dehomogenization of $I(\Sigma)$ is:

$$I(\Sigma)^{\#} = \{F(1, x_1, \ldots, x_n) \mid F \in I(\Sigma)\} \implies$$

$$I(\Sigma \cap (\mathbb{P}^n \setminus H)) = \sqrt{I(\Sigma)^{\#}}.$$

Example 13 (Hypersurfaces)**.** Let $f(X_0, \ldots, X_n)$ be an irreducible homogeneous polynomial. Then (f) is prime, so $f = 0$ defines an irreducible algebraic set in $\mathbb{P}_n(k)$ called a hypersurface.

Example 14 (Twisted cubic in $\mathbb{P}_3(k)$)**.** Start with the ideal

$$A_0 = (xz - y^2, yw - z^2) \subseteq k[x, y, z, w].$$

$V(A_0)$ is the intersection in $\mathbb{P}_3(k)$ of the 2 quadrics $xz = y^2$ and $yw = z^2$. Look in the affine space with coordinates

$$X = x/w, \quad Y = y/w, \quad Z = z/w$$

(the complement of $w = 0$).

In here $V(A_0)$ is the intersection of $XZ = Y^2$ and $Y = Z^2$. This intersection falls into $Y = Z = 0$ and the twisted cubic itself, meaning A_0 is the intersection of the ideal of the line and of the twisted cubic:

$$A_0 = (y, z) \cap (xz - y^2, yw - z^2, xw - yz).$$

So we let $A := (xz - y^2, yw - z^2, xw - yz)$. We define the **twisted cubic** to be $V(A)$.

A is prime by being the kernel of the homomorphism $\varphi : k[x, y, z, w] \to k[s, t]$ sending:

$$x \mapsto s^3, \quad y \mapsto s^2 t, \quad z \mapsto st^2, \quad w \mapsto t^3.$$

1.3 Definition of a Morphism

Example 1. If $V \subseteq k^n$ is an irreducible algebraic set and if $f \in k[X_1, \ldots, X_n]$, then

$$V^* := \{(x_1, \ldots, x_n, f(x_1, \ldots, x_n)) \mid (x_1, \ldots, x_n) \in V\} \subseteq k^{n+1}$$

is an irreducible algebraic set, and the projection

$$(x_1, \ldots, x_{n+1}) \mapsto (x_1, \ldots, x_n)$$

should be an isomorphism $V \to V^*$.

So in particular e.g. the affine line k and the parabola $y = x^2$ in k^2 should be isomorphic.

Example 2. It turns out that an irreducible conic $C \subseteq \mathbb{P}_2(k)$ will be isomorphic to $\mathbb{P}_1(k)$.

One sees this by identifying $\mathbb{P}_1(k)$ with the set of all lines through some point P_0 on C and then defining the map $\alpha : \mathbb{P}_1(k) \to C$ by taking a line ℓ through P_0 and letting $\alpha(\ell)$ be its second intersection point with C.

Example 3. It turns out that $\mathbb{P}_1(k)$ and the twisted cubic in $\mathbb{P}_3(k)$ are isomorphic.

$\mathbb{P}_1(k) = \{(s,t) \in k^2 \setminus \{0\}\}/\sim$ where $(s,t) \sim (\lambda s, \lambda t)$, and the isomorphism is given by

$$\alpha : \mathbb{P}_1(k) \to \mathbb{P}_3(k), \qquad (s:t) \mapsto (s^3 : s^2t : st^2 : t^3).$$

Example 4. If C is a smooth cubic curve in $\mathbb{P}_2(k)$ and $P_0 \in C$, then for any $P \in C$, if ℓ is the line through P and P_0, let $\alpha(P)$ be the third point in which ℓ meets C (counting intersection multiplicity). Then α is an automorphism of C of order 2.

Definition 5. Let $\Sigma_1 \subseteq k^{n_1}$ and $\Sigma_2 \subseteq k^{n_2}$ be two irreducible algebraic sets. A map

$$\alpha : \Sigma_1 \to \Sigma_2$$

will be called a **morphism** if there exist n_2 polynomials $f_1, \ldots, f_{n_2}$ in the variables $X_1, \ldots, X_{n_1}$ such that

$$\alpha(x) = (f_1(x_1, \ldots, x_{n_1}), \ldots, f_{n_2}(x_1, \ldots, x_{n_1}))$$

for all points $x = (x_1, \ldots, x_{n_1}) \in \Sigma_1$.

Remark 6. Note that this definition implies that every morphism α from Σ_1 to Σ_2 is the restriction of a morphism α' from k^{n_1} to k^{n_2}.

Definition 7. Let $\Sigma \subseteq k^n$ be an irreducible algebraic set. Then the **affine ring** $\Gamma(\Sigma)$ is the ring of k-valued functions on Σ given by the polynomials in the coordinates, i.e.

$$k[X_1, \ldots, X_n]/I(\Sigma).$$

Proposition 8. *If Σ_1, Σ_2 are two irreducible algebraic sets, then the set of morphisms from Σ_1 to Σ_2 and the set of k-homomorphisms from $\Gamma(\Sigma_2)$ to $\Gamma(\Sigma_1)$ are canonically isomorphic, i.e.*

$$\mathrm{Hom}(\Sigma_1, \Sigma_2) \cong \mathrm{Hom}_k(\Gamma(\Sigma_2), \Gamma(\Sigma_1)).$$

Corollary 9. *If Σ is an irreducible algebraic set, then $\Gamma(\Sigma)$ is canonically isomorphic to the set of morphisms from Σ to k.*

For any ideal $A \subseteq \Gamma(\Sigma)$, define

$$V(A) := \{x \in \Sigma \mid f(x) = 0 \text{ for all } f \in A\} \subseteq \Sigma$$

Nullstellensatz implies

$$\{f \in \Gamma(\Sigma) \mid f(x) = 0 \text{ for all } x \in V(A)\} = \sqrt{A}.$$

Proposition 10. *The assignment* $\Sigma \mapsto \Gamma(\Sigma)$ *extends to a contravariant functor* Γ*:*

$$\{\textit{Category of irreducible algebraic sets + morphisms}\} \to$$

$$\{\textit{Category of finitely generated integral domains over } k + k\textit{−homomorphisms}\}$$

which is an equivalence of categories.

Example 11. Consider $\mathbb{P}^2$. Its ring of global regular functions is $\Gamma(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2})$. Note: $\mathbb{P}^2 = \mathbb{A}^2 \cup H$ where H is the line at ∞ (= a copy of $\mathbb{P}^1$).

A polynomial $f \in k[x, y]$ on the affine chart $\mathbb{A}^2$ extends regularly across the line at infinity only if it is constant. Hence

$$\Gamma(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}) = k.$$

Thus $\mathbb{P}^2$ and a point have the same ring of global regular functions. This is why global functions alone do not determine a non-affine variety.

Remark 12. In general,

$$\mathbb{P}^n = U_0 \cup H = \mathbb{A}^n \cup \mathbb{P}^{n-1}.$$

Definition 13. A **closed set** in k^n is to be a closed algebraic set $V(A)$. These define a topology in k^n, called the **Zariski topology.** An irreducible algebraic set $\Sigma \subseteq k^n$ is given the induced topology, also called the Zariski topology. The closed sets of Σ are exactly the sets $V(A)$ where A is an ideal in $\Gamma(\Sigma)$.

Remark 14.

1. All morphisms are continuous in the Zariski topology. In fact, this is the weakest topology making morphisms continuous.
2. A basis for the open sets in the Zariski topology on Σ is given by the open sets $\Sigma_f := \{x \in \Sigma \mid f(x) \neq 0\}$ for elements $f \in \Gamma(\Sigma)$.
3. On k itself (for k infinite), the Zariski topology is the cofinite topology: the proper closed sets are finite.
4. Not all continuous maps are morphisms.
5. Zariski topology on the product $\neq$ Zariski product topology

Definition 15. A topological space X is **noetherian** if its closed sets satisfy the descending chain condition. It is equivalent to require that every open subset be quasi-compact.

Since ideals in $k[X_1, \ldots, X_n]$ satisfy ascending chain condition, the Zariski topology is noetherian.

Definition 16. A **closed set in** $\mathbb{P}_n(k)$ is to be a closed algebraic set $V(A)$. These define a topology in $\mathbb{P}_n(k)$, called the **Zariski topology**. Irreducible algebraic sets are again given the induced topology. As in the affine case, a basis for the open sets is given by

$$[\mathbb{P}_n(k)]_f = \{x \in \mathbb{P}_n(k) \mid f(x) \neq 0\}$$

where f is a **homogeneous** polynomial.

Remark 17. The Zariski topology on $\mathbb{P}_n(k)$ is noetherian.

Problem 1.2. Check that $\mathbb{P}_n(k)_{X_0}$ is homeomorphic to k^n under the usual map

$$(x_0 : \cdots : x_n) \mapsto (\frac{x_1}{x_0}, \ldots, \frac{x_n}{x_0})$$

Call this map $\varphi : \mathbb{P}_n(k)_{X_0} \to k^n$.

Well-defined: If $(x_0 : \cdots : x_n) = (\lambda x_0 : \cdots : \lambda x_n)$, then

$$\varphi(x_0 : \cdots : x_n) = (x_1/x_0, \ldots, x_n/x_0) = (\lambda x_1/\lambda x_0, \ldots, \lambda x_n/\lambda x_0) = \varphi(\lambda x_0 : \cdots : \lambda x_n).$$

Injective: If $(x_0 : \cdots : x_n) \neq (y_0 : \cdots : y_n)$, then for all $\lambda \neq 0$, there exists some $i \in \{0, \ldots, n\}$ such that $x_i \neq \lambda y_i$. If we had $\varphi(x_0 : \cdots : x_n) = \varphi(y_0 : \cdots : y_n)$, then

$$(x_1/x_0, \ldots, x_n/x_0) = (y_1/y_0, \ldots, y_n/y_0) \implies x_i/x_0 = y_i/y_0 \text{ for all } i = 1, \ldots, n$$

$$\implies x_i = \left(\frac{x_0}{y_0}\right) y_i \text{ for all } i = 1, \ldots, n,$$

so $(x_0 : \cdots : x_n) = (y_0 : \cdots : y_n)$, contradicting our assumption that the two projective points were distinct. Surjective: If $(x_1, \ldots, x_n) \in k^n$ then $(1 : x_1 : \cdots : x_n) \in \mathbb{P}_n(k)_{X_0}$ and
$\varphi(1 : x_1 : \cdots : x_n) = (x_1, \ldots, x_n)$.

Defining an inverse ψ :
Let

$$\psi : k^n \to \mathbb{P}_n(k)_{X_0}, \quad (x_1, \ldots, x_n) \mapsto (1 : x_1 : \cdots : x_n)$$

Then

$$\psi(\varphi(x_0 : \cdots : x_n)) = \psi(x_1/x_0, \ldots, x_n/x_0) = (1 : x_1/x_0 : \cdots : x_n/x_0) = (x_0 : \cdots : x_n)$$

and

$$\varphi(\psi(x_1, \ldots, x_n)) = \varphi(1 : x_1 : \cdots : x_n) = (x_1/1, \ldots, x_n/1) = (x_1, \ldots, x_n).$$

φ is continuous: Suppose $f \in k[x_1, \ldots, x_n]$ and $V(f) \subseteq k^n$. If $(x_0 : \cdots : x_n) \in \mathbb{P}_n(k)_{X_0}$ then:

$$\varphi(x_0 : \cdots : x_n) \in V(f) \iff f(x_1/x_0, \ldots, x_n/x_0) = 0 \iff$$
$$x_0^{\deg f} f(x_1/x_0, \ldots, x_n/x_0) = 0 \iff F(x_0, \ldots, x_n) = 0$$

where $F(x_0, \ldots, x_n) := x_0^{\deg f} f(x_1/x_0, \ldots, x_n/x_0) = 0$ is homogeneous.

Thus $\varphi^{-1}(V(f)) = V(F) \cap \mathbb{P}_n(k)_{X_0}$ by Problem 1. But $V(F) \cap \mathbb{P}_n(k)_{X_0}$ is closed in the subspace topology $\implies \varphi^{-1}(V(f))$ is closed. So φ is continuous.

ψ is continuous: Let $(F) \cap \mathbb{P}_n(k)_{X_0} \subseteq \mathbb{P}_n(k)_{X_0}$ be closed (F is a homogeneous polynomial). Then $(x_1, \ldots, x_n) \in k^n$ has $\psi(x_1, \ldots, x_n) \in V(F) \cap \mathbb{P}_n(k)_{X_0} \iff F(1, x_1, \ldots, x_n) = 0$. Thus $\psi^{-1}(V(F) \cap \mathbb{P}_n(k)_{X_0}) = V(F(1, x_1, \ldots, x_n))$, which is closed in k^n. So ψ is continuous.

Conclude: the map

$$\mathbb{P}_n(k)_{X_0} \to k^n, \quad (x_0 : \cdots : x_n) \mapsto (x_1/x_0, \ldots, x_n/x_0)$$

is bijective, continuous, and has a continuous inverse

$$(a_1, \ldots, a_n) \mapsto (1 : a_1 : \cdots : a_n)$$

and thus $\mathbb{P}_n(k)_{X_0} \cong k^n$ as topological spaces.

1.4 Sheaves and affine varieties

Definition 1. Let X be a topological space. A **presheaf** F on X consists of

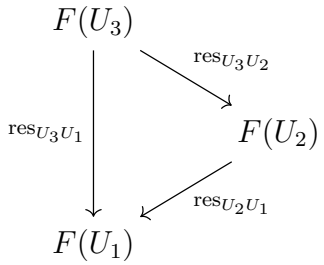
1. for all open $U \subseteq X$, a set $F(U)$
2. for all pairs of open sets $U_1 \subseteq U_2$, a restriction map

$$\mathrm{res}_{U_2 U_1} : F(U_2) \to F(U_1)$$

such that

(a) $\mathrm{res}_{U,U} = \mathrm{id}_{F(U)}$ for all U

(b) If $U_1 \subseteq U_2 \subseteq U_3$, then the following diagram commutes:



Definition 2. If F_1, F_2 are presheaves on X, a **map of presheaves** $\varphi : F_1 \to F_2$ is a collection of maps $\varphi(U) : F_1(U) \to F_2(U)$ for each open U such that if $U \subseteq V$, the following diagram commutes:

$$\begin{array}{ccc} F_1(V) & \xrightarrow{\varphi(V)} & F_2(V) \\ \downarrow{\scriptstyle \mathrm{res}_{V,U}} & & \downarrow{\scriptstyle \mathrm{res}_{V,U}} \\ F_1(U) & \xrightarrow{\varphi(U)} & F_2(U) \end{array}$$

Definition 3. A presheaf F is a **sheaf** if for every collection $\{U_i\}$ of open sets in X with $U = \bigcup_i U_i$, the diagram

$$F(U) \longrightarrow \prod F(U_i) \underset{g}{\overset{f}{\rightrightarrows}} \prod_{i,j} F(U_i \cap U_j)$$

is exact, i.e. the map

$$\mathrm{res}_{U,U_i} : F(U) \to \prod F(U_i)$$

is injective, and its image is the set on which

$$\mathrm{res}_{U_i, U_i \cap U_j} : \prod_i F(U_i) \to \prod_{i,j} F(U_i \cap U_j)$$

and

$$\mathrm{res}_{U_j, U_i \cap U_j} : \prod_j F(U_j) \to \prod_{i,j} F(U_i \cap U_j)$$

agree.

What this really means:

1. If $x_1, x_2 \in F(U)$, and for all i, $\mathrm{res}_{U,U_i} x_1 = \mathrm{res}_{U,U_i} x_2$, then $x_1 = x_2$.
   (i.e. **elements are uniquely determined by local data**)

2. If we have a collection of elements $x_i \in F(U_i)$ such that $\mathrm{res}_{U_i, U_i \cap U_j} x_i = \mathrm{res}_{U_j, U_i \cap U_j} x_j$ for all i, j, then there exists an $x \in F(U)$ such that $\mathrm{res}_{U,U_i} x = x_i$ for all i.
   (i.e. **if we have local data which are compatible, then they patch together to form something in** $F(U)$)

Example 4. Let X, Y be topological spaces. For all open sets U, let

$$F(U) := \{\text{continuous maps } U \to Y\}$$

This is a presheaf with restriction maps being usual restriction.
This is a sheaf, since a function is continuous on $U_i \iff$ its restrictions to each U_i are continuous.

Example 5. Let X, Y be differentiable manifolds. For all open sets U, let

$$G(U) := \{\text{continuous functions } U \to Y \text{ which have relatively compact image}\}$$

This is a subpresheaf of the first example, but need not be a sheaf (we can glue together bounded functions to get an unbounded function). Thus, this fails on gluing, but passes on locality.

Continuity is local, but boundedness is not local.

Example 6. Let X be a topological space. For any open set U, let

$$F(U) := \{\text{locally constant real-valued functions on } U\}/\{\text{constant functions on } U\}$$

This is a presheaf.
This is not a sheaf: Every $s \in F(U)$ goes to zero in $\prod F(U_i)$ for some covering $\{U_i\}$. But if U is not connected, $F(U) \neq (0)$.

Definition 7. Let F be a sheaf on X, $x \in X$. The collection of $F(U)$, U open containing x, is a direct system under restriction toward smaller neighborhoods, and we can form

$$F_x := \varinjlim_{x \in U} F(U)$$

called the **stalk of F at x**.

Example 8. Let

$$F(U) = \{\text{continuous functions } U \to \mathbb{R}\}$$

Then F_x is called the set of **germs of continuous functions at** x, and

$$F_x = \bigcup_{x \in U} F(U)/\sim$$

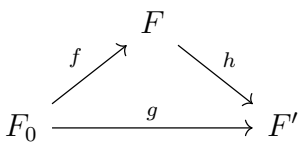
where $f_1 \sim f_2$ if f_1 and f_2 agree in a neighborhood of x.

Example 9. If $\mathcal{F}$ = sheaf of holomorphic functions on $\mathbb{C}$, then the stalk at 0 consists of germs represented by power series with positive radius of convergence at 0, i.e.

$$\mathcal{O}_{\mathbb{C},0} \cong \mathbb{C}\{z\},$$

where $\mathbb{C}\{z\}$ denotes the ring of convergent power series. Question: What about the sheaf of C^∞ functions $\mathbb{R}^2 \to \mathbb{R}$ at the origin?
Two functions can agree on an open set containing the origin and differ on another.
An element of this stalk is a germ of C^∞ functions at the origin.

Sheafification of a presheaf:
Let F_0 be a presheaf on X. Then there is a sheaf F and a map $f : F_0 \to F$ such that if $g : F_0 \to F'$ is any map with F' a sheaf, then there is a unique map $h : F \to F'$ such that the following diagram commutes:



To get F from F_0, you basically just: first identify things which have the same restrictions, then add in all the things which can be patched together.

Example 10. Revisiting $G(U) = \{\text{continuous functions } U \to Y \text{ which have relatively compact image}\}$ (which we said was a presheaf but not a sheaf), if X is locally compact, the sheafification of this presheaf is the sheaf of all continuous functions on X.

Example 11. Revisiting $F(U) = \{\text{locally constant real-valued functions on } U\}/\{\text{constant functions on } U$ the sheafification of this presheaf is (0).

Notation: We write $\Gamma(U, F)$ for $F(U)$ (this is called the set of **sections of** F **over** U) $\Gamma(X, F)$ is the set of **global sections of** F.
In other contexts, we may denote $F(X)$ by $H^0(X, F)$ and call it the **zeroth cohomology group**.

If for all U, $F(U)$ is a group (ring, etc.) and the restriction maps are group (ring, etc.) homomorphisms, then F is called a **sheaf of groups (rings, etc.)**. In this case, F_x is a group (ring, etc.) and so on.

Example 12. Let X be a topological space. Let

$$F_{\text{cont},X}(U) := \{\text{continuous functions } U \to \mathbb{R}\}.$$

Then $F_{\text{cont},X}(U)$ is a sheaf of rings.

Definition 13. If $g : X \to Y$ is a continuous function, the operation $f \mapsto f \circ g$ gives us the following maps: for every open $U \subseteq Y$, a map $F_{\text{cont},Y}(U) \to F_{\text{cont},X}(g^{-1}U)$ such that the following diagram commutes for all open sets $V \subseteq U$:

$$\begin{array}{ccc} F_Y(U) & \longrightarrow & F_X(g^{-1}U) \\ \big\downarrow{\scriptstyle \text{res}} & & \big\downarrow{\scriptstyle \text{res}} \\ F_Y(V) & \longrightarrow & F_X(g^{-1}V) \end{array}$$

This set-up is called a **morphism of the pair** (X, F_X) **to the pair** (Y, F_Y)**.**

Example 14. Let X, Y be differentiable manifolds. Let $F_{\text{diff},X}$ and $F_{\text{diff},Y}$ be subsheaves of $F_{\text{cont},X}$ and $F_{\text{cont},Y}$ of differentiable functions.
Let $g : X \to Y$ be a continuous map. Then g is differentiable $\iff$ for all open sets $U \subseteq Y$, $f \in F_{\text{diff},Y}(U) \implies f \circ g \in F_{\text{diff},X}(g^{-1}U)$.

Example 15. Say X, Y are complex analytic manifolds, Let $F_{\text{an},X}$ and $F_{\text{an},Y}$ be the sheaves of holomorphic functions.

A continuous map $g : X \to Y$ is holomorphic $\iff$ for all open sets $U \subseteq Y$, $f \in F_{\text{an},Y}(U) \implies f \circ g \in F_{\text{an},X}(g^{-1}U)$.

Definition 16. Let $X \subseteq k^n$ be an irreducible algebraic set. Let R be its affine coordinate ring.
X irreducible $\implies$ $I(X)$ is prime and R is a domain. Let $K = \text{Frac}(R)$.

Recall: R has been identified with a ring of functions on X.
For $x \in X$, let

$$m_x := \{f \in R \mid f(x) = 0\}.$$

This is a maximal ideal ($\ker(R \to k, f \mapsto f(x))$. Let

$$\mathcal{O}_{X,x} := R_{m_x} = \{\frac{f}{g} \mid f, g \in R, g(x) \neq 0\} \subseteq K.$$

Now for U open in X, let

$$\mathcal{O}_X(U) := \bigcap_{x \in U} \mathcal{O}_{X,x}.$$

- Each $\mathcal{O}_X(U)$ is a subring of K.
- If $V \subseteq U$, then $\mathcal{O}_X(U) \subseteq \mathcal{O}_X(V)$.

Taking inclusion as the restriction map, this defines a sheaf $\mathcal{O}_X$. This is called the **structure sheaf of** X.

Remark 17. The elements of $\mathcal{O}_X(U)$ can be viewed as functions on U:

If $F \in \mathcal{O}_X(U)$ and $x \in U$, then $F \in \mathcal{O}_{X,x}$ so $F = \frac{f}{g}$ where $g(x) \neq 0$.
Define $F(x) = \frac{f(x)}{g(x)}$.
If $F(x) = 0$ for all $x \in U$, then $F = 0$, so we can identify $\mathcal{O}_X(U)$ with the associated ring of functions on U.

Here $\mathcal{O}(U)$ = polynomial functions on U = regular functions on U. This is a sheaf because being regular is local.

Example 18. $\mathcal{O}(\mathbb{A}^1) = k[X]$
Let $U_1 : x \neq 1$. Then

$$\mathcal{O}(U_1) = k[x, \frac{1}{x-1}] \cong k[x, y]/((x-1)y - 1)$$

and letting $U_2 : x \neq 2$,

$$\mathcal{O}(U_2) = k[x, \frac{1}{x-2}].$$

It turns out: $\mathcal{O}_p = k[x]_{(p)}$.

Example 19. Say $\pi : Y \to X$. For every $U \subseteq X$, consider $\mathcal{F}(U) := \{\text{sections of } \pi|_U\}$. This is a sheaf. It is this example that motivates the name "sheaf."

Proposition 20. *Let X be an irreducible algebraic set. Let $R = \Gamma(X)$. Let $f \in R$, and $X_f = \{x \in X \mid f(x) \neq 0\}$. Then*

$$\mathcal{O}_X(X_f) = R_f.$$

Corollary 21. $\Gamma(X, \mathcal{O}_X) = R$.

So:
The set of global functions of $\mathcal{O}_X$ is

$$\mathcal{O}_X(X) = \bigcap_{x \in X} \mathcal{O}_{X,x} = \bigcap_{x \in X} \left\{ \frac{f}{g} \mid f, g, \in \Gamma(X), g(x) \neq 0 \right\}$$

and indeed equals $\Gamma(X)$.

Remark 22.

1. If $f \in \mathcal{O}_X(U)$ and f vanishes nowhere on U, then $\frac{1}{f} \in \mathcal{O}_X(U)$, since: $f(x) \neq 0 \implies \frac{1}{f} \in \mathcal{O}_{X,x}$.

2. The stalk of $\mathcal{O}_X$ at x is $\mathcal{O}_{X,x}$, since:
   The sets X_f are a basis of the Zariski topology of X, so

   $$\varinjlim_{x \in U} \mathcal{O}_X(U) = \varinjlim_{x \in X_f} \mathcal{O}_X(X_f) = \varinjlim_{f(x) \neq 0} R_f.$$

   Since all restriction maps in our sheaf are injective, this equals

   $$\bigcup_{f(x) \neq 0} R_f = \mathcal{O}_{X,x}.$$

3. The field $K (= \mathrm{Frac}(\Gamma(X)))$ can be recovered from the sheaf $\mathcal{O}_X$:
   X is irreducible $\implies$ the intersection of any two nonempty open sets is nonempty $\implies$ we have an inverse system containing all open sets $\implies$ we can define a **generic stalk** of any sheaf F on X.
   K is the generic stalk of the structure sheaf $\mathcal{O}_X$.

4. If $h \in \mathcal{O}_X(U)$ for some open $U \subseteq X$, then it's not necessarily true that $h = f/g$ with $f, g \in R$ and g vanishing nowhere on U.
   For example:
   Let $X = V(xw - yz) \subseteq k^4$.
   Let $U = X_y \cup X_w$.
   Define $h \in \mathcal{O}_X(U)$ so that $h = x/y$ on X_y and $h = z/w$ on X_w
   One can show (Mumford page 21) that h cannot equal f/g, $g \neq 0$ in U.
   But if U has the form X_g, then this is necessarily the case (by the Proposition).

Remark 23. $V(f)$ is called a principal hypersurface. $X_f = \{f \neq 0\}$ is called a distinguished open / basic open, and is the complement of $V(f)$.

Proposition 24. *Let $X \subseteq k^n, Y \subseteq k^m$ be irreducible algebraic sets. Let $f : X \to Y$ be continuous. TFAE:*

1. *f is a morphism.*
2. *For every open $U \subseteq Y$ and every $g \in \Gamma(U, \mathcal{O}_Y)$, one has $g \circ f \in \Gamma(f^{-1}(U), \mathcal{O}_X)$.*
3. *For every $x \in X$ and every regular function g defined on a neighborhood of $f(x)$, the germ of $g \circ f$ at x belongs to $\mathcal{O}_{X,x}$.*

4. *For every $x \in X$, pullback induces a local homomorphism $\mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$.*

Definition 25. An **affine variety** is a topological space X plus a sheaf of k-valued functions $\mathcal{O}_X$ on X, which is isomorphic to an irreducible algebraic subset of k^n, plus the structure sheaf of X (as in Definition **??**).

Remark 26. Many books do not require irreducible.

Definition 27. The affine variety $(k^n, \mathcal{O}_{k^n})$ is $\mathbb{A}^n$, called **affine n-space**.

Bijective morphisms need not be isomorphisms:

- A bijective continuous map from a compact space to a Hausdorff space is a homeomorphism; analogous statements require additional hypotheses in other categories.
- For differentiable manifolds, a bijective smooth map need not be a diffeomorphism. Consider $f : \mathbb{R} \to \mathbb{R}, x \mapsto x^3$; its inverse is not differentiable at 0.
- This is also not true for affine varieties. Some examples:

  Example 28. If $\mathrm{char}(k) = p \neq 0$, $f : \mathbb{A}^1 \to \mathbb{A}^1, t \mapsto t^p$ is bijective and on the ring level corresponds to $k[X] \hookleftarrow k[X^p]$ which are not equal.

  Example 29. Let k be an algebraically closed field, $f : \mathbb{A}^1 \to \mathbb{A}^2, t \mapsto (t^2, t^3)$ has image the irreducible closed curve $C : X^3 = Y^2$. $f : \mathbb{A}^1 \to C$ is a bijection and corresponds to $k[T] \hookleftarrow k[T^2, T^3]$, which are not equal.

  Example 30. Let $f : \mathbb{A}^1 \to \mathbb{A}^2, X = t^2 - 1, Y = t(t^2 - 1)$. The image of this morphism is the curve $D : Y^2 = X^2(X + 1)$.
  f is bijective between $\mathbb{A}^1$ and D except that $t = 1$ and $t = -1$ both map to the origin. If $X_1 = \mathbb{A}^1 \setminus \{1\}$, then f restricts to a bijection $f' : X_1 \to D$, but this morphism corresponds to $k[T, (T-1)^{-1}] \hookleftarrow k[T^2 - 1, T(T^2 - 1)]$, which are not equal.

Induced affine variety structure on open and closed subsets of affine varieties:
Let Y be an irreducible closed subset of an affine variety $(X, \mathcal{O}_x)$.
Define an induced sheaf $\mathcal{O}_X$ of functions on Y as follows:
If V is open in Y,

$$\mathcal{O}_Y(V) = \{f : V \to k \mid \forall x \in V, \exists \text{ a neighborhood } U \text{ of } x \text{ in } X \text{ and } F \in \mathcal{O}_X(U) \text{ so that } f = F|_{U \cap V}\}$$

Proposition 31. *$(Y, \mathcal{O}_Y)$ is an affine variety.*

Proposition 32. *Let $(X, \mathcal{O}_X)$ be an affine variety. Let $f \in \Gamma(X, \mathcal{O}_X)$. Then $(X_f, \mathcal{O}_X|_{X_f})$ is an affine variety.*

Here we pushed the zeroes of f out to infinity. For example:
Suppose $X = \mathbb{A}^1$, f is the coordinate X_1. Then $B = (1 - X_1X_2) \implies$ the projection of the B down to the axis is an isomorphism with X_f.

Example 33. Not all open subsets of affine varieties are affine varieties. You cannot push the origin in $\mathbb{A}^2$ out to infinity.

$X = \mathbb{A}^2 \setminus \{(0,0\}$ is not an affine variety: if we had some embedding $\mathbb{A}^2 \setminus \{(0,0\} \hookrightarrow k^n$, then since no rational function is well-defined on $\mathbb{A}^2 \setminus \{(0,0\}$ but not at $(0,0)$, the coordinate functions giving this embedding would have to extend to all of $\mathbb{A}^2$. So the restriction map $\Gamma(\mathbb{A}^2, \mathcal{O}_{\mathbb{A}^2}) = k[x,y] \to \Gamma(X, \mathcal{O}_X)$ is bijective.

Then if X were affine, it would mean that since X and $\mathbb{A}^2$ have the same coordinate rings, that they must be isomorphic as varieties, which is not true.

This is similar to the idea that a holomorphic function on $\mathbb{C} \times \mathbb{C} \setminus \{(0,0\}$ is necessarily holomorphic at the origin.

Remark 34. We call an open subset of an affine (which is not itself affine) "quasi-affine."

1.5 Prevarieties and morphisms

Definition 1. A topological space X plus a sheaf $\mathcal{O}_X$ of k-valued functions on X is a **prevariety** if

1. X is connected, and
2. there is a finite open covering $\{U_i\}$ of X such that for all i, $(U_i, \mathcal{O}_X|_{U_i})$ is an affine variety.

Definition 2. An open subset U of X is called an **open affine set** if $(U, \mathcal{O}_X|_U)$ is an affine variety.

The open affine sets are a basis of the topology.

Definition 3. A topological space is **irreducible** if it is not the union of two proper closed subsets (equivalently, the intersection of ay two nonempty sets is nonempty).

Proposition 4. *Every prevariety X is an irreducible topological space.*

So every open set is dense.

Remark 5. If you remove the irreducible requirement, opens need not be dense. Consider the union X of the x and y axes. In here, the x-axis minus the origin is open but not dense, since its closure does not contain the x-axis.

Proposition 6. *If X is a prevariety, then the closed sets of X satisfy DCC, i.e. X is noetherian.*

So every variety is quasi-compact.

Proposition 7. *Let X be a noetherian topological space. Then every closed set Z in X can be written uniquely as an irredundant union of finitely many irreducible closed sets, called the components of Z.*

Remark 8. The stalk at a point is all functions regular at the point. They are actually all regular in some neighborhood of the point. We often specify that we are talking about a set of functions all regular at "some point," and in these situations, we call the point the **generic point.**

Example 9.

$$k(\mathbb{P}^1) = \varinjlim_{\text{nonempty opens in } \mathbb{P}^1} \mathcal{O}_{\mathbb{P}^1}(U)$$

- Why is $k(X)$ (in general from the definition) a field?

  Elements of $k(X)$ are equivalence classes of pairs (U, f). If $f \in k(X)$ is a nonzero element represented by (U, f), then because it is nonzero in the limit, f does not identically vanish on any non-empty open subset. Thus $V(f) \subseteq U$ is a proper closed subset.

  Since X is irreducible, $U_f = U \setminus V(f)$ is a nonempty, dense open set. By definition, $f(x) \neq 0$ for all $x \in U_f$. So $1/f$ is a well-defined regular function on U_f, meaning $(U_f, 1/f) \in k(X)$.

  Intuitively, U irreducible $\implies$ $R = \mathcal{O}_X(U)$ is a domain $\implies$ $k(X) \cong \mathrm{Frac}(R)$, a field.

  More concretely, given f/g with g nonzero, pick $U = \{g \neq 0\}$. U is a dense open. So $f/g \in k(\mathbb{A}^1)$. And conversely, everything in $k(\mathbb{A}^1)$ is in some $\mathcal{O}_X(U)$, so looks like some f/g where f, g are polynomials.

- Why is $k(\mathbb{P}^1) = k(t)$?

  Because $\mathbb{P}^1$ is irreducible, every non-empty open set is dense. The collection of open subsets contained within the standard affine chart $U_0 = \{[x_0 : x_1] \mid x_0 \neq 0\} \cong \mathbb{A}^1$ forms a cofinal system in the poset of all non-empty open sets. Therefore, the direct limit restricts isomorphically to U_0:

  $$\varinjlim_{U \subset \mathbb{P}^1} \mathcal{O}_{\mathbb{P}^1}(U) \cong \varinjlim_{V \subset U_0} \mathcal{O}_{U_0}(V) = k(U_0)$$

  The open subset U_0 is isomorphic to the affine line $\mathbb{A}^1$, meaning its coordinate ring is the polynomial ring in one variable. Choosing the coordinate function $t = x_1/x_0$, we have:

  $$\mathcal{O}_{\mathbb{P}^1}(U_0) \cong k[t]$$

  For an affine variety, the direct limit of the structure sheaf over all non-empty open sets is the fraction field of its coordinate ring. Thus:

  $$k(\mathbb{P}^1) \cong k(U_0) \cong \mathrm{Frac}(\mathcal{O}_{\mathbb{P}^1}(U_0)) \cong \mathrm{Frac}(k[t]) = k(t)$$

Remark 10. This generalizes:

$$k(\mathbb{P}^n) = k(x_1, \cdots, x_n).$$

Example 11. Let $C = \{P(x, y) = 0\}$ be a curve in $\mathbb{A}^2$ (where P is an irreducible polynomial). Then

$$\mathcal{O}(C) = k[x, y]/(P)$$

and

$$k(C) = \text{Frac}(k[x, y]/(P)) = \{f/g \mid f, g \in k[x, y], g \neq 0\} = k(x)[y]/(P).$$

Definition 12. If X is a prevariety, the **function field** $k(X)$ is the generic stalk of $\mathcal{O}_X$, i.e.

$$k(X) = \varinjlim_{\text{non-empty open } U} \mathcal{O}_X(U).$$

- $k(X)$ equals the function field of each open affine set U in X, because of density
- The elements of $k(X)$ are called **rational functions on** X

Definition 13. Let $Y \subseteq X$ be an irreducible closed subset. Define

$$\mathcal{O}_{Y,X} := \varinjlim_{\text{open sets } U \text{ s.t. } U \cap Y \neq \varnothing} \mathcal{O}_X(U).$$

Fix an open affine $U \subseteq X$ which meets Y. Let R be the coordinate ring of U. Let $P = I(Y \cap U)$. Then

$$\mathcal{O}_{Y,X} = \varinjlim_{\text{open sets } U_f, f \in R, f \notin P} \mathcal{O}_X(U_f) = R_P.$$

So $\mathcal{O}_{Y,X}$ is a local ring with fraction field $k(X)$ and residue field $k(Y)$.

Example 14. If $X = \mathbb{A}^2$ and Y is a line in the plane, then taking $\mathcal{O}_{Y,X}$, we get all functions defined in a neighborhood of some point in Y (i.e. on most points in Y), i.e. at a generic point of Y, or at **the generic point**.

Call this "the local ring at the generic point of Y."

Example 15. Let $X = \mathbb{A}^2, Y = x$-axis.

If $Z =$ origin, then $\mathcal{O}_{Z,X} = \mathcal{O}_Z = k[x, y]_{(x,y)}$ (invert everything that does not vanish at the origin)

$\mathcal{O}_{Y,X} =$ functions defined in a neighborhood of $y = 0 = k[x, y]_{(y)}$

Proposition 16. *An open subset of a prevariety is a prevariety.*

Note that this was not true about affine varieties, since $\mathbb{A}^2 \setminus \{(0, 0\}$ is not an affine variety. But since $\mathbb{A}^2$ is a prevariety, $\mathbb{A}^2 \setminus \{(0, 0\}$ is a prevariety; we can write is as $\mathbb{A}^2 \setminus \{(y = 0\} \cup \mathbb{A}^2 \setminus \{(x = 0\}$.

Let $Y \subseteq X$ is a closed irreducible subset and X is a prevariety. If V is open in Y, define

$\mathcal{O}_Y(V) := \{f : V \to k \mid \forall x \in V \exists$ a neighborhood U of x in $X, F \in \mathcal{O}_X(U)$ so that $f = F|_{U \cap V}\}$

Proposition 17. *$(Y, \mathcal{O}_Y)$ is a prevariety.*

The last two propositions allow us to give a prevariety structure to any locally closed subset of a previariety X. Call the set of prevarieties obtained in this way **sub-prevarieties** of X.

Example 18. We can obtain $\mathbb{P}^1$ by gluing two copies U, V of $\mathbb{A}^1$:

If u, v are the coordinates on U, V, let $U_0 = \{u \neq 0\} \subseteq U$ and $V_0 = \{v \neq 0\} \subseteq V$.
Then $\Gamma(U\mathcal{O}_U) = k[u]$, so $\Gamma(U\mathcal{O}_{U_0}) = k[u, u^{-1}]$ and similarly for V, V_0.
Define

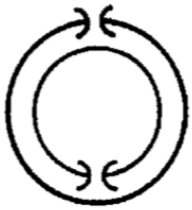
$$\varphi : U_0 \to V_0, u = a \mapsto v = \frac{1}{a}.$$

This induces

$$\varphi^* : k[v, v^{-1}] \to k[u, u^{-1}], v \mapsto u^{-1}, v^{-1} \mapsto u.$$

φ^* is an isomorphism of rings, so φ is an isomorphism of varieties.

Patch U and V together via φ, i.e. form $U \cup V$ with U_0 and V_0 identified via φ. This space is homeomorphic to $\mathbb{P}^1$. We call this the variety $\mathbb{P}^1$:



If we had patched via $v \mapsto u, v^{-1} \mapsto u^{-1}$, we would get $\mathbb{A}^1$ with a point doubled, which is a prevariety but not a variety. It is not "separated" (this idea is the analog of Hausdorff in algebraic geometry).

Improve the following exposition:
Definition of Projective Varieties:
Let $P \subseteq k[X_0, \ldots, X_n]$ be a homogeneous prime ideal. Let $X = V(P) \subseteq \mathbb{P}_n(k)$.

The elements of $k[X_0, \ldots, X_n]$ are not well-defined (because we can rescale anything, so there are not unique coordinates) but the ratio of any two having the same degree is a function (rescaling in the numerator and denominator cancels out).

P homogeneous $\implies$ $R = k[X_0, \ldots, X_m]/P$ is a graded ring $\bigoplus_{n=0}^{\infty} R_n$ and a domain. Let $k(X)$ be the zero-th piece of the localization of R with respect to homogeneous elements, i.e.

$$k(X) = \left\{ \frac{f}{g} \mid f, g \in R_n \text{ for the same } n \right\}.$$

If $x \in X$ and $g \in R_n$, saying $g(x) \neq 0$ still makes sense. Define a ring in $k(X)$:

$$\mathcal{O}_{X,x} := \left\{ \frac{f}{g} \in k(X) \mid g(x) \neq 0 \right\}$$

and the maximal ideal

$$m_x := \left\{ \frac{f}{g} \in k(X) \mid g(x) \neq 0, f(x) = 0 \right\} \subseteq \mathcal{O}_{X,x}.$$

$\mathcal{O}_{X,x}$ is a local ring. Now define a sheaf: for all open $U \subseteq X$,

$$\mathcal{O}_X(U) := \bigcap_{x \in U} \mathcal{O}_{X,x}$$

We can identify $\mathcal{O}_X$ with a sheaf of k-valued functions:
If $x \in U$ and $a \in \Gamma(U, \mathcal{O}_X)$, then $a = f/g$ where $f, g \in R_n$, $g(x) \neq 0$.
Define $a(x) := f(x)/g(x)$ (i.e. lift f, g to F, G homogeneous polynomials of degree n in $k[X_0, \ldots, X_n]$, let $\tilde{x}$ be a set of homogeneous coordinates of x, and take $F(\tilde{x})/G(\tilde{x})$)
Check: if $a \in \Gamma(U, \mathcal{O}_X)$ and $a(x) = 0$ for all $x \in U$, then $a = 0$. This will come out of the next step.

For all i , $0 \leq i \leq n$,

$$(X \cap \mathbb{P}_n(k))_{X_i}, \text{ restriction of } \mathcal{O}_X)$$

is an affine variety (so in particular, $(X, \mathcal{O}_X)$ is locally isomorphic to an affine variety. Suffices to check this only for X_0:

For every homogeneous polynomial $F \in P$, $F/X_0^{\deg F}$ can be written as a polynomial F' in the variables $\frac{X_1}{X_0}, \ldots, \frac{X_n}{X_0}$.
Let $P' \subseteq k[Y_1, \ldots, Y_n]$ be the ideal generated by all these F'.
Map $k[Y_1, \ldots, Y_n]$ into a subring of $k(X)$ via $Y_i \mapsto X_i/X_0$. The kernel of this map is P' so P' is prime.
We get an isomorphism

$$\varphi : (X \cap \mathbb{P}_n(k))_{X_0} \to X' = V(P') \subseteq k^n \text{ via } x \mapsto \left(\frac{x_1}{x_0}, \ldots, \frac{x_n}{x_0} \right).$$

For $x \in (X \cap \mathbb{P}_n(k))_{X_0}$, $\mathcal{O}_{X,x} = \{f/g \in k(X) \mid f, g, \text{ in same } R_m, g(x) \neq 0\}$.
X' has affine coordinate ring $R' = k[Y_1, \ldots, Y_n]/P'$.
$\mathcal{O}_{X',\varphi(x)} = \{F/G \in k(X') \mid F, G \in R', G(\varphi(x)) \neq 0\}$.
φ extends to an isomorphism $k(X') \to k(X)$.
Claim: the isomorphism takes $\mathcal{O}_{X',\varphi(x)}$ precisely onto $\mathcal{O}_{X,x}$. Proof:
It clearly (?) maps $\mathcal{O}_{X',\varphi(x)}$ into $\mathcal{O}_{X,x}$. And if $f, g \in R_m, g(x) \neq 0$, then

$$\frac{f}{g} = \frac{f/X_0^m}{g/X_0^m}$$

in $k(X)$ and $f/X_0^m, g/X_0^m$ come from F, G in R' with $G(\varphi(x)) \neq 0$.
So the local rings correspond, and since the sheaves are defined by intersecting local rings, they correspond.

Definition 19. Let X, Y be prevarieties. A map $f : X \to Y$ is a **morphism** if f is continuous and for all open sets V in Y,

$$g \in \Gamma(V, \mathcal{O}_Y) \implies g \circ f \in \Gamma(f^{-1}(V), \mathcal{O}_X).$$

Proposition 20. *Let $f : X \to Y$ be any map. Let $\{V_i\}$ be a collection of open affine subsets covering Y. Suppose that $\{U_i\}$ is an open covering of X such that 1) $f(U_i) \subseteq V_i$, and 2) f^* maps $\Gamma(V_i, \mathcal{O}_Y)$ into $\Gamma(U_i, \mathcal{O}_X)$. Then f is a morphism.*

Example 21. Let C be the plane cubic curve defined in homogeneous coordinates by

$$zy^2 = x(x^2 - z^2).$$

In $C \cap \mathbb{P}_2(k)_x$, with affine coordinates $X = x/z, Y = y/z$, the equation of C becomes $Y^2 = X(X^2 - 1)$. For all lines ℓ through the origin, interchange the two points in $\ell \cap C$ (other than the origin).

When we write this out (Mumford page 31), we do not get polynomials. The problems are $P_1 = (0 : 1 : 0)$ (point at ∞), $P_2 = (0 : 0 : 1)$ (origin), $Q_1 = (1 : 0 : 1)$ and $Q_2 = (-1 : 0 : 1)$ (other points on X-axis).

But we can restrict to different subsets of C and get morphisms there, and then, using the Proposition, f will be a morphism (see Mumford page 32).

Note: On a cubic curve, there are 9 inflection points. Using the operation we defined above, we get a group, and the 9 inflection points become 9 3-torsion points.

Problem 1.3. Generalize the result by which we covered $\mathbb{P}_n$ by open affine sets as follows: for all homogeneous polynomials $H \in k[X_0, \ldots, X_n]$ of positive degree, show that $\mathbb{P}_n(k)_H$ is an affine variety.

Let $H \in k[x_0, \ldots, x_n]$ be a homogeneous polynomial of positive degree $d > 0$. $\mathbb{P}_n(k)_H = \{x \in \mathbb{P}_n(k) \mid H(x) \neq 0\}$. Want to show: there exists an isomorphism from $\mathbb{P}_n(k)_H$ to some closed algebraic set inside some affine space $\mathbb{A}^N(k)$.

Regular functions on $\mathbb{P}_n(k)_H$ can be represented by fractions G/H^m where G is a homogeneous polynomial of degree $m \cdot d$ ($d = \deg H$).

Let $M_1, \ldots, M_N$ be all monomials of degree d in $x_0, \ldots, x_n$. (so $N = \binom{n+d}{d}$)
H homogeneous polynomial of degree $d \implies H = \sum_{i=1}^N c_i M_i$ some $c_i \in k$. Define

$$\psi : \mathbb{P}_n(k)_H \to \mathbb{A}^N(k), \quad (x_0 : \cdots : x_n) \mapsto \left(\frac{M_1(x)}{H(x)}, \ldots, \frac{M_N(x)}{H(x)} \right)$$

This is well-defined: since M_i and H both have degree d, replacing x_i with λx_i causes the λ^d terms to cancel.
This is regular: $H(X)$ is nonzero on $\mathbb{P}_n(k)_H$.

The image of ψ is a closed affine variety:

- Let $(Y_1, \ldots, Y_N)$ be the coordinates of $\mathbb{A}^N$.
- Notice that $\sum_{i=1}^N c_i \frac{M_i}{H} = \frac{\sum c_i M_i}{H} = H/H = 1$, so the image of ψ must lie entirely on the hyperplane defined by $\sum_{i=1}^N c_i Y_i = 1$ (*)
- Also, any algebraic relations that hold between M_i will translate into polynomial equations among the Y_i.
- The vanishing set of all these polynomial relations (including (*)) defines a closed algebraic set $V \subseteq \mathbb{A}^N$.

So Im$\psi \subseteq V$.

Now, let $\varphi : V \to \mathbb{P}_n(k)_H$ be defined as:
Let the coordinates of $\mathbb{A}^N$ be labeled by monomials of degree d, so we have $Y_0, \ldots, Y_n$ corresponding to $M_1 = x_0^d, \ldots, M_n = x_n^d$, and $y_{j,i}$ corresponding to $x_j x_i^{d-1}$ for all $j \neq i$.

On $Y_i \neq 0$, $\varphi(Y) := (Y_{0,i} : \cdots : Y_{n,i})$.
Recall that

$$Y_{j,i} = \frac{x_j x_i^{d-1}}{H(x)}, \quad Y_i = \frac{x_i^d}{H(x)}$$

so

$$(\varphi \circ \psi)(x_0 : \cdots : x_n) = \varphi(Y) = \varphi\left(\left(\frac{x_0 x_i^{d-1}}{H(x)} : \cdots : \frac{x_n x_i^{d-1}}{H(x)}\right)\right) = (x_0 : \cdots : x_n).$$

Thus φ is an inverse morphism (given by polynomials) for ψ. Thus ψ is an isomorphism onto V, so $\mathbb{P}_n(k)_H$ is isomorphic to the affine variety V.

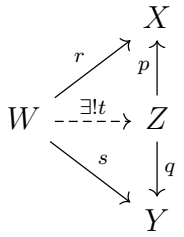
We call the map ψ the **d-uple / Veronese embedding (see Hartshorne exercise 2.12)**.

1.6 Products and the Hausdorff Axiom

Product of Zariski topologies in $\mathbb{A}^n$ and $\mathbb{A}^m$ is too weak to give the Zariski topology in $\mathbb{A}^{n+m}$. For example, in $\mathbb{A}^1 \times \mathbb{A}^1$, the only closed sets in the product topology are finite unions of horizontal and vertical lines.

Definition 1. Let $\mathcal{C}$ be a category. Let X, Y be objects in $\mathcal{C}$. An object Z plus two morphisms $p : Z \to X$ and $q : Z \to Y$ is a **product** if it has the following UMP: for all objects W and morphisms $r : W \to X, s : W \to Y$, there is a unique morphism $t : W \to Z$ such that

$r = p \circ t$, $s = q \circ t$, i.e. such that the following diagram commutes:



The induced morphism t will be denoted (r, s).
We call p, q the **projections** of the product onto its factors.
A product, if it exists, is unique up to a unique isomorphism commuting with the projections.

The requirement of the definition can be rephrased as:

$$\mathrm{Hom}(W, Z) \cong \mathrm{Hom}(W, X) \times \mathrm{Hom}(W, Y)$$

We know: The underlying set must be the set-theoretic product $X \times Y$.
To see this, plug $W = \{\mathrm{pt}\}$ into $\mathrm{Hom}(W, Z) \cong \mathrm{Hom}(W, X) \times \mathrm{Hom}(W, Y)$. Get:

$$Z = \mathrm{Hom}(\{\mathrm{pt}\}, Z) \cong \mathrm{Hom}(\{\mathrm{pt}\}, X) \times \mathrm{Hom}(\{\mathrm{pt}\}, Y) = X \times Y$$

Proposition 2. *If X, Y are affine varieties with coordinate rings R, S:*

1. *There is a product prevariety $X \times Y$.*
2. *$X \times Y$ is affine with coordinate ring $R \otimes_k S$.*
3. *A basis of the topology is given by the open sets defined by $\sum f_i(x) g_i(y) \neq 0$, $f_i \in R, g_i \in S$.*
4. *$\mathcal{O}_{(x,y)}$ is the localization of $\mathcal{O}_x \otimes_k \mathcal{O}_y$ at $m_x \mathcal{O}_y + \mathcal{O}_x m_y$.*

Recall (relevant): for irreducible algebraic sets, the sheaf of functions equals the ring of functions, i.e. if X is an irreducible algebraic set and R is the coordinate ring $\Gamma(X)$ of X, then for $f \in R$, $\mathcal{O}_X(X_f) = R_f$. So the structure sheaf $\mathcal{O}_X$ is completely determined by the ring R, and its global sections recover R.

Theorem 3. *For any prevarieties X and Y over k, the product $X \times Y$ exists.*

Remark 4.
1. If $U \subset X$ is an open subprevariety, then $U \times Y$ is an open subprevariety of $X \times Y$.
2. If $Z \subset X$ is a closed subprevariety, $Z \times Y$ is a closed subprevariety of $X \times Y$.

Terminology: a **ringed space** is a pair $(X, \mathcal{O}_X)$ where X is any topological space and $\mathcal{O}_X$ is any sheaf of rings.

Theorem 5. *The product of two projective varieties is a projective variety.*

- Proven by embedding $\mathbb{P}_n \times \mathbb{P}_m$ into $\mathbb{P}_{nm+n+m}$ using the Segre embedding.
- Map: $(x_0 : \cdots : x_n) \times (y_0 : \cdots : y_m) \mapsto (U_{ij} = x_i y_j)$.
- The image is a closed subvariety (intersection of quadrics).

Example 6. Consider the embedding

$$I : \mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^3,$$

$$(u : v) \times (s : t) \mapsto (x = us : \quad y = vt : \quad z = ut : \quad w = vs)$$

- The image of I is $xy = zw$ (a quadric)
- Our quadric is isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$
- The images $\{(u_0 : v_0)\} \times \mathbb{P}^1$ and $\mathbb{P}^1 \times \{(s_0 : t_0)\}$ give two rulings of lines of our quadric.
- Over an algebraically closed field, all nondegenerate quadrics (those corresponding to nondegenerate quadric forms) are isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$.

Definition 7 (Hausdorff axiom)**.** A prevariety X is a **variety** if for all prevarieties Y and morphisms $f, g : Y \to X$, the set $\{y \in Y | f(y) = g(y)\}$ is closed in Y.

Example 8. Let X be a variety, $Y = X \times X$, $f = p_1, g = p_2$. The diagonal morphism is

$$\Delta : X \to X \times X, \quad x \mapsto (x, x)$$

So

$$\Delta(X) = \{z \in X \times X \mid p_1(z) = p_2(z)\}$$

is closed by the Hausdorff axiom.

The converse also holds:

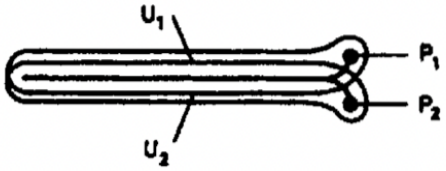
Proposition 9. *Let X be a prevariety: X is a variety $\iff$ $\Delta(X) \subset X \times X$ is closed.*

Remark 10. People typically use the above as the definition for variety. Also, people would typically say X "is separated", rather than saying X "satisfies the Hausdorff axiom."

It is more common to refer to Mumford's prevarieties as varieties and refer to Mumford's varieties as separated varieties.

Example 11. The "line with two origins" X (patching two copies of $\mathbb{A}^1$) is not a variety, because if $i_1, i_2 : \mathbb{A}^1 \to X$, then $\{y \in \mathbb{A}^1 \mid i_1(y) = i_2(y)\} = \mathbb{A}^1 \setminus \{0\}$ is not closed in $\mathbb{A}^1$ (closed sets in $\mathbb{A}^1$ are either all of $\mathbb{A}^1$ or finite sets of points)

Remark 12.
1. Subprevarieties of a variety are varieties.
2. A product of 2 varieties is a variety.



3. Affine varieties are varieties.

4. If $f, g : X \to Y$ are morphisms of prevarieties, $\{y \in Y \mid f(y) = g(y)\}$ is always locally closed.

5. If $f : X \to Y$ is a morphism of prevarieties and Y is a variety, then the graph Γ_f of f is closed. And it is a closed subprevariety of $X \times Y$ isomorphic to X.

Proposition 13. *Let X be a prevariety, Assume that for all $x, y \in X$, there is an open affine U containing both x and y. Then X is a variety.*

Corollary 14. *Every projective variety X is a variety.*

To see this: for every $x, y \in \mathbb{P}_n(k)$ there is a hyperplane H not containing x or y, so take $\mathbb{P}_n(k)_H$ as your open set.

Proposition 15. *Let X be a variety, and let U, V be affine open subsets with coordinate rings R, S. Then $U \cap V$ is an affine open subset with coordinate ring $R \cdot S \subseteq k(X)$.*

Remark 16. As discussed in class, the ring compositum $\neq$ the field compositum:

$$k(x) \cdot k(y) = k(x, y)$$

is the field compositum, but

$$k(x) \otimes_k k(y)$$

is the ring compositum, and these are not equal because e.g. $\frac{1}{x+y}$ lies in $k(x, y)$ but not $k(x) \otimes_k k(y)$.
Observe:

$$\mathrm{Frac}k[x] \otimes \mathrm{Frac}k[y] = k(x) \otimes_k k(y) \neq k(x, y) = \mathrm{Frac}k[x, y]$$

The function field of $\mathbb{A}^1 \times \mathbb{A}^1$ is $k(x, y)$, and this does not equal the tensor product of the function fields $k(x)$, $k(y)$ of each copy of $\mathbb{A}^1$.

Example 17. Let $X = \mathbb{A}^2$, $U = \{x \neq 0\}$, $V = \{y \neq 0\}$. Then if R is the coordinate ring of U, $R = k[x, y, x^{-1}]$, and if S is the coordinate ring of V, $S = k[x, y, y^{-1}]$.

Now the proposition says that the coordinate ring of $U \cap V$ is $R \cdot S = k[x, y, x^{-1}, y^{-1}] \subsetneq k(x, y) = k(\mathbb{A}^2)$ (so here it was important that we took the ring compositum, not the field compositum).

Problem 1.4. Prove the following converse: let X be a prevariety, $\{U_i\}$ an affine open covering of X. Let R_i be the coordinate ring of U_i. Then if $U_i \cap U_j$ is an affine subset of X with coordinate ring $R_i \cdot R_j$, X is a variety.

Want to show: $\Delta \subseteq X \times X$ is closed.
$\{U_i\}$ affine open cover of $X \implies$

$$X \times X = \bigcup_{i,j} (U_i \times U_j)$$

A subset of a space is closed $\iff$ its intersection with every member of an open cover is closed.

Suffices to show: $\Delta \cap (U_i \times U_j)$ is closed in $U_i \times U_j$ for all i, j. So fix i, j:
By hypothesis, $U_i \cap U_j$ is affine with coordinate ring $R_i \cdot R_j$. Let $\delta : U_i \cap U_j \to U_i \times U_j, \quad x \mapsto (x, x)$. Every function in R_i and every function in R_j restricts to a regular function on $U_i \cap U_j$, so there is a natural homomorphism (actually induced by δ)

$$R_i \otimes R_j \to R_{ij}, \quad f \otimes g \mapsto fg.$$

Since $R_{ij} = R_i \cdot R_j$, this map is surjective. Thus $R_{ij} \cong (R_i \otimes R_j)/I$ for some ideal I. By Chapter 1, $(R_i \otimes R_j)/I$ is the coordinate ring of the closed subvariety $V(i) \subseteq U_i \times U_j$.

δ induces an isomorphism

$$\Gamma(V(I)) \cong \Gamma(U_i \cap U_j)$$

so $U_i \cap U_j \cong V(I)$ as affine varieties. Thus $\delta(U_i \cap U_j)$ is closed in $U_i \times U_j$. Then Δ itself is closed in $X \times X$, since $\{U_i \times U_j\}$ form an open cover. So X is a variety.

Definition 18. Let $\mathcal{O} \subseteq \mathcal{O}'$ be local rings. We say $\mathcal{O}'$ **dominates** $\mathcal{O}$ ($\mathcal{O}' > \mathcal{O}$) if $m' \supset m$ ($\iff m' \cap \mathcal{O} = m$) where m, m' are the maximal ideals of $\mathcal{O}, \mathcal{O}'$ respectively.

Example 19. Let $\mathcal{O} = k[x, y]_{(x,y)}$. This embeds into $\mathcal{O}' = k[x, z]_{(x,z)}$ via $x \mapsto x, y \mapsto xz$.

$m = (x, y) = (x, xz)$ and $m' = (x, z)$, so $m \subseteq m'$.

$\mathcal{O}'$ dominates $\mathcal{O}$.

Local Criterion: Let X be a prevariety. Then X is a variety $\iff$ for all $x, y \in X$ such that $x \neq y$, there is no local ring $\mathcal{O} \subseteq k(X)$ such that $\mathcal{O} > \mathcal{O}_x$ and $\mathcal{O} > \mathcal{O}_y$.

The fact that this holds for varieties will be proven later.

(Chevalley-Nagata) A variety is a finitely generated field extension K of k plus a collection of local subrings of K satisfying various conditions.

Problem 1.5. Let $X \subseteq \mathbb{P}^n$ be defined by a homogeneous ideal $P \subseteq k[X_0, \ldots, X_n]$. Let P define the affine variety $X^* \subseteq \mathbb{A}^{n+1}$. Show that for all i,

$$\{\textbf{Subset of } X^* \textbf{ where } X_i \neq 0\} \cong (\mathbb{A}^1 \setminus \{0\}) \times \{\textbf{Subset of } X \textbf{ where } X_i \neq 0\}$$

Let $W_i := \{a \in X^* \mid a_i \neq 0\}$.
Let $U_i := \{(a_0, \ldots, a_n) \in \mathbb{P}^n \mid a_i \neq 0\} \cong \mathbb{A}^n$.
Let $X_i := X \cap U_i$.
Want to show: $W_i \cong (\mathbb{A}^1 \setminus \{0\}) \times X_i$.
Define

$$\Phi : W_i \to (\mathbb{A}^1 \setminus \{0\}) \times X_i, \quad (a_0, \ldots, a_n) \mapsto (a_i, (a_0 : \cdots : a_n))$$

This lands in $(\mathbb{A}^1 \setminus \{0\}) \times X_i$:
$(a_0, \ldots, a_n) \in W_i \implies a_i \neq 0 \implies (a_0 : \cdots : a_n)$ is a valid projective point
$a \in X^* \implies$ every homogeneous polynomial in P vanishes at $a \implies (a_0 : \cdots : a_n) \in X$ (and $a_i \neq 0$ so $(a_0 : \cdots : a_n) \in X_i$)

Constructing an inverse:
Let $\psi : (\mathbb{A}^1 \setminus \{0\}) \times X_i \to W_i$ be the map so that if $(t, p) \in (\mathbb{A}^1 \setminus \{0\}) \times X_i$, then $p \in X_i \implies p = (b_0 : \cdots : b_n)$ with $b_i \neq 0$. And $p \in U_i \implies \exists$ unique representative satisfying $b_i = 1$: $p = (b_0 : \cdots : b_{i-1} : 1 : b_{i+1} : \cdots : b_n)$. Send:

$$(t, p) \mapsto (tb_0, \ldots, tb_{i-1}, t, tb_{i+1}, \ldots, tb_n)$$

This lands in W_i:
If $f \in P$ is homogeneous of degree d, $p \in X \implies f(b_0, \ldots, 1, \ldots, b_n) = 0$. And f homogeneous $\implies f(tb_0, \ldots, t, \ldots, tb_n) = t^d f(b_0, \ldots, 1, \ldots, b_n) = 0$. So every polynomial of P vanishes on $\psi(t, p)$, meaning $\psi(t, p) \in X^*$. If $t \neq 0$, its i-th coordinate equals t, so it lies in W_i.

This is an inverse:
$a \in W_i \implies \Phi(a) = (a_i, (a_0 : \cdots : a_n))$, and $(a_0 : \cdots : a_n) = (a_0/a_i : \cdots : 1 : \cdots : a_n/a_i)$, so

$$\psi(\Phi(a)) = \left(a_i \frac{a_0}{a_i}, \ldots, a_i, \ldots, a_i \frac{a_n}{a_i}\right) = (a_0, \ldots, a_n) \implies \psi \circ \Phi = \mathrm{id}.$$

Conversely, if $(t, p) \in k^\times \times X_i$, write $p = (b_0 : \cdots : 1 : \cdots : b_n)$.

$$\psi(t, p) = (tb_0, \ldots, t, \ldots, tb_n) \implies$$

$$\Phi(\psi(t, p)) = (t, (tb_0 : \cdots : t : \cdots : tb_n)) = (t, (b_0 : \cdots : 1 : \cdots : b_n)) = (t, p)$$

so $\Phi \circ \psi = \mathrm{id}$.

These maps are regular:
We can identify X_i with affine variety having coordinates $y_j = x_j/x_i, j \neq i$ (this is from the $\mathbb{P}_n(k)_{X_i} \cong \mathbb{A}^n$ isomorphism).
So under these coordinates, $\Phi(a) = (a_i, a_0/a_i, \ldots, a_{i-1}/a_i, a_{i+1}/a_i, \ldots, a_n/a_i)$ and $a_i \neq 0 \implies$ these are regular on W_i.
Similarly, $\psi(t, y_0, \ldots, y_{i-1}, y_{i+1}, \ldots, y_n) = (ty_0, \ldots, t, \ldots, ty_n)$, which is polynomial.
Thus, they define an isomorphism of varieties.

1.7 Dimension

Definition 1. Let X be a variety. $\dim X := \operatorname{trdeg}_k k(X)$.

If U is an open nonempty set in X, then $\dim U = \dim X$ since $k(U) = k(X)$.

Since k is algebraically closed, TFAE:

1. $\dim X = 0$
2. $k(X) = k$
3. X is a point.

Example 2. If $k = \mathbb{R}$, and $X = V(x^2 + 1) = \varnothing$ in $\mathbb{A}^1_{\mathbb{R}}$, then $\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$ is a field, so $\dim X = 0$ but $X = \varnothing$. Thus when k is not algebraically closed, $\dim X = 0 \not\implies X$ is a point.

Example 3.
1. $k(\mathbb{A}^n) = k(x_1, \ldots, x_n) \implies \dim \mathbb{A}^n = n$.
2. $\dim(\mathbb{A}^m \times \mathbb{A}^n) = m + n$
3. $\mathbb{P}^1 \times \mathbb{P}^1 \hookrightarrow \mathbb{P}^3$, image $xy = zw$
   $\dim(\mathbb{P}^1 \times \mathbb{P}^1) = 1 + 1 = 2$ so quadric surfaces have dimension 2.
4. $C = V(y^2 - x^3) \subseteq \mathbb{A}^2 \implies k(C) \cong k(t)$ via $x = t^2, y = t^3 \implies \dim C = 1$.
5. $k(\mathbb{P}^n) = k(x_1/x_0, \ldots, x_n/x_0) \implies \dim \mathbb{P}^n = n$.

Remark 4.
- Dimension of a variety is a birational invariant. Recall:
  - Rational map = equivalence class of morphisms that agree on a dense open subset of a domain
  - Birational map = rational map with rational inverse
- Removing a proper closed subset does not change dimension.

Proposition 5. *Let Y be a proper closed subvariety of X. Then* $\dim Y < \dim X$.

The proof uses this lemma:

Lemma 6. *Let R be an integral domain over k, $P \subseteq R$ a prime. Then* $\operatorname{trdeg}_k R \geq \operatorname{trdeg}_k R/P$, *with equality only if $P = 0$ or both sides are ∞.*

In the situation of the proposition, call $\dim X - \dim Y$ the **codimension** of Y in X.

Theorem 7 (Geometric Version of Krull's Principal Ideal Theorem)**.** *Let X be a variety, $U \subseteq X$ open, $g \in \Gamma(U, \mathcal{O}_X)$, Z an irreducible component of $\{x \in U \mid g(x) = 0\}$. Then if $g \neq 0$, $\dim Z = \dim X - 1$.*

Theorem 8 (Algebraic Version of Krull's Principal Ideal Theorem)**.** *Let R be a finitely generated integral domain over k, $f \in R$, P an isolated prime ideal of (f) (i.e. minimal among the prime ideals containing it). Then if $f \neq 0$, $\operatorname{trdeg}_k R/P = \operatorname{trdeg}_k R - 1$.*

Example 9. $V_{\mathbb{R}}(x^2 + y^2) = (0, 0)$ so has codimension 2 in $\mathbb{A}^2_{\mathbb{R}}$. This does not contradict Theorem **??** because $\mathbb{R}$ is not algebraically closed.
$V_{\mathbb{C}}(x^2 + y^2) = \{x = \pm yi\} = \{x = yi\} \cup \{x = -yi\}$ has codimension 1 in $\mathbb{A}^2_{\mathbb{C}}$.
$x^2 + y^2$ is irreducible over $\mathbb{R}$, reducible over $\mathbb{C}$.

$V_{\mathbb{R}}(x^2 + y^2)$ is irreducible but not geometrically irreducible.

Example 10. $V(xy) = V(x) \cup V(y)$ is reducible and pure codimension 1.

Example 11. Claim: There is no surjective morphism

$$f : \mathbb{P}^2 \to \mathbb{P}^1.$$

Recall (Bézout's Theorem):
If two projective plane curves of positive degree have no common irreducible component, then they intersect.

Suppose that

$$f : \mathbb{P}^2 \to \mathbb{P}^1$$

is a surjective morphism. Then

$$f = (F : G),$$

where F and G are homogeneous polynomials of the same degree with no common zero.
Choose two distinct points

$$(a : b), (c : d) \in \mathbb{P}^1.$$

Their inverse images are the curves

$$f^{-1}(a : b) = V(bF - aG), \qquad f^{-1}(c : d) = V(dF - cG).$$

Since f is surjective, both curves are nonempty.

If these curves met at a point p, then

$$(F(p) : G(p)) = (a : b) = (c : d),$$

contradicting the fact that $(a : b) \neq (c : d)$. Hence the two curves are disjoint.

They also cannot have a common irreducible component, since this would force them to intersect. Therefore, by Bézout's theorem, they must intersect, a contradiction.

Definition 12. Let $f : X \to Y$ be a morphism of affine varieties. Let R and S be the coordinate rings of X and Y, and let $f^* : S \to R$ be the induced homomorphism. Then f is **finite** if R is integral over $f^*(S)$.

Remark 13. The restriction of a finite morphism from X to Y to a closed subvariety Z of X is also finite.

Example 14.
1. If char$k = p \neq 0$, the map $f : \mathbb{A}^1 \to \mathbb{A}^1, t \mapsto t^p$ is finite. This induces $k[X^p] \hookrightarrow k[X]$.
2. If k is algebraically closed, $f : \mathbb{A}^1 \to \mathbb{A}^2, t \mapsto (t^2, t^3)$ is finite. This induces $k[T^2, T^3] \hookrightarrow k[T]$.
3. $f : \mathbb{A}^1 \to \mathbb{A}^2$ defined by $X = t^2 - 1, \quad Y = t(t^2 - 1)$ is NOT finite. This induces $k[T^2 - 1, T(T^2 - 1)] \hookrightarrow k[T, (T - 1)^{-1}]$.

Proposition 15. *Let $f : X \to Y$ be a finite morphism of affine varieties.*

1. *Then f is a closed map.*
2. *For all $y \in Y, f^{-1}(y)$ is a finite set.*
3. *f is surjective $\iff$ the corresponding map f^* of coordinate rings is injective.*

Geometric form of Noether's normalization lemma.
Let X be an affine variety of dimension n. Then there exists a finite surjective morphism $\pi : X \to \mathbb{A}^n$.

Definition 16. Let X be a variety, and let $Z \subseteq X$ be a closed subset. Then Z has **pure dimension** r if each of its components has dimension r (similarly for **pure codimension** r).

The conclusion of Theorem **??** may be stated as: $V(g)$ has pure codimension 1, for any nonzero $g \in \Gamma(X, \mathcal{O}_X)$.

The converse holds: if Z is an irreducible closed subset of a variety X of codimension 1, then for all open sets U with $Z \cap U \neq \varnothing$ and for all nonzero functions f on U vanishing on Z, $Z \cap U$ is a component of $f = 0$.

Corollary 17. *Let X be a variety and Z a maximal closed irreducible subset, smaller than X itself. Then* $\dim Z = \dim X - 1$.

Corollary 18 (Topological characterization of dimension)**.** *Suppose*

$$\varnothing \neq Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_r = X$$

is any maximal chain of closed irreducible subsets of X. Then $\dim X = r$.

Corollary 19. *Let X be a variety and let Z be a component of $V(f_1, \ldots, f_r)$, where $f_1, \ldots, f_r \in \Gamma(X, \mathcal{O}_X)$. Then* $\text{codim} Z \leq r$.

Equality need not hold: e.g. take $f_1 = \cdots = f_r, r > 1$.

Corollary 20. *Let U be an affine variety, Z a closed irreducible subset. Let $r = \operatorname{codim} Z$. Then there exist $f_1, \ldots, f_r$ in $R = \Gamma(U, \mathcal{O}_U)$ such that Z is a component of $V(f_1, \ldots, f_r)$.*

Recall: every noetherian local ring $\mathcal{O}$ has a Krull dimension, which is defined as:

1. the length r is the longest chain of prime ideals $P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_r = M$ where M is the maximal ideal of $\mathcal{O}$, or
2. the least integer n such that there exist elements $f_1, \ldots, f_n \in M$ for which $M = \sqrt{(f_1, \ldots, f_n)}$

Also recall that every irreducible closed subset Z of every variety X has a local ring $\mathcal{O}_{Z,X}$.

Corollary 21. *The Krull dimension of $\mathcal{O}_{Z,X}$ is $\dim X - \dim Z$.*

Suppose $Z \subseteq X$ is irreducible and of codimension 1. Question: is it true that for all $y \in Z$, there is some neighborhood U of y in X and some function $f \in \Gamma(U, \mathcal{O}_X)$ such that $Z \cap U$ is not just a component of $f = 0$ but equal to it.

More generally, if $Z \subseteq X$ is a closed subset of pure codimension r, is it true that for all $y \in Z$, there is a neighborhood U of y and functions $f_1, \ldots, f_r \in \Gamma(U, \mathcal{O}_X)$ such that $Z \cap U = V(f_1, \ldots, f_r)$?

Even if Z is irreducible of codimension 1, this is not always true. A closed set Z with this property is referred to as a **local set-theoretic complete intersection**.

Proposition 22. *Let X be an affine variety with coordinate ring R. Assume R is a UFD. Then every closed subset $Z \subseteq X$ of pure codimension 1 equals $V(f)$ for some $f \in R$.*

Proposition 23. $\dim X \times Y = \dim X + \dim Y$.

We also have projective versions of all of these:
Let $X \subseteq \mathbb{P}_n(k)$ be a projective variety and let $I(X) \subseteq k[X_0, \ldots, X_n]$ be its ideal.

Theorem 24. *If $f \in k[X_0, \ldots, X_n]$ is homogeneous and not a constant and $f \notin I(X)$, then $X \cap V(f)$ is non-empty and of pure codimension 1 in X, unless $\dim X = 0$.*

Corollary 25. *If $f_1, \ldots, f_r \in k[X_0, \ldots, X_n]$ are homogeneous and not constants, then all components of $X \cap V(f_1, \ldots, f_r)$ have codimension at most r in X. And if $\dim X \geq r$, then $X \cap V(f_1, \ldots, f_r)$ is non-empty.*

Corollary 26. *If $Y \subseteq X$ is a closed subvariety (resp. Y is the empty subset), then there exist homogeneous non-constant elements $f_1, \ldots, f_r \in k[X_0, \ldots, X_n]$ where $r =$ codimension of Y (resp. $r = \dim X + 1$) such that Y is a component of $X \cap V(f_1, \ldots, f_r)$ (resp. $X \cap V(f_1, \ldots, f_r) = \varnothing$).*

This implies that every space curve (i.e. one-dimensional subvariety of $\mathbb{P}_3(k)$) is a component of an intersection $H_1 \cap H_2$ of 2 surfaces.

Proposition 27. *Every closed subset of $\mathbb{P}_n(k)$ of pure codimension 1 is a hypersurface, i.e. equals $V(f)$ for some homogeneous element $f \in k[X_0, \ldots, X_n]$.*

Proposition 28. *Suppose $f : \mathbb{P}^n \to \mathbb{P}^m$ is a morphism. Assume $W = f(\mathbb{P}^n)$ is closed (this turns out to always be true). Then either W is a single point, or $\dim W = n$.*

Example 29. The projection map $\mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^1$ has dimension 1 image.

Hence $\mathbb{P}^1 \times \mathbb{P}^1$ is very different from $\mathbb{P}^2$. But $k(\mathbb{P}^2) = k(\mathbb{P}^1 \times \mathbb{P}^1) = k(x, y)$, and each contains $\mathbb{A}^2$ as a dense open.

Example 30. Compute the total ring of fractions of $\mathbb{Z}[t]/(t^2)$. Observe: e.g. $a+t$ (for $a \neq 0$) is inverted after inverting all nonzero integers, since

$$(a + t)(a - t) = a^2 - t^2 = a^2$$

so $\frac{a-t}{a^2}$ is the inverse. Conclude

$$\mathrm{Tot}(\mathbb{Z}[t]/(t^2)) = \mathbb{Q}[t]/(t^2)$$

and another example of total ring of fractions (this time for a disconnected space):

$$\mathrm{Tot}(k[x] \times k[y, z]) = k(x) \times k(y, z) = \Gamma(\mathbb{A}^1) \times \Gamma(\mathbb{A}^2) = \Gamma(\mathbb{A}^1 \sqcup \mathbb{A}^2)$$

$k[x] \times k[y, z]$ has idempotents $(1, 0)$ and $(0, 1)$. Idempotents $\implies$ disconnected. However, while disconnected $\implies$ not domain, not domain $\not\implies$ disconnected.

Similarly, the total ring of fractions of $k[x, t]/(t^2)$ is $k(x)[t]/(t^2)$. Its spectrum has one underlying point, but the ring is nonreduced because it contains the nilpotent class of t.

1.8 The fibers of a morphism

Let $f : X \to Y$ be a morphism of varieties.

Definition 1. A morphism $f : X \to Y$ is **dominating** if its image is dense in Y, i.e. $Y = \overline{f(X)}$.

Proposition 2. *If $f : X \to Y$ is any morphism, let $Z = \overline{f(X)}$. Then Z is irreducible, the restricted morphism $f' : X \to Z$ is dominating, and f' induces an injection*

$$f'^* : k(Z) \hookrightarrow k(X).$$

So when we study fibers of an arbitrary morphism, we can focus on the case of dominating morphisms.
Our situation thus is: given $f : X \to Y$, we get $k(Y) \hookrightarrow k(X)$. But this is not well-defined unless f is dominating (e.g. given a morphism from a point to a line, there is no induced homomorphism of function fields $k(Y) \to k(X)$ because the generic point of Y is not in the image). For this reason, we focus on the dominating case.

If $f : X \to Y$ is a map between affine varieties, it induces a map between the coordinate rings of the varieties. But if we remove the "affine" conditions, we get a map between the function fields. Homomorphisms between fields are always injective. So we end up with a well-defined injective morphism.

Note that a finite morphism is dominating $\iff$ it is surjective (because finite morphisms are closed).

Theorem 3. *Let $f : X \to Y$ be a dominating morphism of varieties, and let $r = \dim X - \dim Y$. Let $W \subseteq Y$ be a closed irreducible subset, and let Z be a component of $f^{-1}(W)$ that dominates W. Then*

$$\dim Z \geq \dim W + r$$

or

$$\operatorname{codim}(Z \text{ in } X) =: \operatorname{codim}_X Z \leq \operatorname{codim}_Y W.$$

Corollary 4. *If Z is a component of $f^{-1}(y)$ for some $y \in Y$, then $\dim Z \geq r$.*

Example 5. Given $f : \mathbb{A}^2 \to \mathbb{A}^2, (x, z) \mapsto (x, y)$ via $y = xz$. Here $r = 0$. And $f^{-1}((0, 0))$ is a line, which has dimension 1. So the whole axis goes to a point. So here we get strict inequality $\dim Z > r$.

This is a morphism. It is an isomorphism on a dense open subset (=birational isomorphism). But not an isomorphism, since $f^{-1}((0, 0)) =$ line.

Theorem 6. *Let $f : X \to Y$ be a dominating morphism of varieties and let $r = \dim X - \dim Y$. Then there exists a nonempty open set $U \subseteq Y$ such that*

1. *$U \subseteq f(X)$*
2. *For all irreducible closed subsets $W \subseteq Y$ such that $W \cap U \neq \varnothing$, and for all components Z of $f^{-1}(W)$ such that $Z \cap f^{-1}(U) \neq \varnothing$,*

$$\dim Z = \dim W + r$$

or

$$\operatorname{codim}_X Z = \operatorname{codim}_Y W.$$

Corollary 7. *Let f be as above. Then there is a nonempty open set $U \subseteq Y$ such that for all $y \in U$, $f^{-1}(y)$ is a nonempty "pure" r-dimensional set, i.e. all its components have dimension r.*

Definition 8. Let X be a variety. A subset A of X is **constructible** if it is a finite union of locally-closed subsets of X.

The constructible sets form an algebra; they are the smallest algebra containing all open sets.

Example 9. Let $Z = \{\mathbb{A}^2 \setminus V(x)\} \cup \{(0,0)\}$.

Then Z is a constructible set, but is not locally closed.

This is because $\mathbb{A}^2 \setminus V(x)$ is open, and $\{(0,0)\}$ is closed—so Z is a finite union of locally closed sets.

But Z is not locally closed. A set is locally closed $\iff$ it is open in its closure. The closure $\overline{Z}$ of Z is $\mathbb{A}^2$, since the closure of the open part is $\mathbb{A}^2$. If Z were open in $\mathbb{A}^2$, then $\mathbb{A}^2 \setminus Z$ would have to be closed. And $\mathbb{A}^2 \setminus Z = V(x) \cap (\mathbb{A}^2 \setminus \{(0,0)\}) =$ the y-axis minus the origin. This set is not closed.

Corollary 10 (Chevalley)**.** *Let $f : X \to Y$ be a morphism. Then the image of f is a constructible set in Y. More generally, f maps constructible sets in X to constructible sets in Y.*

Corollary 11 (Upper semi-continuity of dimension)**.** *Let $f : X \to Y$ be any morphism. For all $x \in X$, define*

$$e(x) = \max\{\dim Z \mid Z \text{ is a component of } f^{-1}(f(x)) \text{ containing } x\}$$

Then e is upper semi-continuous, i.e. for all integers n,

$$S_n(f) = \{x \in X \mid e(x) \geq n\}$$

is closed.

Definition 12. A morphism $f : X \to Y$ is **birational** if it is dominating and the induced map $f^* : k(Y) \to k(X)$ is an isomorphism.

Theorem 13. *If $f : X \to Y$ is a birational morphism, then there is a nonempty open set $U \subseteq Y$ such that f restricts to an isomorphism from $f^{-1}(U)$ to U.*

Example 14. Let

$$f : \mathbb{A}^2 \to \mathbb{A}^2, \quad (x,y) \mapsto (xy, y).$$

1. The image of f is the union of $(0,0)$ and $(\mathbb{A}^2)_y = \mathbb{A}^2 \setminus \{y = 0\}$.
   This set is not locally closed (we showed this in a previous example).

2. f is birational since $k(x,y) \cong k(yx,y)$, and if $U = \mathbb{A}^2_y$, then $f^{-1}(U) = \mathbb{A}^2_y$ and the restriction of f to a map from $f^{-1}(U)$ to U is an isomorphism.

3. $f^{-1}((0,0))$ is the whole line of points $(x,0)$.

4. $S_0(f) = \mathbb{A}^2, \quad S_1(f) = \{(x,0)\}, \quad S_2(f) = \varnothing$.
   (where $e(x) = \max\{\dim Z \mid Z$ is a component of $f^{-1}(f(x))$ containing $x\}$ and $S_n(f) = \{x \in X \mid e(x) \geq n\}$)

Example 15 (Previous examples revisited)**.** Given the finite birational morphism $f : \mathbb{A}^1 \to C$, where C is the affine plane curve $X^3 = Y^2$, then if $U = C \setminus \{(0,0)\}$, then $f^{-1}(U) = \mathbb{A}^1 \setminus \{(0)\}$, and $f^{-1}(U) \cong U$.
On rings, $k[T^2, T^3] \subsetneq k[T]$, but $k[T, T^{-1}] = k[T^2, T^3, T^{-2}]$.

Given the finite birational morphism $f : \mathbb{A}^1 \to D$ where D is the curve $Y^2 = X^2(X+1)$, its restriction $f' : \mathbb{A}^1 \setminus \{1\} \to D$ is a bijection, and if $U = D \setminus \{(0,0)\}$, then $f'^{-1}(U) = \mathbb{A}^1 \setminus \{\pm 1\}$ and $f^{-1}(U) \cong U$.

1.9 Complete varieties

An affine variety can be embedded in a projective variety by a birational inclusion. Projective varieties cannot be embedded birationally in anything even bigger. This makes them *complete* varieties. This is our algebraic geometric analog of compact spaces in topology.

Note: Over $\mathbb{C}$, complete = compact.

Example 1. $V = V(x^2 + y^2 - 1)$ is not complete because over $\mathbb{C}$, it's clearly not compact (a change of variables makes it look like a hyperbola).

Example 2. Look at $\mathbb{P}^1(k)$. $R = k[x]_{(x)} \subseteq k(x) = k(\mathbb{P}^1)$ is a valuation ring.

Assertion: $\mathbb{P}^1$ complete $\implies$ R contains $\mathcal{O}_P$ for some P. This is indeed true: take P to be the point where $x = 0$.

Example 3. $\mathbb{P}^1 \setminus P \cong \mathbb{A}^1$ has $k(\mathbb{P}^1 \setminus P) = k(x)$. $k(x)$ still contains $k[x]_{(x)}$. But $k[x]_{(x)}$ is not the local ring of a point now, since we threw out P.

Similarly, let $\mathbb{A}^1 = \mathbb{P}^1 \setminus \{\infty\}$. Now $k[x]_{(x)}$ corresponds to 0. But $k(x) = k(x^{-1}) \supseteq k[x^{-1}]_{(x^{-1})}$ does not correspond to any point because it would have to be ∞.

Definition 4. A variety X is **complete** if for all varieties Y, the projection morphism $p_2 : X \times Y \to Y$ is a closed map.

Some properties:

1. If $f : X \to Y$ is a morphism with X complete, then $f(X)$ is closed in Y and is complete.
2. If X and Y are complete, then $X \times Y$ is complete.
3. If X is complete and $Y \subseteq X$ is a closed subvariety, then Y is complete.
4. An affine variety X is complete only if $\dim X - 0$, i.e. X consists of a single point.

Complete means universally closed, i.e. closed in anything you put it into, and for all Y, $X \times Y \to Y$ is closed.
Notice that this is similar to topology, where compact $\iff$ any time it's embedded into a Hausdorff space, it's closed there.

Example 5. $\mathbb{A}^1$ is not complete, because $\mathbb{A}^1 \times \mathbb{P}^1 \to \mathbb{P}^1$ is not closed:

Consider $\Delta \subseteq \mathbb{A}^1 \times \mathbb{P}^1$ (thinking of $\mathbb{A}^1 \hookrightarrow \mathbb{P}^1$).
The image is $\mathbb{A}^1$, which is not closed in $\mathbb{P}^1$.

Remark 6. Given $f : X \to Z$ a morphism of varieties, we can ask: after every base change $Y \to Z$, is the induced map $X \times_Z Y \to Y$ closed? (in this case f is universally closed or **proper**)

Say this is true: taking $Z = \{\text{pt}\}$, X being complete means $X \to \{\text{pt}\}$ is proper.

Theorem 7. *$\mathbb{P}^n$ is complete.*

Let Y be a variety, let $U_i = (\mathbb{P}^n)_{X_i} \times Y$, and let Z be a closed subset of $\mathbb{P}^n \times Y$. Let $R = \Gamma(Y, \mathcal{O}_Y)$, let $S = R[X_0, \ldots, X_n]$, let S_m be the graded piece of S of degree m, and let $A_m \subseteq S_m$ be the vector space of homogeneous polynomials $f(X_0, \ldots, X_n)$ of degree m with coefficients in R such that for all i,

$$f\left(\frac{X_0}{X_i}, \ldots, \frac{X_n}{X_i}\right) \in I(Z \cap U_i).$$

The proof of Theorem **??** uses the following lemma:

Lemma 8. *For all i and $g \in I(Z \cap U_i)$ there is a polynomial $f \in A_m$ for some m such that*

$$g = f\left(\frac{X_0}{X_i}, \ldots, \frac{X_n}{X_i}\right).$$

The proof also uses Nakayama:

Lemma 9 (Nakayama's Lemma)**.** *Let M be a finitely generated R-module, and let $A \subseteq R$ be an ideal such that*

$$M = A \cdot M.$$

Then there is an element $f \in 1 + A$ which annihilates M.

Putting the theorem and property (3) together, every projective variety is complete.
Not all complete varieties are projective, but every complete variety can be dominated by a projective variety with the same function field.

Another way to prove Theorem **??**:
Valuative Criterion. A variety X is complete $\iff$ for all valuation rings $R \subseteq k(X)$ containing k and with quotient field $k(X)$, $R \supseteq \mathcal{O}_{X,x}$ for some $x \in X$.

Remark 10.

- Projective varieties are complete.
- For curves and regular surfaces, complete implies projective (though this is not always true)

1.10 Complex varieties

When k is a topological field, there is a unique way to endow all varieties X over k with a new topology called the **strong topology**, such that the following properties hold:

1. the strong topology is stronger than the Zariski topology (i.e. a Zariski-closed subset $Z \subseteq X$ is always strongly closed, and an open subset $Z \subseteq X$ is always strongly open).
2. all morphisms are strongly continuous
3. if $Z \subseteq X$ is a locally closed subvariety, then the strong topology on Z is the one induced by the strong topology on X
4. the strong topology on $X \times Y$ is the product of the strong topologies on X and Y (product topology)
5. the strong topology on $\mathbb{A}^1$ is exactly the given topology on k

Note: recall
Locally closed = open subset of closed set = intersection of open and closed set
(for example $\mathbb{A}^1 \setminus \{0\}$ is locally closed)
Constructible = finite union of locally closed. So locally closed $\implies$ constructible, but the converse does not hold:
(for example, $(\mathbb{A}^2 \setminus V(x)) \cup \{(0,0)\}$ is constructible but not locally closed)

Remark 1. All varieties X are Hausdorff spaces in their strong topology ($\Delta(X)$ is strongly closed)
Here by Hausdorff we mean "separated"

Now suppose $k = \mathbb{C}$ with its usual topology. Then there is a unique collection of sheaves (in the strong topology) of strongly continuous $\mathbb{C}$-valued functions $\{\Omega_X\}$, one for each variety X, such that:

1. for each Zariski-open set $U \subseteq X$, $\mathcal{O}_X(U) \subseteq \Omega_X(U)$
2. all morphisms $f : X \to Y$ are holomorphic, i.e. f^* takes sections of Ω_Y to sections of Ω_X
3. if $Z \subseteq X$ is a locally closed subvariety, then Ω_Z is the sheaf of $\mathbb{C}$-valued functions on Z induced by the sheaf Ω_X on X
4. if $X = \mathbb{A}^n =$ (as a set) $\mathbb{C}^n$, then Ω_X is the usual sheaf of holomorphic functions on $\mathbb{C}^n$

Note: Ω_X is not standard notation. Typically this denotes differential 1-forms. Instead people use $\mathcal{H}$ or $\mathcal{O}_{\text{an}}$.

Theorem 2. *Let X be a variety and U a nonempty open subvariety. Then U is strongly dense in X.*

Corollary 3. *If $Z \subseteq X$ is a constructible subset of a variety, then the Zariski closure and the strong closure of Z are the same.*

Theorem 4. *Let X be a variety over $\mathbb{C}$. Then X is complete $\iff$ X is compact in its strong topology.*

Example 5. Over $\mathbb{R}$,$x^2+y^2=1$ looks compact, but over $\mathbb{C}$ its points can go off to infinity: for large x, one can take $y = i\sqrt{x^2-1}$. Thus the affine curve is not complete/proper; its projective closure adds the missing points at infinity.

Remark 6. In topology, a compact space Y has the property that for every space Z, the projection

$$Y \times Z \to Z$$

is a closed map; this is the topological model for being universally closed. In algebraic geometry, a morphism $Y \to X$ is universally closed if after every base change $Z \to X$, the projection

$$Y \times_X Z \to Z$$

is closed. Taking $X = \{\text{pt}\} = \operatorname{Spec} k$, this is one of the conditions for Y to be proper (or, in Mumford's terminology, complete).

Lemma 7 (Chow's Lemma)**.** *Let X be a complete variety (over any algebraically closed field k). Then there exists a closed subvariety Y of $\mathbb{P}_n(k)$ for some n and a surjective birational morphism $\pi : Y \to X$.*

Chow's lemma says: complete is almost projective.
Note: A projective variety is complete, because closed in $\mathbb{P}^n \implies$ since $\mathbb{P}^n$ is complete, it's complete.
For curves, complete implies projective. For surfaces, smooth and complete implies projective. In all other cases, there are counterexamples.

Lemma 8 (Sublemma)**.** *Let S and T be varieties with isomorphic open subsets $V_S \subseteq S$, $V_T \subseteq T$. For simplicity, identify V_S with V_T and look at the morphism $\Delta : V \to V \times V \subseteq S \times T$. If $\overline{V}$ is the closure of the image, then $\overline{V} \cap (S \times V) = \overline{V} \cap (V \times T) = \Delta(V)$.*

Remark 9 (GAGA). We know that from the structure sheaf, given an open set $U \subseteq X$, we associate a ring $\mathcal{O}(U)$.

Suppose you are given a sheaf $\mathcal{M}$ such that $\mathcal{M}(U)$ is an $\mathcal{O}(U)$-module for every open set U. If the module structures are compatible with the restriction maps, then we call $\mathcal{M}$ a **sheaf of $\mathcal{O}$-modules**. In particular, $\mathcal{M}$ is also a sheaf of abelian groups.

Examples of coherent sheaves of $\mathcal{O}$-modules include:

- $\mathcal{M} = \Omega^1_X$, the sheaf of differential 1-forms;
- if X is a curve and P is a point, $\mathcal{O}(P)$, which associates to an open set U the functions on U allowed to have a pole at P of order at most 1;
- more generally, if D is a divisor (a linear $\mathbb{Z}$-combination of points on a curve), the sheaf $\mathcal{O}(D)$.

A sheaf $\mathcal{M}$ of $\mathcal{O}$-modules is called **quasicoherent** if, locally on affine open sets $U = \operatorname{Spec} A$, it comes from an A-module. In particular, if

$$V = D(f) = \{f \neq 0\} \subseteq U,$$

then

$$\mathcal{M}(V) = \mathcal{M}(U)_f.$$

It is called **coherent** if, locally on affine open sets, the corresponding modules are finitely generated (more precisely, finitely presented). For a Noetherian scheme, this is equivalent to being quasicoherent and locally finitely generated.

So given a space X and a structure sheaf $\mathcal{O}$, we get a category of coherent sheaves.

Now suppose X is a projective variety over $\mathbb{C}$. We can view X either algebraically or as a complex analytic space X^{an}. GAGA gives an equivalence of categories

$$\{\text{coherent algebraic sheaves on } X\} \simeq \{\text{coherent analytic sheaves on } X^{an}\}.$$

Moreover, GAGA says that the corresponding cohomology groups are the same:

$$H^i(X, \mathcal{F}) \cong H^i(X^{an}, \mathcal{F}^{an}).$$

2 Preschemes

2.1 $\operatorname{Spec}(R)$

For a commutative ring R, define $\operatorname{Spec}(R) :=$ the set of prime ideals $P \subsetneq R$.
Write $[P]$ for the element of $\operatorname{Spec}(R)$ given by the prime ideal P.

Define a topology on $\operatorname{Spec}(R)$ called the **Zariski topology**:
Closed sets = sets of the form $\{[P] \mid P \supseteq A\}$ for some ideal $A \subseteq R$. (Denote this set $V(A)$)
One can check that

1. $V\left(\sum_\alpha A_\alpha\right) = \cap_\alpha V(A_\alpha)$
2. $V(A \cap B) = V(A) \cup V(B)$

So the collection of closed subsets $\{V(A)\}$ does define a topology.

Now define $\operatorname{Spec}(R)_f = \{[P] \mid f \notin P\}$.
Since $\operatorname{Spec}(R)_f = \operatorname{Spec}(R) \setminus V(f)$, $\operatorname{Spec}(R)_f$ is an open subset of $\operatorname{Spec}(R)$. We refer to these open subsets as **distinguished open subsets**.
These form a basis of the open sets of $\operatorname{Spec}(R)$, because

$$\operatorname{Spec}(R) \setminus V(A) = \bigcup_{f \in A} \operatorname{Spec}(R)_f.$$

The closure of $[P]$ is exactly $V(P)$, i.e. $\{P' \mid P' \supseteq P\}$. Thus $[P]$ is a closed point $\iff$ P is a maximal ideal.
When R is a domain, (0) is a prime ideal and $[(0)]$ is called the **generic point** of $\operatorname{Spec}(R)$ since its closure is all of $\operatorname{Spec}(R)$.

Definition 1. Suppose Z is an irreducible closed subset of $\mathrm{Spec}(R)$. Then a point $z \in Z$ is a **generic point** of Z if Z is the closure of z, i.e. every open subset Z_0 of Z contains z.

Proposition 2. *If $x \in \mathrm{Spec}(R)$, then the closure of $\{x\}$ is irreducible and x is a generic point of this set. Conversely, every irreducible closed subset $X \subseteq \mathrm{Spec}(R)$ equals $V(P)$ for some prime ideal $P \subseteq R$, and $[P]$ is its unique generic point.*

Proposition 3. *Let $\{f_\alpha \mid \alpha \in S\}$ be a set of elements of R. The*

$$\mathrm{Spec}(R) = \bigcup_{\alpha \in S} \mathrm{Spec}(R)_{f_\alpha} \iff 1 \in (\dots, f_\alpha, \dots).$$

Notice: $1 \in (\dots, f_\alpha, \dots) \iff$ there is a finite set $f_{\alpha_1}, \dots, f_{\alpha_n}$ of the f_α's and elements $g_1, \dots, g_n$ of R such that

$$1 = \sum_{i=1}^{n} g_i \cdot f_{\alpha_i}.$$

Corollary 4. $\mathrm{Spec}(R)$ *is quasi-compact.*

You can generalize this to show $\mathrm{Spec}(R)_f$ is also quasi-compact.
Unless R is e.g. noetherian, there may be some open sets in $\mathrm{Spec}(R)$ which are not quasi-compact.

We can endow $\mathrm{Spec}(R)$ with a sheaf of rings $\mathcal{O}_{\mathrm{Spec}(R)}$, called its **structure sheaf**:
Let $X = \mathrm{Spec}(R)$.
Some properties of distinguished opens:

1. $X_f \cap X_g = X_{fg}$ for all $f, g \in R$.
2. $X_f \supseteq X_g \iff g \in \sqrt{(f)}$. (In fact, since $\sqrt{(f)} = \cap\{P \mid f \in P\}$, $g \notin \sqrt{(f)} \iff \exists P, f \in P, g \notin P \iff \exists P, [P] \notin X_f, [P] \in X_g$)

If $X_g \subseteq X_f$ then there is a canonical map $R_f \to R_g$ ($g^n = h \cdot f$ and we map $a/f^m \mapsto (a \cdot h^m)/g^{nm}$). Further, if $[P] \in X_f$, then $f \notin P$ and there is a natural map $R_f \to R_P$ since $\{f, f^2, \dots\} \subseteq R \setminus P$. One can check that

$$R_P = \varinjlim_{f \in R \setminus P} R_f = \varinjlim_{f \text{ such that } [P] \in X_f} R_f.$$

Lemma 5. *Suppose $X_f = \bigcup_{\alpha \in S} X_{f_\alpha}$. If an element $g \in R_f$ has image 0 in all the rings R_{f_α}, then $g = 0$.*

Lemma 6. *Suppose $X_f = \bigcup_{\alpha \in S} X_{f_\alpha}$. Suppose we have elements $g_\alpha \in R_{f_\alpha}$ such that g_α and g_β have the same image in $R_{f_\alpha f_\beta}$. Then there is an element $g \in R_f$ which has image g_α in R_{f_α} for all α.*

For each open $U \subseteq X$, let

$$\Gamma(U, \mathcal{O}_X) := \left\{ \{s_P\} \in \prod_{[P] \in U} R_P \right\}$$

such that $\exists$ a covering of U by distinguished open sets X_{f_α} together with $s_\alpha \in R_{f_\alpha}$ such that s_P equals the image of s_α in R_P whenever $[P] \in X_{f_\alpha}$.

This is a ring. If $V \subseteq U$, then the coordinate projection

$$\prod_{[P] \in U} R_P \to \prod_{[P] \in V} R_P$$

takes $\Gamma(U, \mathcal{O}_X)$ into $\Gamma(V, \mathcal{O}_X)$. Taking this as the restriction map, we get a presheaf $\mathcal{O}_X$. Some facts:

1. $\mathcal{O}_X$ is a sheaf.
2. $\Gamma(X_f, \mathcal{O}_X) = R_f$.
3. The stalk of $\mathcal{O}_X$ at $[P]$ is R_P.

Example 7. Let k be a field. Then $\mathrm{Spec}(k)$ has just one point $[(0)]$, and the structure sheaf is just k sitting on that point.

More generally, let R be any commutative ring with DCC. Then $R \cong \prod_{i=1}^n R_i$ is a finite product of Artinian local rings.

$$\mathrm{Spec}(R) = \bigsqcup_{i=1}^n \mathrm{Spec}(R_i).$$

where each $\mathrm{Spec}(R_i)$ is open in $\mathrm{Spec}(R)$. Each Artinian local ring R_i has a unique prime ideal, namely its maximal ideal

$$M_i = \{x \in R_i \mid x \text{ is nilpotent}\},$$

and $\mathrm{Spec}(R_i)$ is one point. So $\mathrm{Spec}(R)$ itself consists of n points with the discrete topology. The structure sheaf consists of R_i sitting as the stalk on the i-th point.

Geometrically, the presence of nilpotents in M_i means the i-th point is surrounded by some infinitesimal normal neighborhood.

Example 8. The **affine line over** k is $\mathrm{Spec}(k[X])$. This is denoted $\mathbb{A}^1_k$.
$k[X]$ has two types of prime ideals: (0) and $(f(X))$ for f an irreducible polynomial. So $\mathrm{Spec}(k[X])$ has one closed point for each monic irreducible polynomial, and one generic point $[(0)]$ whose closure is all of $\mathrm{Spec}(k[X])$.

If k is algebraically closed, then the closed points are all of the form $[(X-a)]$. We call this the point $X=a$. In that case, $\mathbb{A}^1_k$ is the ordinary X-line together with a generic point. And proper closed sets are finite unions of closed points.

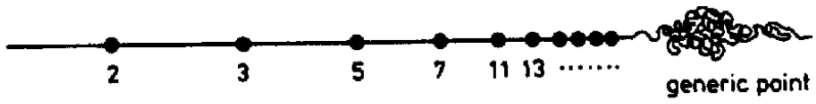
The stalk of $\mathcal{O}_{\mathbb{A}^1}$ at $[(X-a)]$ is

$$k[X]_{(X-a)} = \left\{ \frac{f(X)}{g(X)} \mid f, g \text{ polynomials}, \quad g(a) \neq 0 \right\}.$$

The stalk at $[(0)]$ is $k(X)$, which we called before the function field.

Example 9. $\mathbb{Z}$ is a PID, like $k[X]$. Spec($\mathbb{Z}$) is usually visualized as a line.

There is one closed point for each prime number, plus a generic point.



The stalk at $[(p)]$ is $\mathbb{Z}_{(p)}$, and at $[(0)]$, it is $\mathbb{Q}$, so $\mathbb{Q}$ is the "function field" of Spec($\mathbb{Z}$).

The nonempty open sets of Spec($\mathbb{Z}$) are obtained by throwing away finitely many primes. If $m = \prod p_i$, then this is the distinguished open set Spec$(\mathbb{Z})_m$, and

$$\Gamma(\text{Spec}(\mathbb{Z})_m, \mathcal{O}_{\text{Spec}(\mathbb{Z})}) = \{a/m^k \mid a \in \mathbb{Z}, k \geq 0\}.$$

The residue fields of the stalks $\mathcal{O}_x$ are $\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/3\mathbb{Z}, \ldots, \mathbb{Q}$.

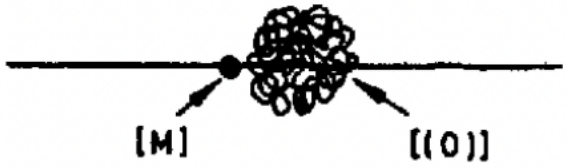
Example 10. Almost identical comments apply to Spec(R) for any Dedekind domain R. All prime ideals are maximal or (0). So again there is a line of closed points plus a generic point.

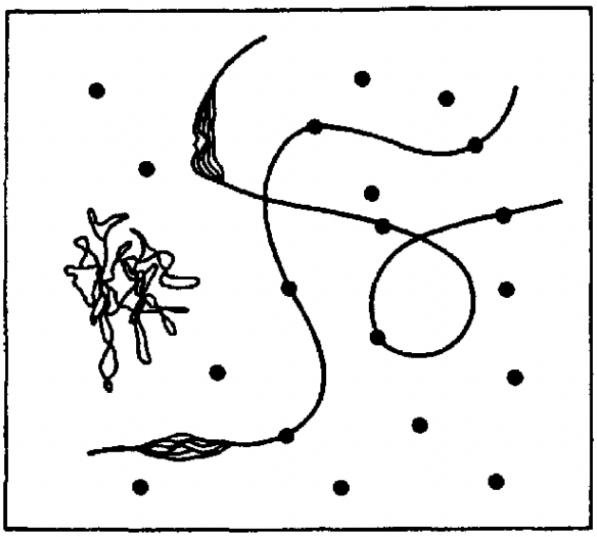
If R is not a PID, we cannot conclude as before that all non-empty open sets are distinguished; we only know that they are obtained by throwing out a finite set of closed points.

When R is a discrete valuation ring (of rank 1), R has a unique maximal ideal M , hence Spec(R) has two points $[(0)]$ and $[M]$. Then $[(0)]$ is an open point and $[M]$ is closed.

Example 11. $\mathbb{A}^2_k = \text{Spec}(k[X,Y])$, k algebraically closed.
We get the maximal ideal $(X-a, Y-b)$, the principal primes (f) for f irreducible, and (0). The set of maximal ideals gives a set of closed points isomorphic to the usual X,Y-plane. Then add one generic point, and for every irreducible curve, a point generic in that curve but not sticking out of it.

A proper closed set is obtained by taking a finite number of irreducible curves (including





their generic points) plus a finite number of closed points.

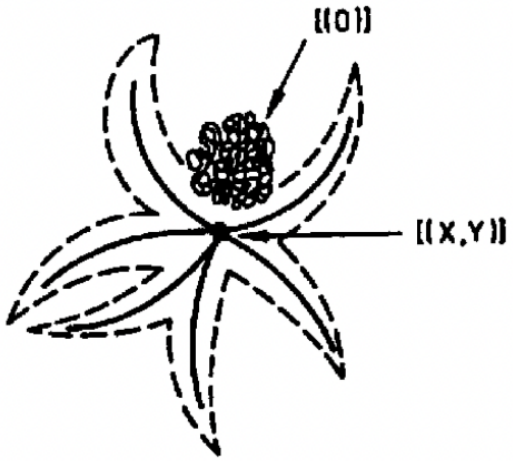
Given a closed subset of the set of closed points in the old topology, there is a unique set of non-closed points to add to get a closed set in our new plane.

Example 12. Let

$$\mathcal{O} = \left\{ \frac{f(X,Y)}{g(X,Y)} \mid f, g \in k[X,Y], g(0,0) \neq 0 \right\}.$$

Then $\mathcal{O}$ is the stalk of $\mathcal{O}_X$ at $(0,0)$ if $X = \mathrm{Spec}(k[X,Y])$.
$\mathcal{O}$ has the maximal ideal (X,Y), the principal prime ideals (f) where f is irreducible and $f(0,0) = 0$, and (0).
Thus $\mathrm{Spec}(\mathcal{O})$ has only one closed point: If $K = k(X)$, this scheme is $\mathrm{Spec}(K[Y]_S)$ for some $S \subseteq K[Y]$, i.e. is part of the affine line over K.
But if you throw away the closed point and the Y-axis, you get the distinguished open set $X \neq 0$, which contains none of the original closed points.

Example 13 (Arithmetic surface)**.** The prime ideals of $\mathbb{Z}[X]$ are (0), the principal primes (f) where f is either a prime p or a $\mathbb{Q}$-irreducible polynomial with coefficients having gcd 1, or the maximal ideals (p, f) where p is a prime and f is a monic integral polynomial irreducible mod p.



Remark 14. • A point of $\operatorname{Spec}\mathbb{Z}[X]$ is a prime ideal of $\mathbb{Z}[X]$. There are three kinds visible in the picture:

1. The generic point (0). Its closure is $V(0) = \operatorname{Spec}\mathbb{Z}[X]$.
2. Height-one primes (p) and (f).
   (p) is prime since $\mathbb{Z}[X]/(p) \cong \mathbb{F}_p[X]$ is a domain.
   Likewise if $f \in \mathbb{Z}[X]$ is primitive and irreducible, then (f) is prime.
   These are represented by the vertical and horizontal curves in the picture.
3. Closed points (p, g). These are the maximal ideals.

- The columns are the fibers over the closed points of $\operatorname{Spec}\mathbb{Z}$. The inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}[X]$ gives $\pi : \operatorname{Spec}\mathbb{Z}[X] \to \operatorname{Spec}\mathbb{Z}$.

  Since $\operatorname{Spec}\mathbb{Z} = \{(0), (2), (3), (5), \dots\}$, the fiber over (p) is

  $$\pi^{-1}((p)) = \operatorname{Spec}(\mathbb{Z}[X] \otimes_{\mathbb{Z}} \mathbb{F}_p) = \operatorname{Spec}\mathbb{F}_p[X].$$

  Equivalently, as a closed subset of $\operatorname{Spec}\mathbb{Z}[X]$, $\pi^{-1}((p)) = V(p)$. So for example, (2) is the unique generic point of $V(2)$, and $V(2)$ is exactly the fiber over $(2) \in \operatorname{Spec}\mathbb{Z}$, i.e. the column labeled $V((2))$ is $V(2) = \operatorname{Spec}\mathbb{F}_2[X]$, its generic point is (2), and its closed points correspond to irreducible polynomials in $\mathbb{F}_2[X]$:

  $$(2, X), \quad (2, X+1), \quad (2, X^2+X+1).$$

  These are the vertical fibers.

- Look at the fiber over the generic point $(0) \in \operatorname{Spec}\mathbb{Z}$. It is $\operatorname{Spec}\mathbb{Q}[X]$. The closed points of $\operatorname{Spec}\mathbb{Q}[X]$ correspond to irreducible polynomials over $\mathbb{Q}$:

  $$(X), \quad (X^2+1), \quad (X^2-2), \dots$$

  Take one of those points and take the closure inside $\operatorname{Spec}\mathbb{Z}[X]$. For example, $(X) \in \operatorname{Spec}\mathbb{Q}[X]$ spreads out to $V(X) \subseteq \operatorname{Spec}\mathbb{Z}[X]$. These are the horizontal curves in

the drawing. So the horizontal row marked $[(X^2+1)]$ represents the irreducible closed subset $V(X^2+1)$. Its unique generic point is $[(X^2+1)]$. Note: $V(X^2+1) = \operatorname{Spec}\mathbb{Z}[X]/(X^2+1) \cong \operatorname{Spec}\mathbb{Z}[i]$.

- Since $V(I) \cap V(J) = V(I+J)$, the intersection between the p-column and the X^2+1 row is $V(p) \cap V(X^2+1) = V(p, X^2+1)$. So we determine the intersection simply by reducing X^2+1 mod p and factoring it in $\mathbb{F}_p[X]$.

- So e.g. reducing X^2+1 mod 3, the possible squares mod 3 are 0 and 1, so there is no solution to $X^2 = -1 = 2$. Thus X^2+1 is irreducible in $\mathbb{F}_3[X]$. Hence $V(3) \cap V(X^2+1)$ consists of one point: $(3, X^2+1)$.

  Its residue field is

  $$\mathbb{Z}[X]/(3, X^2+1) \cong \mathbb{F}_3[X]/(X^2+1) \cong \mathbb{F}_9.$$

  So the picture has one intersection over 3. The prime is inert, not ramified.

- If you reduce X^2+1 mod 5, $2^2 = 4 = -1 \pmod 5$, so $X^2+1 = (X-2)(X+2)$ in $\mathbb{F}_5[X]$. So $X^2+1 = (X+3)(X+2)$ and then we get two different intersection points:

  $$(5, X+2), \quad (5, X+3).$$

  This is why the picture splits into 2 branches near the 5-column. This means (5) splits in $\mathbb{Z}[i]$.

- Modulo 2, $X^2+1 = X^2-1 = (X-1)^2$. So there is only one underlying point $(2, X+1)$. But the factor occurs with multiplicity 2. This is why there is a loop at $p=2$. Scheme-theoretically,

  $$V(2, X^2+1) = \operatorname{Spec}\mathbb{F}_2[X]/((X+1)^2)$$

  and this ring is not reduced. Since

  $$\mathbb{F}_2[X]/((X+1)^2) \cong \mathbb{F}_2[\varepsilon]/(\varepsilon^2),$$

  there is one geometric point but it has multiplicity 2. So the prime in the X^2+1 row lying over 2 is ramified.

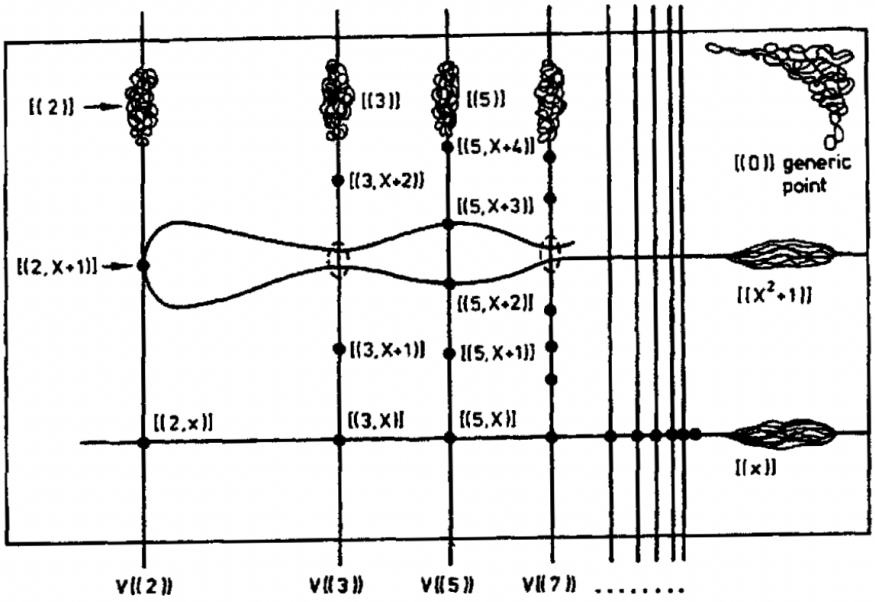
- $V(X^2+1)$ intersects every vertical fiber $V(p)$.
  For an odd prime p, X^2+1 either splits or stays irreducible modulo p. The complete rule is: if $p=2$ it ramifies. If $p \equiv 1 \pmod 4$, it splits. If $p \equiv 3 \pmod 4$, it is unramified/inert (one degree 2-point).

- The bottom row $V(X)$ is much simpler: $V(X) = \operatorname{Spec}\mathbb{Z}[X]/(X) \cong \operatorname{Spec}\mathbb{Z}$.
  at every prime p, $V(X) \cap V(p) = V(p, X)$, so there is exactly one point (p, X).

- For any prime ideal P, $\overline{\{P\}} = V(P)$ so the point P is the generic point of $V(P)$.

- This is an arithmetic surface, $\mathbb{A}^1_{\mathbb{Z}}$. The vertical direction is arithmetic: 2, 3, 5, 7,... and the horizontal direction is polynomial/geometric: $X, X^2+1, \ldots$

$\operatorname{Spec}(\mathbb{Z}[X])$ looks like:



Problem 2.1. What is $V((p)) \cap V((f))$ where f is a $\mathbb{Q}$-irreducible polynomial? What is $V((f)) \cap V((g))$ where f, g are distinct $\mathbb{Q}$-irreducible polynomials?

Recall: $V(I) \cap V(J) = V(I + J)$.

1) $V((p)) \cap V((f)) = V((p) + (f)) = V(p, f)$.
A point here is a prime ideal $I \subseteq \mathbb{Z}[X]$ containing both p and f. So since it contains p, reduce mod p: $\mathbb{Z}[X]/(p) \cong \mathbb{F}_p[X]$. Let $\overline{f}$ denote $f \pmod p$. Write

$$\overline{f} = h_1^{e_1} \cdots h_r^{e_r}$$

where h_i are distinct irreducible polynomials in $\mathbb{F}_p[x]$.

Then $V((p)) \cap V((f)) = \{(p, h_1), \ldots, (p, h_r)\}$.
Geometrically, the horizontal curve $V(f)$ meets the vertical fiber $V(p)$ at the points corresponding to the irreducible factors of $f \pmod p$.

2) $V(f) \cap V(g) = V(f, g)$. Since f, g are distinct irreducible polynomials over $\mathbb{Q}$, $\gcd(f, g) = 1$. By Bezout, we can write

$$A(x)f(x) + B(x)g(x) = 1, \quad A, B \in \mathbb{Q}[X].$$

Clearing denominators, there exist $A_0, B_0 \in \mathbb{Z}[X]$ and a nonzero integer N such that

$$A_0 f + B_0 g = N.$$

Thus $N \in (f, g)$.

Suppose $P \in V(f) \cap V(g)$. Then $(f, g) \subseteq P \implies N \in P \implies P$ contains some prime number p dividing N. So every intersection point lies over one of finitely many rational primes p. For such a p, reduce f mod p:
A closed point (p, h) lies on both $\iff h \mid \overline{f}$ and $h \mid \overline{g}$ in $\mathbb{F}_p[X]$.

Thus $V(f) \cap V(g) = \{(p, h) \mid h \text{ irreducible and } h \mid \gcd(\overline{f}, \overline{g})\}$.
Conclude:

$$V(p) \cap V(f) = \{\text{irreducible factors of } f \text{ mod } p.\}$$

$$V(f) \cap V(g) = \{\text{common irreducible factors of } f \text{ mod } p, g \text{ mod } p \text{ for finitely many } p\}.$$

Each closed subset $V((p))$ is a copy of $\mathbb{A}^1_{\mathbb{Z}/p\mathbb{Z}}$.
All maximal chains of irreducible proper subsets have length 2. And if $\mathcal{O}$ is the local ring at a closed point x, then $\mathcal{O}$ has Krull dimension 2.

The elements of R can always be interpreted as functions on $\mathrm{Spec}(R)$. For all $x = [P] \in \mathrm{Spec}(R)$, let

$$\kappa(x) = R_P/(P \cdot R_P) = \text{ quotient field } (R/P) = \text{ residue field}(\mathcal{O}_x).$$

If $a \in R$, a induces an element of R_P, and then in $\kappa(x)$. We call this the **value** $a(x)$ **of** a **at** x**.** These values lie in different fields at different points.

For instance, if $m \in \mathbb{Z}$, its value at $[(p)]$ is "$m \pmod p$" in $\mathbb{Z}/p\mathbb{Z}$.

More generally, if $a \in \Gamma(U, \mathcal{O}_X)$, and $x \in U$, we can let $a(x)$ be the image of a in $\kappa(x)$. For varieties we embedded $\mathcal{O}_X$ in the sheaf of k-valued functions on X, but this does not generalize: if $a \in R$, $a(x) = 0 \forall x \in \mathrm{Spec}(R) \iff a \in P$ for all prime ideals $P \subseteq R \iff a$ is nilpotent.

Example 15. Let $Z = \mathrm{Spec}(k[X]/(X^2))$ for k algebraically closed. Then Z is a single point. Z can also be embedded in $\mathbb{A}^1$ so that its unique point goes to the origin and so that its sheaf (which is just $k[X]/(X^2)$ can be obtained by taking all functions $f \in \mathcal{O}_{0,\mathbb{A}^1}$ modulo functions vanishing to second order at 0.

Then the function X which just vanishes to first order is still nonzero as a function on Z, and if f is any polynomial in Z, then one can compute from the restriction of f to the subscheme Z its value at 0 as well as its first derivative at 0.

More generally, $\mathrm{Spec}(k[X_1, \ldots, X_n]/A)$ can be interpreted as $V(A) \subseteq \mathbb{A}^n_k$ but with a sheaf of rings on it suitably fattened so as to include the first few terms of the Taylor expansion of polynomials in directions normal to $V(A)$.

If R is any ring with nilpotents, one can visualize $\mathrm{Spec}(R)$ as containing some extra normal material.

Proposition 16. *Let R be a ring and let $f \in R$. Then the topological space $\mathrm{Spec}(R)_f$ together with the restriction of the sheaf of rings $\mathcal{O}_{\mathrm{Spec}(R)}$ to $\mathrm{Spec}(R)_f$ is isomorphic to $\mathrm{Spec}(R_f)$ together with the sheaf of rings $\mathcal{O}_{\mathrm{Spec}(R_f)}$.*

2.2 The category of preschemes

Definition 1. A **prescheme** is a topological space X, plus a sheaf of rings $\mathcal{O}_X$ on X, provided that there exists an open covering $\{U_\alpha\}$ of X such that each pair $(U_\alpha, \mathcal{O}_X|_{U_\alpha})$ is isomorphic to $(\mathrm{Spec}(R_\alpha), \mathcal{O}_{\mathrm{Spec}(R_\alpha)})$ for some commutative ring R.

Note: modern language calls Mumford's preschemes "schemes" and his schemes "separated schemes."

Definition 2. An **affine scheme** is a prescheme $(X, \mathcal{O}_X)$ isomorphic to $(\mathrm{Spec}(R), \mathcal{O}_{\mathrm{Spec}(R)})$ for some ring R.

Note: An open subset of a prescheme is also a prescheme.

Definition 3. If $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ are 2 preschemes, a **morphism** from X to Y is a continuous map $f : X \to Y$ plus a collection of homomorphisms

$$f_V^* : \Gamma(V, \mathcal{O}_Y) \to \Gamma(f^{-1}(V), \mathcal{O}_X)$$

one for each open set $V \subseteq Y$, such that

1. Whenever $V_1 \subseteq V_2$ are two open sets in Y, the diagram

$$\begin{array}{ccc} \Gamma(V_2, \mathcal{O}_Y) & \xrightarrow{f_{V_2}^*} & \Gamma(f^{-1}(V_2), \mathcal{O}_X) \\ \downarrow \text{res} & & \downarrow \text{res} \\ \Gamma(V_1, \mathcal{O}_Y) & \xrightarrow{f_{V_1}^*} & \Gamma(f^{-1}(V_1), \mathcal{O}_X) \end{array}$$

   commutes, and

2. If $V \subseteq Y$ is open and $x \in f^{-1}(V)$ and $a \in \Gamma(V, \mathcal{O}_Y)$, then $a(f(x)) = 0$ implies $f_V^*(a)(x) = 0$.

Condition (2) may be expressed in other ways too. If $x \in X$ and $y = f(x)$, for all open neighborhoods U, V of x, y respectively (such that $f(U) \subseteq V$), we have a homomorphism

$$\Gamma(V, \mathcal{O}_Y) \underset{f_V^*}{\longrightarrow} \Gamma(f^{-1}(V), \mathcal{O}_X) \underset{\text{res}}{\longrightarrow} \Gamma(U, \mathcal{O}_X)$$

Passing to the limit over such U and V we get a homomorphism between stalks $f_x^* : \mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$. These stalks are always local rings. Thus condition (2) asserts that f_x^* is a local homomorphism, i.e. $f_x^*(m_y) \subseteq m_x$. When this is the case f_x^* also induces a homomorphism

$$\kappa(y) = \mathcal{O}_{Y,y}/m_y \xrightarrow{\overline{f_x^*}} \mathcal{O}_{X,x}/m_x = \kappa(x).$$

Call this induced residue-field map $\kappa(f(x)) \to \kappa(x)$. This induced map on residue fields has the property:
(2'): For all $a \in \Gamma(V, \mathcal{O}_Y)$, for some neighborhood V of y, $f_V^*(a)(x)$ is the image of $a(y)$ under $\kappa(y) \to \kappa(x)$.

Theorem 4. *Let X be a prescheme and let R be a ring. To every morphism $f : X \to \mathrm{Spec}(R)$, associate the homomorphism:*

$$R \cong \Gamma(\mathrm{Spec}(R), \mathcal{O}_{\mathrm{Spec}(R)}) \xrightarrow{f^*} \Gamma(X, \mathcal{O}_X).$$

Then this induces a bijection between $\mathrm{Hom}(X, \mathrm{Spec}(R))$ *and* $\mathrm{Hom}(R, \Gamma(X, \mathcal{O}_X))$.

Recall: given a map on schemes, if we restrict to affines, we get a map of rings in the opposite direction. If we take the direct limit, we get a map of local rings (between stalks of corresponding points), and this is how you get f_x^*.
To get f^*, if $a \in \Gamma(V, \mathcal{O}_Y)$, then $a \in \mathcal{O}_Y(V) \implies a$ is a function on V so that if a vanishes at $f(x)$, f^*a vanishes at x.

Remark 5.
- $\mathbb{P}_2, \mathbb{P}_1 \times \mathbb{P}_1$, and $\mathbb{P}_1$ have the same ring of functions (=constants)
- Requiring the morphism to map into affines allows the correspondence, since affines have large enough rings of functions.

Corollary 6. *The category of affine schemes is isomorphic to the category of commutative rings with unit, with arrows reversed.*

Corollary 7. $\mathrm{Spec}\,\mathbb{Z}$ *is the final object in the category of preschemes, i.e. for every prescheme X, there is a unique morphism $f : X \to \mathrm{Spec}\,\mathbb{Z}$.*

Every prescheme is a fibered object, with one fiber for each prime p, and one over the generic point of $\mathrm{Spec}\,\mathbb{Z}$. The fibering is given by $x \mapsto \mathrm{char}(\kappa(x))$.

Proposition 8. *Let X be a prescheme, and $Z \subseteq X$ an irreducible closed subset. Then there is one and only one point $z \in Z$ such that $Z = \overline{\{z\}}$.*

The point z will be called the **generic point** os Z. So there is a 1-1 correspondence between the points of X and the irreducible closed subsets of X.

If $x \in X$, there is a 1-1 correspondence between:

1. irreducible closed subsets $Z \subseteq U$ such that $x \in Z$
2. points $z \in X$ such that $x \in \overline{\{z\}}$
3. prime ideals $P \subseteq \mathcal{O}_{X,x}$

Example 9. Let $\mathcal{O}$ be a noetherian local ring with maximal ideal M. Thus $\mathrm{Spec}(\mathcal{O})$ has a unique closed point $x = [M]$.
Let X be the open subscheme $\mathrm{Spec}(\mathcal{O}) \setminus \{x\}$. The closed points of X correspond to prime ideals $P \subsetneq M$ such that there are no prime ideals between P and M.

X is rarely affine itself: to cover X, choose $f_1, \ldots, f_n \in M$ such that $M = \sqrt{(f_1, \ldots, f_n)}$. Then

$$X = \bigcup_{i=1}^{n} \mathrm{Spec}(\mathcal{O}_{f_i})$$

where $\mathrm{Spec}(\mathcal{O}_{f_i})$ and $\mathrm{Spec}(\mathcal{O}_{f_j})$ are patched along $\mathrm{Spec}(\mathcal{O}_{f_i f_j})$.

For instance if $\mathcal{O} = \mathbb{C}[[X_1, \ldots, X_n]]$ then X turns out to have topological properties identical to the $(2n-1)$-sphere.

Example 10 ($\mathbb{P}^n_{\mathbb{Z}}$)**.** Consider the union of $n+1$ copies $U_0, \ldots, U_n$ of $\mathbb{A}^n_{\mathbb{Z}} = \mathrm{Spec}\,\mathbb{Z}[X_1, \ldots, X_n]$.

Introduce $X_{ij}, 0 \leq i, j \leq n$ and $i \neq j$ and set

$$U_i = \mathrm{Spec}\,\mathbb{Z}[X_{0i}, X_{1i}, \ldots, X_{i-1,i}, X_{i+1,i}, \ldots, X_{ni}].$$

Then since

$$(U_i)_{X_{ji}} = \mathrm{Spec}\,\mathbb{Z}[X_{0i}, \ldots, X_{ni}, X_{ji}^{-1}]$$
$$(U_j)_{X_{ij}} = \mathrm{Spec}\,\mathbb{Z}[X_{0j}, \ldots, X_{nj}, X_{ij}^{-1}]$$

these two preschemes can be identified by

$$X_{kj} = X_{ki}/X_{ji}, \quad 0 \leq k \leq n, k \neq i, j$$

$$X_{ij} = 1/X_{ji}$$

with inverse

$$X_{ki} = X_{kj}/X_{ij}, \quad 0 \leq k \leq n, k \neq i, j$$
$$X_{ji} = 1/X_{ij}.$$

This is usually abbreviated by introducing $X_0, \ldots, X_n$ and replacing X_{ij} by X_i/X_j. Then

$$U_i = \mathrm{Spec}\,\mathbb{Z}[X_0/X_i, \ldots, X_n/X_i]$$

and the identification of U_i and U_j is given by the identity map between $\mathbb{Z}[X_0/X_i, \ldots, X_n/X_i, X_i/X_j]$ and $\mathbb{Z}[X_0/X_j, \ldots, X_n/X_j, X_j/X_i]$.

And then if $F(X_0, \ldots, X_n)$ is a homogeneous integral polynomial, we can talk about the set $F = 0$ where by definition

$$\{F = 0\} \cap U_i = V((F/X_i^d))$$

where $d = \deg F$.

Recall:

Definition 11. Given morphisms $r : X \to S$ and $s : Y \to S$, a **fiber product** is a commutative diagram

$$\begin{array}{ccc} X \times_S Y & \xrightarrow{p_2} & Y \\ \downarrow{\scriptstyle p_1} & & \downarrow{\scriptstyle s} \\ X & \xrightarrow{r} & S \end{array}$$

with universal property so that given any commutative diagram

$$\begin{array}{ccc} Z & \xrightarrow{q_2} & Y \\ \downarrow{\scriptstyle q_1} & & \downarrow{\scriptstyle s} \\ X & \xrightarrow{r} & S \end{array}$$

there is a unique morphism $t : Z \to X \times_S Y$ such that $q_1 = p_1 \circ t, \quad q_2 = p_2 \circ t$.

The fiber product is unique up to canonical isomorphism.

Remark 12.

- The product is a special case of fiber product (take $S = \{\text{pt}\}$)
- Fiber product is also called pullback (reversal of arrows gives the pushback)
- Direct products are products, direct sums in familiar categories are pushouts, tensor products are pushouts
- The pullback of the fiber product in the category of affine schemes = pushout in category of rings
- When not specified, $\times$ just means fiber product over $\operatorname{Spec}\mathbb{Z}$

t

Theorem 13. *If A and B are C-algebras, let the diagram of affine schemes*

$$\begin{array}{ccc} \operatorname{Spec}(A \otimes_C B) & \xrightarrow{p_2} & \operatorname{Spec}(B) \\ \downarrow{\scriptstyle p_1} & & \\ \operatorname{Spec}(A) & \xrightarrow{r} & \operatorname{Spec}(C) \end{array}$$

be defined by the canonical homomorphisms $C \to A, C \to B, A \to A \otimes_C B, B \to A \otimes_C B$. This makes $\operatorname{Spec}(A \otimes_C B)$ into a fiber product.

Theorem 14. *Given any morphisms $r : X \to S$, $s : Y \to S$, a fiber product exists.*

Note: for example, if $\mathbb{A}^n_{\mathbb{Z}} = \operatorname{Spec}(\mathbb{Z}[X_1, \ldots, X_n])$, then for all rings R,

$$\mathbb{A}^n_{\mathbb{Z}} \times \operatorname{Spec}(R) = \operatorname{Spec}(R[X_1, \ldots, X_n]).$$

Definition 15. $\mathbb{A}^n_R := \operatorname{Spec}(R[X_1, \ldots, X_n])$.

Example 16. Since $\mathbb{P}^n_{\mathbb{Z}}$ is the union of the $n+1$ affines

$$U_i = \operatorname{Spec}\mathbb{Z}[X_0/X_i, \ldots, X_n/X_i],$$

$\mathbb{P}^n_{\mathbb{Z}} \times \operatorname{Spec}(R)$ is the union of the affines

$$U_i' = \operatorname{Spec} R[X_0/X_i, \ldots, X_n/X_i]$$

with the standard patching. All homogeneous polynomials $F(X_0, \ldots, X_n)$ over R define closed subsets $F = 0$ of $\mathbb{P}^n_{\mathbb{Z}} \times \operatorname{Spec}(R)$.

Definition 17. $\mathbb{P}^n_R := \mathbb{P}^n_{\mathbb{Z}} \times \operatorname{Spec}(R)$.

Problem 2.2. For all homogeneous $F \in R[X_0, \ldots, X_n]$, show that $\{x \in \mathbb{P}^n_R \mid F(X) \neq 0\}$ is isomorphic to

$$\operatorname{Spec} R[X_0^d/F, \ldots, (\prod X_i^{d_i})/F, \ldots, X_n^d/F]$$

where $\sum d_i = d = \deg(F)$.

Let

$$S = R[X_0, \ldots, X_n],$$

and let $F \in S$ be homogeneous of degree d. Set

$$U = \{x \in \mathbf{P}^n_R \mid F(x) \neq 0\}.$$

We will show that

$$U \cong \operatorname{Spec} A,$$

where

$$A = R\left[\frac{X_0^{d_0} \cdots X_n^{d_n}}{F} \;\middle|\; d_0 + \cdots + d_n = d\right].$$

For each i, let

$$U_i = \{X_i \neq 0\} \subseteq \mathbf{P}^n_R.$$

Recall that

$$U_i \cong \operatorname{Spec} R\left[\frac{X_0}{X_i}, \ldots, \frac{\widehat{X_i}}{X_i}, \ldots, \frac{X_n}{X_i}\right].$$

Since F is homogeneous of degree d, the expression

$$f_i = \frac{F}{X_i^d}$$

is a polynomial in the affine coordinates X_j/X_i. Hence

$$U \cap U_i = D(f_i),$$

so

$$U \cap U_i \cong \operatorname{Spec} R\left[\frac{X_0}{X_i}, \ldots, \frac{\widehat{X_i}}{X_i}, \ldots, \frac{X_n}{X_i}\right]_{f_i}.$$

Now put

$$t_i = \frac{X_i^d}{F} \in A.$$

We claim that

$$A_{t_i} = R\left[\frac{X_0}{X_i}, \ldots, \frac{\widehat{X_i}}{X_i}, \ldots, \frac{X_n}{X_i}\right]_{f_i}.$$

Indeed, for every j we have

$$\frac{X_j}{X_i} = \frac{X_j X_i^{d-1}/F}{X_i^d/F} = \frac{X_j X_i^{d-1}/F}{t_i},$$

and $X_j X_i^{d-1}/F$ is one of the generators of A. Thus

$$\frac{X_j}{X_i} \in A_{t_i}.$$

Moreover,

$$\frac{1}{f_i} = \frac{X_i^d}{F} = t_i.$$

Therefore

$$R\left[\frac{X_0}{X_i}, \ldots, \widehat{\frac{X_i}{X_i}}, \ldots, \frac{X_n}{X_i}\right]_{f_i} \subseteq A_{t_i}.$$

Conversely, consider a generator

$$\frac{X_0^{d_0} \cdots X_n^{d_n}}{F}$$

of A, where $\sum_j d_j = d$. We may write

$$\frac{X_0^{d_0} \cdots X_n^{d_n}}{F} = \frac{(X_0/X_i)^{d_0} \cdots (X_n/X_i)^{d_n}}{F/X_i^d} = \frac{(X_0/X_i)^{d_0} \cdots (X_n/X_i)^{d_n}}{f_i}.$$

Hence every generator of A belongs to

$$R\left[\frac{X_0}{X_i}, \ldots, \widehat{\frac{X_i}{X_i}}, \ldots, \frac{X_n}{X_i}\right]_{f_i}.$$

This proves the claimed equality of rings, and therefore

$$D(t_i) \cong U \cap U_i.$$

It remains to check that the $D(t_i)$ cover $\operatorname{Spec} A$. Write

$$F = \sum_{|\alpha|=d} c_\alpha X^\alpha.$$

Dividing by F gives

$$1 = \sum_{|\alpha|=d} c_\alpha \frac{X^\alpha}{F}.$$

Thus the elements X^α/F, $|\alpha| = d$, generate the unit ideal of A.

Let

$$J = (t_0, \ldots, t_n) = \left(\frac{X_0^d}{F}, \ldots, \frac{X_n^d}{F} \right).$$

For any $\alpha = (\alpha_0, \ldots, \alpha_n)$ with $|\alpha| = d$,

$$\left(\frac{X^\alpha}{F} \right)^d = \prod_{i=0}^{n} \left(\frac{X_i^d}{F} \right)^{\alpha_i} \in J.$$

Hence every X^α/F lies in $\sqrt{J}$. Since these elements generate the unit ideal, we obtain

$$\sqrt{J} = A.$$

Therefore

$$D(t_0) \cup \cdots \cup D(t_n) = \operatorname{Spec} A.$$

The isomorphisms

$$D(t_i) \cong U \cap U_i$$

agree on overlaps, since they are induced by the usual projective coordinate changes

$$\frac{X_j}{X_i} = \frac{X_j/X_k}{X_i/X_k}.$$

Consequently they glue to an isomorphism of preschemes

$$\boxed{\{x \in \mathbf{P}_R^n \mid F(x) \neq 0\} \cong \operatorname{Spec} R \left[\frac{X_0^{d_0} \cdots X_n^{d_n}}{F} \;\middle|\; d_0 + \cdots + d_n = d \right].}$$

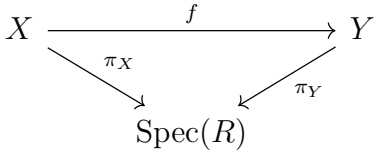
Proposition 18. *If $r : X \to S$ is surjective, then $p_2 : X \times_S Y$ is surjective.*

2.3 Varieties and preschemes

Definition 1. Let R be a ring. Then a **prescheme** X **over** R is a morphism $\pi : X \to \operatorname{Spec}(R)$.

Note: This is the same as giving $\Gamma(X, \mathcal{O}_X)$ an R-algebra structure.

Definition 2. Let X, Y be preschemes over R. An R**-morphism** from X to Y is a morphism $f : X \to Y$ such that



commutes.

This just means that the maps f_V^* ($V \subseteq Y$ open) are all R-algebra homomorphisms. This gives us the **category of preschemes over** R.

Definition 3. A prescheme X over R is of **finite type over** R if X is quasicompact and for all open affine subsets $U \subseteq X$, $\Gamma(U, \mathcal{O}_X)$ is a finitely-generated R-algebra.

Example 4. $R[[X]]$ is not of finite type but is complete.

Note: We typically mean f.g. as an algebra, not as a module. e.g. $R[1/f]$ is not a f.g. R-module.
Finite = f.g. as a module
Finite type = f.g. as an algebra

Proposition 5. *Let X be a prescheme over R. If there exists a finite affine open covering $\{U_i\}$, $1 \leq i \leq n$ of X such that $R_i = \Gamma(U_i, \mathcal{O}_X)$ is a finitely generated R-algebra, then X is of finite type over R.*

Corollary 6. *If X is an affine scheme and $U \subseteq X$ is an open affine subscheme, then $\Gamma(U, \mathcal{O}_X)$ is finitely generated over $\Gamma(X, \mathcal{O}_X)$.*

Warning: X may be of finite type over R but $\Gamma(X, \mathcal{O}_X)$ may not be a finitely generated R-algebra. This is the fact that Hilbert's 14th Problem was false.

Definition 7. A prescheme X is **reduced** if $\mathcal{O}_X$ contains no nilpotent sections, i.e. $\Gamma(U, \mathcal{O}_X)$ has no nilpotent elements for all open sets $U \subseteq X$.

This holds if and only if all the stalks have no nilpotents. And that holds if there is a covering of X by open affines U_i so that $\Gamma(U_i, \mathcal{O}_X)$ has no nilpotents.

Example 8. Consider $\mathbb{C}[x,t]/(t^2)$. This has (t) as a prime. $[(t)]$ is a generic point ((t) is the unique minimal prime; t^2 is in every prime). All other primes $(x-a,t)$ are maximal. Suppose the image of t does not vanish. It vanishes at all $(x-a,t)$. So it vanishes at every point, but it is not 0. Having nilpotents is what allows this.

Recall: the value of something at a point (= maximal ideal = closed point) is its image in the ring modulo that maximal ideal. So for $\operatorname{Spec}\mathbb{Z}$, the value at (p) is the value mod p.

Theorem 9. *Let k be an algebraically closed field. Then there is an equivalence of categories between*

1. *The category of reduced, irreducible preschemes of finite type over k, and k-morphisms*
2. *The category of prevarieties over k and morphisms of these (as in Ch. 1)*

Remark 10. We have maximal ideals, and then we have the extra primes, which are the generic points of irreducible subvarieties. The minimal prime in a domain is unique. To go from a scheme to a prevariety, we forget the generic points.

And $\operatorname{Spec} R$ is reduced and irreducible (=integral) $\iff$ R is a domain.

Lemma 11. *Let X be a prescheme of finite type over k, and let $x \in X$. Then x is closed $\iff$ the composition*

$$k \to \mathcal{O}_{X,x} \to \kappa(x)$$

is surjective.

So for preschemes X of finite type over k, a point $x \in X$ is closed $\iff$ it is closed in some open neighborhood of itself. This is false for general preschemes.

Example 12. Consider $\mathbb{A}^1_{\mathbb{C}}$. (x) is a closed point. Looking at the composition

$$k \to \mathcal{O}_{X,x} \to \mathcal{O}_{X,x}/m_x$$

we get

$$\mathbb{C} \to \mathbb{C}[x]_{(x)} \to \mathbb{C}[x]_{(x)}/(x) \cong \mathbb{C}$$

which turns out to be the identity. But if we take a non-closed point like (0), we get

$$\mathbb{C} \to \mathbb{C}(x) \to \mathbb{C}(x)/(0)$$

which cannot be surjective.

Lemma 13. *Let X be a prescheme of finite type over k. Then the closed points of X are dense in every closed subset of X.*

Note:

Definition 14 (Redefinition)**.** If k is an algebraically closed field, a **prevariety over** k is a reduced and irreducible prescheme of finite type over k.

Proposition 15. *The product of two prevarieties X and Y over k, in the category of prevarieties over k, is the same as their fiber product $X \times_{\operatorname{Spec}(k)} Y$, in the category of all preschemes.*

An application of the generic point of a variety:
Suppose X and Y are two prevarieties over k. Suppose we're given k-isomorphisms $\alpha : k(X) \to K$, $\beta : k(Y) \to K$ of their function fields with a third field K. Then α and β define morphisms $A : \operatorname{Spec} K \to X$, $B : \operatorname{Spec} K \to Y$ (by requiring that the images of A, B be the generic points x, y of X and Y and that $A^* : \mathcal{O}_{X,x} = k(X) \to K$ be α and that $B^* : \mathcal{O}_{Y,y} = k(Y) \to K$ be β.

By the proposition, we get a morphism $(A, B) : \operatorname{Spec} K \to X \times_k Y$.

Let t be the image point and let $T = \overline{\{t\}}$, a closed subset of $X \times Y$. T is irreducible, as it is the closure of a point, so T is a closed subprevariety of $X \times Y$. And $k(T)$ is isomorphic via $(A, B)^*$ to K.

We describe this situation by saying that $\beta^{-1} \circ \alpha$ is a **birational correspondence** between X and Y and that T is its **graph**.

Example 16. Let $X = \mathbb{P}_2$ and let x_0, x_1, x_2 and y_0, y_1, y_2 be homogeneous coordinates in X and Y.

Let $U_0 \subseteq X$ and $V_0 \subseteq Y$ be defined as the open sets $x_0 \cdot x_1 \cdot x_2 \neq 0$ and $y_0 \cdot y_1 \cdot y_2 \neq 0$. Define an isomorphism between U_0 and V_0 by the map:

$$y_0 = 1/x_0, \quad y_1 = 1/x_1, \quad y_2 = 1/x_2.$$

This is an extension of the isomorphism of function fields

$$k(x_1/x_0, x_2/x_0) \to k(y_1/y_0, y_2/y_0), \quad x_1/x_0 \mapsto y_1/y_0, x_2/x_0 \mapsto y_2/y_0.$$

Now we look at the closure $T \subseteq \mathbb{P}_2 \times \mathbb{P}_2$ of this isomorphism. Specifically, we'll just look at one piece of T. Let

$$U = \operatorname{Spec} k[x_1/x_0, x_2/x_0] = X_{x_0}$$
$$U = \operatorname{Spec} k[y_0/y_1, y_2/y_1] = Y_{y_1}.$$
$$T \cap (U \times V) = \operatorname{Spec} k[x_1/x_0, x_2/x_0, y_0/y_1, y_2/y_1]/P = \operatorname{Spec} k[A, B, C, D]/P$$

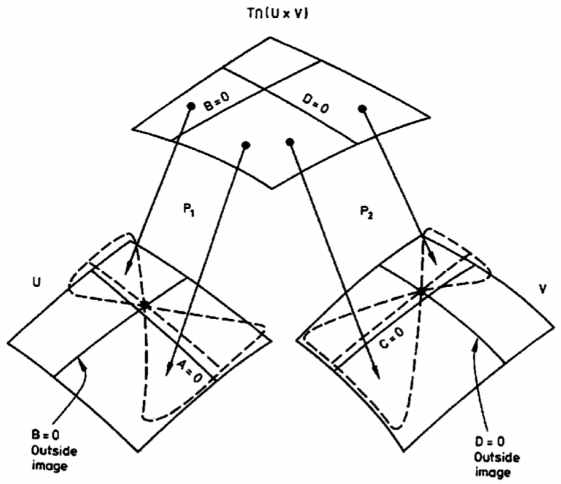
where P is the ideal of relations we get by putting all functions in the same function field (so $A - C \in P$ and $BD - C \in P$). Dividing out by $A - C$ and $BD - C$ gives $k[B, D]$ which is a domain of the correct dimension $\implies P = (A - C, BD - C)$ and

$$T \cap (U \times V) = \operatorname{Spec}(k[B, D]).$$

And the projections $p_1 : T \cap (U \times V) \to U$, $p_2 : T \cap (U \times V) \to V$ give

$$p_1^* : k[A, B] \to k[B, D], \quad p_2^* : k[C, D] \to k[B, D]$$

where $p_1^*(A) = B \cdot D$ and $p_2^*(C) = B \cdot D$. So the line $B = 0$ in $T \cap (U \times V)$ collapses to the point $A = B = 0$ in U, and the line $D = 0$ in $T \cap (U \times V)$ collapses to the point $C = D = 0$ in V.



2.4 Fields of definition

Let k be an algebraically closed field and let X be a closed subvariety of $\mathbb{A}^n$ defined by equations $f_i(X_1, \ldots, X_n) = 0$ for $1 \leq i \leq m$. Say k_0 is a subfield of k. Then if all coefficients of the f_i lie in k_0, we say X is **defined over** k_0. Let

$$A = (f_1, \ldots, f_m) = I(X).$$

Then if the coefficients of the f_i lie in k_0, $A = k \cdot A_0$ where $A_0 = A \cap k_0[X_1, \ldots, X_n]$, and if $R_0 = k_0[X_1, \ldots, X_n]/A_0$, then the affine ring $R = k[X_1, \ldots, X_n]/A$ of X is of the form $R_0 \otimes_{k_0} k$.

So when X is defined over k_0, there is an affine scheme $X_0 = \mathrm{Spec}(R_0)$ of finite type over k_0 such that $X = X_0 \times_{\mathrm{Spec}(k_0)} \mathrm{Spec}(k)$.

Conversely, if X is an affine variety over k and X_0 is a prescheme of finite type over k_0 such that $X = X_0 \times_{\mathrm{Spec}(k_0)} \mathrm{Spec}(k)$, then let $R_0 = \Gamma(X_0, \mathcal{O}_{X_0})$ and write $R_0 = k_0[X_1, \ldots, X_n]/(f_1, \ldots, f_m)$. Then if $R = \Gamma(X, \mathcal{O}_X)$, then $R \cong k[X_1, \ldots, X_n]/(f_1, \ldots, f_m)$, hence X is isomorphic to the closed subvariety of $\mathbb{A}^n$ defined by $f_i, 1 \leq i \leq m$. And since the coefficients of the f_i are in k_0, X is defined over k_0.

Often write $X = X_0 \times_{k_0} k$ for $X_0 \times_{\mathrm{Spec}(k_0)} \mathrm{Spec}(k)$.
If X_0 is the union of open affines $(U_0)_i = \mathrm{Spec}(R_i)$, then X is the union of the affine schemes $U_i = \mathrm{Spec}(R_i \otimes_{k_0} k)$.

Definition 1. Let σ be an automorphism of k over k_0. The conjugation map $\sigma_X : X \to X$ is the underlying map of the morphism

$$X = X_0 \times_{\mathrm{Spec}(k_0)} \mathrm{Spec}(k) \xrightarrow{1_{X_0} \times \phi} X = X_0 \times_{\mathrm{Spec}(k_0)} \mathrm{Spec}(k)$$

where $\phi : \mathrm{Spec}(k) \to \mathrm{Spec}(k)$ is defined by taking σ^{-1} as the homomorphism $\phi^* : k \to k$.

Check: $(\sigma \circ \tau)_X = \sigma_X \circ \tau_X$, i.e. the Galois group of k/k_0 is acting on the topological space X. So each σ_X is a homeomorphism of X.

If we have:

- $X_0 = \mathrm{Spec}\, k_0[X_1, \ldots, X_n]/(f_1, \ldots, f_m)$
- $X = \mathrm{Spec}\, k[X_1, \ldots, X_n]/(f_1, \ldots, f_m)$
- $x \in X$ is the closed point $X_1 = \alpha_1, \ldots, X_n = \alpha_n$ where $\alpha_1, \ldots, \alpha_n \in k$ and $f_i(\alpha_1, \ldots, \alpha_n) = 0 \forall i$

then $\sigma_X(x)$ is the closed point $X_1 = \sigma(\alpha_1), \ldots, X_n = \sigma(\alpha_n)$.

Theorem 2. *Let X_0 be a prescheme over k_0, let $X = X_0 \times_{k_0} k$, and let $p : X \to X_0$ be the projection. Assume k is an algebraic closure of k_0. Then*

1. *p is surjective and both open and closed*
2. *For all $x, y \in X$, $p(x) = p(y) \iff x = \sigma_X(y)$ for some $\sigma \in \mathrm{Gal}(k/k_0)$, i.e. for all $x \in X_0, p^{-1}(x)$ is an orbit of $\mathrm{Gal}(k/k_0)$. Moreover, $p^{-1}(x)$ is a finite set.*

Corollary 3. *X_0, as a topological space, is the quotient of X by the action of $\mathrm{Gal}(k/k_0)$.*

Definition 4. The **k_0-topology** on X is the set of $\mathrm{Gal}(k/k_0)$-invariant open sets, i.e. the set of open sets $\{p^{-1}(U) \mid U \text{ open in } X_0\}$.

Theorem 5. *Let X_0 be a prescheme over k_0, let $X = X_0 \times_{k_0} k$ and let $p : X \to X_0$ be the projection. Assume X_0 is of finite type over k_0 (hence X is of finite type over k).*

1. *For all $U \subseteq X_0$ open, the canonical map*

$$\Gamma(U, \mathcal{O}_{X_0}) \otimes_{k_0} k \to \Gamma(p^{-1}(U), \mathcal{O}_X)$$

is bijective. Moreover, the functions $f \in \Gamma(U, \mathcal{O}_{X_0})$ satisfy

$$f(\sigma_X(x)) = \sigma(f(x))$$

for all closed points $x \in p^{-1}(U)$, all $\sigma \in \mathrm{Gal}(k/k_0)$.

2. *If k_0 is perfect and X is reduced, then $\Gamma(U, \mathcal{O}_{X_0})$ is exactly the subring of elements $f \in \Gamma(p^{-1}(U), \mathcal{O}_X)$ satisfying "$f(\sigma_X(x)) = \sigma(f(x))$ for all closed points $x \in p^{-1}(U)$, all $\sigma \in \mathrm{Gal}(k/k_0)$."*

So X_0 as a prescheme over k_0 can be reconstructed from:

1. the prescheme X over k
2. the action of $\mathrm{Gal}(k/k_0)$ on X via conjugation
3. the subsheaf of $\mathcal{O}_X$, defined only on the k_0-open sets, which is $\mathcal{O}_{X_0}$.

Call $\mathcal{O}_{X_0}$ (viewed as a sheaf in the k_0-topology on X) the **sheaf of k_0-rational functions**. Presenting X as a fiber product $X_0 \times_{k_0} k$ is the same thing as endowing X with a **k_0-structure** consisting of the conjugation action $\{\sigma_X\}$ and the subsheaf $\mathcal{O}_{X_0}$ of k_0-rational functions.

Note: if $x \in X$ is closed, then $p(x)$ is closed in X_0, and if $y \in X_0$ is closed, then $p^{-1}(y)$ is a finite set of conjugate closed points of X.

Proposition 6. *Let $y \in X_0$ be a closed point. Then $\kappa(y)$ is a finite algebraic extension of k_0, and there is a natural bijection between the set of points $x \in p^{-1}(y)$ and the set of k_0-embeddings of $\kappa(y)$ into k.*

Example 7. Let $k_0 = \mathbb{R}$ and $k = \mathbb{C}$. Let $X_0 = \mathbb{A}^1_{\mathbb{R}}$ and $X = \mathbb{A}^1_{\mathbb{C}}$.
Take $y \in X_0$ closed (say $y = [(x^2+1)]$). Then $\kappa(y) = \mathbb{R}[x]/(x^2+1)$. The Proposition says: there are two embeddings of $\mathbb{R}[x]/(x^2+1)$ into $\mathbb{C}$. The embeddings are actually $x \mapsto i$ and $x \mapsto -i$.

But also let $p : \mathbb{A}^1_{\mathbb{C}} \to \mathbb{A}^1_{\mathbb{R}}$. Then $p^{-1}(y) = \{x - i, x + i\} = \{x = \pm i\}$, so the points are in bijection with the embeddings.

Example 8. Take a point where the residue field has q^d elements. We get d points upstairs. We get d embeddings of the residue field into $\overline{\mathbb{F}_q}$. And $\mathrm{Gal}(\mathbb{F}_{q^d}/\mathbb{F}_q)$ has d elements. There are d embeddings $\mathbb{F}_{q^d} \hookrightarrow \overline{\mathbb{F}_q}$.

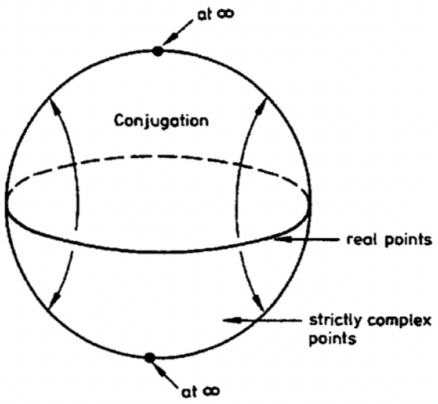
Definition 9. A closed point $y \in X_0$ is **rational over** k_0 if $\kappa(y) \cong k_0$.

Corollary 10. *If k_0 is perfect and $x \in X$, then $p(x)$ is rational over $k_0 \iff x$ is left fixed by all conjugations.*

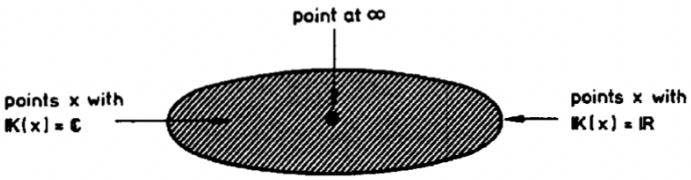
Example 11. Take $k = \mathbb{C}$, $k_0 = \mathbb{R}$,

$$X_0 = \mathrm{Spec}(\mathbb{R}[X, Y]/(X^2 + Y^2 - 1)).$$

Then X is a complex affine conic with two points at infinity:



And X_0 looks like:



The boundary of the disc is the circle of points rational over $\mathbb{R}$ corresponding to the maximal ideals

$$(X - \alpha, Y - \beta) \quad \alpha, \beta \in \mathbb{R} \text{ and } \alpha^2 + \beta^2 = 1.$$

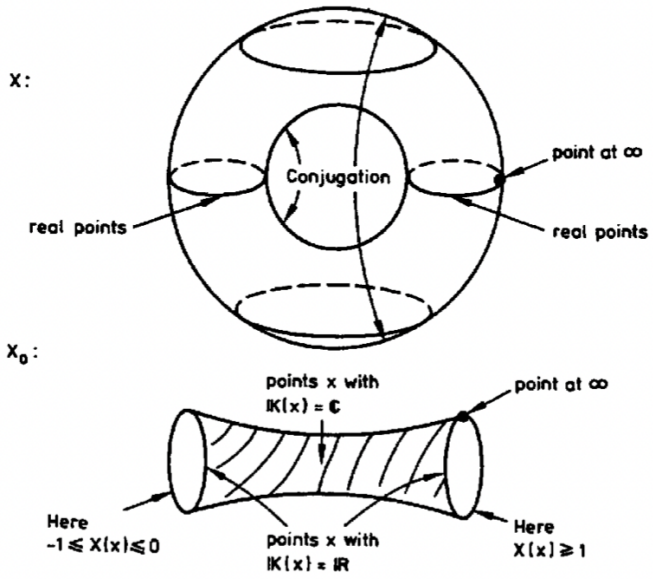
The interior points $\alpha, \beta \in \mathbb{R}$ and $\alpha^2 + \beta^2 < 1$ correspond to the maximal ideals

$$(X^2 + Y^2 - 1, \alpha X + \beta Y - 1)$$

and are not rational over $\mathbb{R}$.

For example, $x = \sqrt{2}$ meets the circle (in $\mathbb{C}$) at $(\sqrt{2}, i)$ and $(\sqrt{2}, -i)$. These correspond to $(x - \sqrt{2}, y^2 + 1)$, one closed point with residue field $\mathbb{C}$.

Example 12. $k_0 = \mathbb{R}$, $k = \mathbb{C}$, X_0 the real plane curve $Y^2 = X(X^2 - 1)$:



Example 13. $k_0 = \mathbb{R}$, $k = \mathbb{C}$, $X_0 = \mathrm{Spec}(\mathbb{R}[X,Y]/(X^2+Y^2))$. Since $\mathbb{R}[X,Y]/(X^2+Y^2)$ is a domain, X_0 is reduced and irreducible. But

$$X = \mathrm{Spec}(\mathbb{C}[X,Y]/(X+iY)(X-iY))$$

is the union of two lines, so X is reduced but not irreducible. Also X_0 has exactly one point rational over $\mathbb{R}$ (the intersection of two lines).

$k_0(X_0) = \mathbb{R}(X_0) \cong \mathbb{C}(t)$. $\mathbb{R}$ is not separably algebraically closed in $\mathbb{C}(t)$.

So X_0 is reduced and irreducible. The ring is not integrally closed, is a domain. X is reduced, not irreducible.

Example 14. If k_0 is imperfect of characteristic p and $a \in k_0 \setminus k_0^p$, then let $\ell(X,Y)$ be a linear polynomial and let

$$X_0 = \mathrm{Spec}(k_0[X,Y]/(\ell(X,Y)^p - a))$$

$$X = \mathrm{Spec}(k[X,Y]/((\ell(X,Y) - a^{1/p})^p))$$

X_0 is reduced and irreducible but X is just a p-fold line (set-theoretically just $\ell = a^{1/p}$, but $\ell - a^{1/p}$ is a nonzero nilpotent function on X)

$k_0(X_0) = \mathrm{Frac}(k_0[X,Y]/(\ell^p - a)) \cong k_0(a^{1/p})(t)$ is not separable over k_0.

Proposition 15. *Let X_0 be a reduced and irreducible prescheme over k_0. Let k be an algebraically closed extension of k_0. Let $x \in X_0$ be its generic point and let $k_0(X_0)$ be the stalk of $\mathcal{O}_{X_0}$ at x. Then*

1. *$X_0 \times_{k_0} k$ is reduced $\iff$ $k_0(X_0)$ is separable over k_0*

2. $X_0 \times_{k_0} k$ is irreducible $\iff$ k_0 is separably algebraically closed in $k_0(X_0)$.

Definition 16. Let X_0 be a prescheme of finite type over k_0. Then X_0 is a **prevariety over** k_0 if it is reduced and irreducible and if a) its function field $k_0(X_0)$ is separable over k_0, and b) k_0 is algebraically closed in $k_0(X_0)$.

A prevariety over k_0 is a geometrically integral prescheme of finite type over k_0.

2.5 Closed subpreschemes

Let $\mathcal{F}, \mathcal{G}$ be sheaves of abelian groups on the topological space X, $\varphi : \mathcal{F} \to \mathcal{G}$ a morphism. For U open, let $\mathcal{K}(U) = \ker(\mathcal{F}(U) \to \mathcal{G}(U))$. Can check: $\mathcal{K}$ is a sheaf, which we call the **kernel** of φ.

We say φ is **injective** if $\mathcal{K} = 0$ (i.e. if $\mathcal{F}(U) \to \mathcal{G}(U)$ is injective for all U ($\iff$ $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$ is injective for all x.) This *does not imply* that φ_U is surjective for all U.

Let $\mathcal{L}^*$ be the presheaf given by $\mathcal{L}^*(U) = \mathcal{G}(U)/\varphi(\mathcal{F}(U))$. Let $\mathcal{L}$ be the associated sheaf and call it the **cokernel** of φ.

We say φ is **surjective** if $\mathcal{L} = 0$ ($\iff$ $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$ is surjective for all x.) This *does not imply* that φ_U is surjective for all U.

Remark 1. The category of sheaves of abelian groups on X is an abelian category.

Proposition 2. *If $\varphi : \mathcal{F} \to \mathcal{G}$ is injective and surjective, then it is an isomorphism.*

Remark 3. If X is a topological space and $Y \subseteq X$ is a closed subset, then the sheaves of abelian groups on Y correspond bijectively to the sheaves of abelian groups on X such that $\mathcal{F}_x = \{0\}$ if $x \notin Y$:
If $\mathcal{F}_0$ is a sheaf on Y, **extend it by zero** by defining, for $U \subseteq X$ open,

$$\mathcal{F}(U) = \begin{cases} \{0\}, & U \cap Y = \varnothing \\ \mathcal{F}_0(U \cap Y), & U \cap Y \neq \varnothing \end{cases}$$

And the inverse to this is given by restricting a sheaf on X to one on Y (precise definition recalled in Mumford's footnote on p.104)

Suppose $Y \subseteq X$ is a closed subset. If X were just an ordinary variety with ordinary functions, you might think {functions on X} $\to$ {functions on Y} should be obtained simply by restricting functions. But for a prescheme, the underlying points do not capture all the information in the structure sheaf. For example: $\operatorname{Spec} k[t]/(t^2)$ has exactly one point, just like $\operatorname{Spec} k[t]/(t)$. But these are not the same prescheme. The first has a nonzero nilpotent t.

Definition 4. A **closed subprescheme** of X is

1. A closed set $Y \subseteq X$
2. A sheaf of rings $\mathcal{O}_Y$ on Y such that $(Y, \mathcal{O}_Y)$ is a prescheme

3. A surjective homomorphism $\pi : \mathcal{O}_X \to \mathcal{O}_Y$ (extended by zero)

Let $\mathcal{Q} = \ker \pi$. $\mathcal{Q}$ is a **sheaf of ideals** in $\mathcal{O}_X$,
i.e. $\mathcal{Q}$ is a subsheaf of $\mathcal{O}_X$ such that for all U, $\mathcal{Q}(U)$ is an ideal in $\mathcal{O}_X(U)$. We will call such an object a $\mathcal{O}_X$**-ideal.**

Remark 5. Observe:

- We have a stalk sequence $0 \to \mathcal{Q}_x \to \mathcal{O}_{X,x} \to \mathcal{O}_{Y,x} \to 0$
- If $\mathcal{Q}_x = \mathcal{O}_{X,x}$, the map to $\mathcal{O}_{Y,x}$ is zero, so x is not actually in the support of the prescheme.
- And if $\mathcal{Q}_x \neq \mathcal{O}_{X,x}$, then something survives at x.

Conclude that $Y = \{x \in X \mid \mathcal{Q}_x \neq \mathcal{O}_{X,x}\}$, and $\mathcal{O}_Y \cong \operatorname{coker}(\mathcal{Q} \to \mathcal{O}_X)$. Thus $\mathcal{Q} = \ker \pi$ determines the closed subpreschemes up to canonical isomorphism, because:

1. $\mathcal{O}_Y = \mathcal{O}_X/\mathcal{Q}$
2. $Y = \{x \in X \mid \mathcal{Q}_x \neq \mathcal{O}_{X,x}\}$

So from $\mathcal{Q}$ and X, we can recover $(Y, \mathcal{O}_Y)$.

Definition 6. A morphism of preschemes $f : Y \to X$ is a **closed immersion** if

1. f is injective
2. f is closed, i.e. $Z \subseteq Y$ closed $\implies$ $f(Z)$ closed
3. $f_y^* : \mathcal{O}_{X,f(y)} \to \mathcal{O}_{Y,y}$ is surjective for all $y \in Y$.

Note: a closed immersion is exactly a morphism that identifies Y with a closed subprescheme of X.

Proposition 7. *Let R be a ring, $A \subseteq R$ an ideal. The canonical map $\pi : R \to R/A$ defines a morphism $f : \operatorname{Spec}(R/A) \to \operatorname{Spec}(R)$. Then f is a closed immersion and the corresponding sheaf of ideals $\mathcal{Q}$ satisfies:*

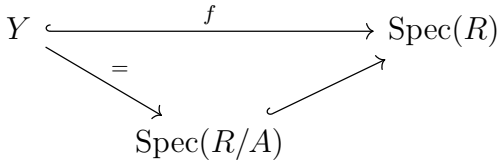
1. $\Gamma(\operatorname{Spec}(R)_f, \mathcal{Q}) = A \cdot R_f$
2. $\mathcal{Q}_x = A \cdot \mathcal{O}_{\operatorname{Spec}(R),x}$

Thus an ideal produces a closed subprescheme, via

$$A \subseteq R \mapsto \operatorname{Spec}(R/A) \subseteq \operatorname{Spec} R$$

because $\operatorname{Spec}(R/A) \to \operatorname{Spec} R$ is a closed immersion. And for the defining ideal sheaf $\mathcal{Q}$ of $\operatorname{Spec}(R/A)$: On a principal affine open $D(f) = (\operatorname{Spec} R)_f$, $\mathcal{Q}(D(f)) = AR_f$, and at a point $x = [P]$, $\mathcal{Q}_x = AR_P$.

Theorem 8. *Let $X = \operatorname{Spec}(R)$ and let $Y \subseteq X$ be a closed subprescheme, $f : Y \to X$ the inclusion. Let $\mathcal{Q}$ be the $\mathcal{O}_X$-ideal defining Y, $A = \Gamma(X, \mathcal{Q})$. Then Y is canonically isomorphic to $\operatorname{Spec}(R/A)$, i.e. there is a commutative diagram*



Thus every affine closed subprescheme comes from an ideal. Given a closed subprescheme $Y \subseteq X = \operatorname{Spec} R$, Y has its defining ideal sheaf $\mathcal{Q} = \ker(\mathcal{O}_X \to \mathcal{O}_Y)$. We can get an ideal via

$$(Y \subseteq X) \mapsto (A = \Gamma(X, \mathcal{Q}) \subseteq \Gamma(X, \mathcal{O}_X) = R).$$

Further, $Y \cong \operatorname{Spec}(R/A)$.

Remark 9. Nullstellensatz gave us:

$$\{\text{radical ideals}\} \leftrightarrow \{\text{closed subsets}\}$$

Our results from II.5 give:

$$\{\text{all ideals}\} \leftrightarrow \{\text{closed subpreschemes}\}$$

Corollary 10. *Let $f : Y \to X$ be a morphism of preschemes. Then TFAE:*

(i) f is a closed immersion;

(ii) for all affine open sets $U \subseteq X$, $f^{-1}(U)$ is affine and the map

$$\Gamma(U, \mathcal{O}_X) \longrightarrow \Gamma(f^{-1}(U), \mathcal{O}_Y)$$

is surjective;

(iii) there exists an affine open covering $\{U_i\}$ of X such that $f^{-1}(U_i)$ is affine and the map

$$\Gamma(U_i, \mathcal{O}_X) \longrightarrow \Gamma(f^{-1}(U_i), \mathcal{O}_Y)$$

is surjective for all i.

Corollary 11. *Let X be a prescheme and let $\mathcal{Q}$ be an $\mathcal{O}_X$-ideal. Let*

$$Y = \{y \in X \mid 1 \notin \mathcal{Q}_y\},$$

a closed subset of X. The cokernel of $0 \longrightarrow \mathcal{Q} \longrightarrow \mathcal{O}_X$ is 0 outside of Y, so consider it as a sheaf of rings $\mathcal{O}_Y$ on Y. Then:

$$(Y, \mathcal{O}_Y) \text{ is a subprescheme of } (X, \mathcal{O}_X) \iff$$

For all $y \in X, \exists$ a neighborhood U of y and sections $\{s_\alpha\}$ in $\Gamma(U, \mathcal{Q})$ such that $\underbrace{\mathcal{Q}_x = \sum_\alpha \operatorname{res}_x(s_\alpha)\mathcal{O}_{X,x}}_{\text{quasi-coherence condition}}$

Example 12 (Closed subschemes of $\operatorname{Spec}(k[t])$)**.** Say k is algebraically closed. Since $k[t]$ is a PID, all nonzero ideals are of the form

$$A = \left(\prod_{i=1}^{n} (t - a_i)^{r_i} \right).$$

The corresponding subscheme $Y = \operatorname{Spec}(k[t]/A)$ of $\mathbb{A}^1 = \operatorname{Spec}(k[t])$ has support $\{a_1, \ldots, a_n\}$, and at a_i, its structure sheaf is

$$\mathcal{O}_{Y,a_i} = \mathcal{O}_{\mathbb{A}^1,a_i}/\mathfrak{m}_i^{r_i},$$

where

$$\mathfrak{m}_i = \mathfrak{m}_{\mathbb{A}^1,a_i} = (t - a_i).$$

Focus on the schemes that are supported just by $\{0\}$. Since

$$V((t^n)) = V(\sqrt{(t^n)}) = V((t)) = \{0\},$$

these are the schemes

$$Y_n = \operatorname{Spec}\left(k[t]/(t^n)\right).$$

Then

$$Y_1 \subset Y_2 \subset Y_3 \subset \cdots \subset Y_n \subset \cdots \subset \mathbb{A}^1.$$

Suppose

$$g(t) = a_0 + a_1 t + a_2 t^2 + \cdots.$$

Modulo t^n, every term of degree n or higher disappears:

$$[g(t)] = a_0 + a_1 t + \cdots + a_{n-1} t^{n-1}.$$

So $g|_{Y_n}$ remembers exactly the first n coefficients of g. In general

$$a_j = \frac{g^{(j)}(0)}{j!}.$$

Thus

$$g(t) \equiv g(0) + g'(0)t + \frac{g''(0)}{2!}t^2 + \cdots + \frac{g^{(n-1)}(0)}{(n-1)!}t^{n-1} \pmod{t^n}.$$

So if you restrict a function g on $\mathbb{A}^1$ to Y_n, it remembers the derivatives

$$g(0), g'(0), \ldots, g^{(n-1)}(0).$$

Example 13 (Closed subschemes of $\operatorname{Spec}(k[x,y])$)**.** Let k be algebraically closed. Let

$$X = \mathbb{A}^2_k = \operatorname{Spec} k[x,y], \qquad P = (0,0), \qquad \mathfrak{m}_P = (x,y).$$

For $(\alpha, \beta) \neq (0,0)$, consider

$$A_{\alpha,\beta} = (\alpha x + \beta y,\ x^2,\ xy,\ y^2) = (\alpha x + \beta y) + \mathfrak{m}_P^2.$$

The corresponding closed subprescheme is

$$Y_{\alpha,\beta} = \operatorname{Spec}\left(\frac{k[x,y]}{(\alpha x + \beta y, x^2, xy, y^2)}\right).$$

Since $\sqrt{A_{\alpha,\beta}} = (x,y)$, the underlying closed set is just

$$|Y_{\alpha,\beta}| = \{P\}.$$

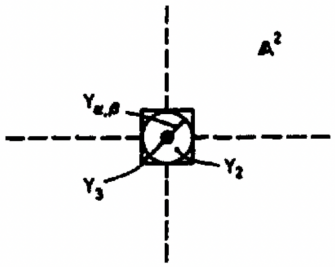
But $Y_{\alpha,\beta}$ is not reduced: it retains one infinitesimal direction at P.

All of the following have underlying space $\{P\}$:

$$Y_{\text{red}} = \operatorname{Spec}\frac{k[x,y]}{(x,y)}, \quad Y_{\alpha,\beta} = \operatorname{Spec}\frac{k[x,y]}{(\alpha x + \beta y, x^2, xy, y^2)},$$

$$Y_2 = \operatorname{Spec}\frac{k[x,y]}{(x^2, xy, y^2)} = \operatorname{Spec}\frac{k[x,y]}{\mathfrak{m}_P^2}, \quad \text{and } Y_3 = \operatorname{Spec}\frac{k[x,y]}{(x^2, y^2)}.$$

- Y_{red} remembers only the value $g(P)$.
- $Y_{\alpha,\beta}$ remembers the value and one directional first-order variation.
- Y_2 remembers $g(P)$, $\frac{\partial g}{\partial x}(P)$, $\frac{\partial g}{\partial y}(P)$.
- Y_3 retains still more infinitesimal information, since $xy \neq 0$ in $k[x,y]/(x^2,y^2)$.



Now consider

$$A = (x^2, xy) = x(x,y) \subseteq k[x,y].$$

Then

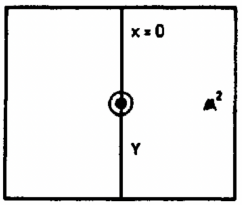
$$Y = \operatorname{Spec}\frac{k[x,y]}{(x^2,xy)}.$$

Its underlying closed set is $V(x^2,xy) = V(x)$ because $\sqrt{(x^2,xy)} = (x)$. Thus the support is the line $\{x = 0.\}$ But the scheme has extra infinitesimal structure at the origin: $x \neq 0, x^2 = 0,$ and $xy = 0$ in its coordinate ring.

Proposition 14. *Let X be a prescheme and $Z \subseteq X$ a closed subset. Consider the various closed subschemes*

$$Z_1 = (Z, \mathcal{O}_X/\mathcal{I}_1)$$

that can be defined with support Z.



1. *Among all these, there is a unique reduced subscheme (call it $Z_0 = (Z, \mathcal{O}_X/\mathcal{I}_0)$).*
2. *Furthermore, if $Z_1 = (Z, \mathcal{O}_X/\mathcal{I}_1)$ is any other, then $\mathcal{I}_1 \subseteq \mathcal{I}_0$ (so Z_0 is a subscheme of Z_1) and $\mathcal{I}_0 = \sqrt{\mathcal{I}_1}$, i.e.,*
$$\Gamma(U, \mathcal{I}_0) = \sqrt{\Gamma(U, \mathcal{I}_1)}$$
for U affine.

So if $Z \subseteq X$ is a closed subset, then among all the different closed subpreschemes having support Z, there is a unique reduced one.

Example 15 (Fibers of a morphism)**.** Let $f : X \to Y$ be a morphism and let $y \in Y$ be a closed point.
Consider the fiber product $X_y = X \times_Y \operatorname{Spec}(\kappa(y))$ in the diagram

$$\begin{array}{ccc} X_y & \xrightarrow{j} & X \\ \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle f} \\ \operatorname{Spec}(\kappa(y)) & \xrightarrow{i} & Y \end{array}$$

where i is the morphism with image y and $i^* : \mathcal{O}_{Y,y} \to \kappa(y)$ is the canonical map to the residue field.
Since y is closed,
$$i : \operatorname{Spec} \kappa(y) \hookrightarrow Y$$
is a closed immersion. Closed immersions are preserved under base change, so

$$\boxed{j : X_y \hookrightarrow X \text{ is a closed immersion.}}$$

Thus the fiber $X_y = X \times_Y \operatorname{Spec}(\kappa(y))$ is naturally a closed subprescheme of X supported on the set-theoretic fiber $f^{-1}(y)$.
If

- $V = \operatorname{Spec}(S)$ is an affine open neighborhood of y,
- $\{U_i = \operatorname{Spec}(R_i)\}$ covers $f^{-1}(V)$,
- and $y = [M]$ where $M \subseteq S$ is maximal

then by our results about fiber products, $X_y = X \times_Y \operatorname{Spec}(\kappa(y))$ is covered by open affine pieces
$$X_y^{(i)} = \operatorname{Spec}(R_i \otimes_S \kappa(y)),$$

and since $\kappa(y) \cong S/M$,

$$X_y^{(i)} \cong \operatorname{Spec}(R_i/M \cdot R_i).$$

Thus, scheme-theoretic fibers can be nonreduced, because the ideal MR_i need not be radical. For example, let

$$X = \operatorname{Spec}(k[X_1, \ldots, X_n]/(f_1, \ldots, f_m))$$

be an affine scheme of finite type over k. Define

$$\varphi : \mathbb{A}_k^n \to \mathbb{A}_k^m$$

such that $\varphi^* : k[Y_1, \ldots, Y_m] \to k[X_1, \ldots, X_n]$ maps $Y_i \mapsto f_i$.

Then X can be realized as the fiber of φ over the origin.

And if $(f_1, \ldots, f_m)$ is not radical then the fiber need not be reduced.

For example: $X_0 = \operatorname{Spec}(k[t]/(t^2))$ can be realized as the fiber of $f : \mathbb{A}_k^1 \to \mathbb{A}_k^1, t \mapsto t^2$ over 0, and this ends up being a nonreduced double point.

Problem 2.3. Show that the closed subpreschemes of $\mathbb{P}_k^n$ correspond bijectively with the homogeneous ideals

$$A \subseteq k[X_0, \ldots, X_n]$$

such that

$$A : (X_0, \ldots, X_n) = A$$

(i.e., if $f \in k[X_0, \ldots, X_n]$, $X_i \cdot f \in A$ for all i, then $f \in A$.) This condition says that A is biggest among those ideals that induce the same ideals in each affine piece of $\mathbb{P}^n$.)

Let

$$S = k[X_0, \ldots, X_n], \qquad \mathfrak{m} = (X_0, \ldots, X_n).$$

We prove that closed subpreschemes of $\mathbb{P}_k^n$ are in bijection with homogeneous ideals $A \subseteq S$ satisfying

$$A : \mathfrak{m} = A.$$

Recall that $\mathbb{P}_k^n$ is covered by the standard affine open sets

$$U_i = \{X_i \neq 0\} \cong \operatorname{Spec} k \left[\frac{X_0}{X_i}, \ldots, \widehat{\frac{X_i}{X_i}}, \ldots, \frac{X_n}{X_i} \right].$$

Step 1: A homogeneous ideal determines a closed subprescheme.
Let $A \subseteq S$ be homogeneous. On the affine chart U_i, define

$$A_i = \left\{ \frac{f}{X_i^d} \;\middle|\; f \in A \text{ homogeneous of degree } d \right\}.$$

Thus A_i is an ideal in

$$k\left[\frac{X_0}{X_i}, \ldots, \frac{\widehat{X_i}}{X_i}, \ldots, \frac{X_n}{X_i}\right].$$

It therefore defines a closed subprescheme

$$Y_i = V(A_i) \subseteq U_i.$$

On the overlap $U_i \cap U_j$, the ideals A_i and A_j agree after localization, since both come from the same homogeneous ideal A. Hence the Y_i agree on overlaps and glue to a closed subprescheme

$$Y(A) \subseteq \mathbb{P}^n_k.$$

Step 2: The condition $A : \mathfrak{m} = A$ makes A the largest ideal giving the same affine ideals. Suppose $f \in S$ is homogeneous of degree d and

$$X_i f \in A \qquad \text{for every } i.$$

Then on U_i,

$$\frac{f}{X_i^d} = \frac{X_i f}{X_i^{d+1}} \in A_i.$$

Thus adjoining f to A would not change the ideal induced on any of the affine charts U_i.

Therefore, if A is to be the largest homogeneous ideal inducing the given ideals A_i, we must have

$$X_i f \in A \text{ for all } i \quad \Longrightarrow \quad f \in A.$$

This is exactly the condition

$$A : \mathfrak{m} = A.$$

Conversely, suppose $A : \mathfrak{m} = A$. Then, by induction,

$$A : \mathfrak{m}^r = A$$

for every $r \geq 1$.
Indeed, if

$$\mathfrak{m}^r f \subseteq A,$$

then for each i,

$$\mathfrak{m}^{r-1}(X_i f) \subseteq A.$$

Inductively $X_i f \in A$ for every i, and since $A : \mathfrak{m} = A$, it follows that $f \in A$.

Thus if some homogeneous polynomial can be added to A without changing any of the ideals on the affine charts, it must already belong to A.

Step 3: Recover an ideal from a closed subprescheme.
Conversely, let

$$Y \subseteq \mathbb{P}^n_k$$

be a closed subprescheme.

For each i, the intersection

$$Y \cap U_i$$

is a closed subprescheme of the affine scheme U_i, so it is defined by a unique ideal

$$I_i \subseteq k\left[\frac{X_0}{X_i}, \ldots, \frac{\widehat{X_i}}{X_i}, \ldots, \frac{X_n}{X_i}\right].$$

Since the I_i arise from the same closed subprescheme, they agree on the overlaps $U_i \cap U_j$.

Define a homogeneous ideal $A(Y) \subseteq S$ as follows. For a homogeneous polynomial f of degree d, declare

$$f \in A(Y)$$

if and only if

$$\frac{f}{X_i^d} \in I_i \qquad \text{for every } i.$$

This defines a homogeneous ideal of S.

By construction, the ideal induced by $A(Y)$ on each U_i is I_i. Hence

$$Y(A(Y)) = Y.$$

We now check that

$$A(Y) : \mathfrak{m} = A(Y).$$

Suppose f is homogeneous of degree d and

$$X_i f \in A(Y) \qquad \text{for all } i.$$

Fix i. Since $X_i f \in A(Y)$, we have

$$\frac{X_i f}{X_i^{d+1}} = \frac{f}{X_i^d} \in I_i.$$

This holds for every i, so by the definition of $A(Y)$,

$$f \in A(Y).$$

Therefore

$$A(Y) : \mathfrak{m} = A(Y).$$

Step 4: The two constructions are inverse.
We have already shown that

$$Y(A(Y)) = Y.$$

Now let A be a homogeneous ideal satisfying

$$A : \mathfrak{m} = A.$$

Certainly

$$A \subseteq A(Y(A)).$$

For the reverse inclusion, let f be homogeneous and suppose

$$f \in A(Y(A)).$$

Then on every chart U_i,

$$\frac{f}{X_i^{\deg f}} \in A_i.$$

By the definition of localization, for each i there exists an integer $r_i \geq 0$ such that

$$X_i^{r_i} f \in A.$$

Choose N large enough that every monomial of degree N is divisible by at least one of

$$X_0^{r_0}, \ldots, X_n^{r_n}.$$

For example, one may take

$$N = r_0 + \cdots + r_n.$$

It follows that

$$\mathfrak{m}^N f \subseteq A.$$

But, as observed above,

$$A : \mathfrak{m}^N = A,$$

so

$$f \in A.$$

Hence

$$A(Y(A)) = A.$$

Thus the assignments

$$A \longmapsto Y(A) \qquad \text{and} \qquad Y \longmapsto A(Y)$$

are mutually inverse.

Therefore the closed subpreschemes of $\mathbb{P}^n_k$ correspond bijectively to the homogeneous ideals

$$A \subseteq k[X_0, \ldots, X_n]$$

such that

$$\boxed{A : (X_0, \ldots, X_n) = A.}$$

Equivalently, these are exactly the homogeneous ideals that are largest among all homogeneous ideals inducing the same ideals on each standard affine piece of $\mathbb{P}^n_k$.

2.6 The functor of points of a prescheme

Example 1. The point set of a fiber product $X \times_S Y$ does not necessarily map injectively into the set-theoretic product of X and Y.

For example,

$$\operatorname{Spec}(\mathbb{C}) \times_{\operatorname{Spec}(\mathbb{R})} \operatorname{Spec}(\mathbb{C}) \cong \operatorname{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}) .$$

Now

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}[X]/(X^2 + 1) \cong \mathbb{C}[X]/((X - i)(X + i)) \cong \mathbb{C} \times \mathbb{C}.$$

Thus

$$\operatorname{Spec}(\mathbb{C}) \times_{\operatorname{Spec}(\mathbb{R})} \operatorname{Spec}(\mathbb{C}) \cong \operatorname{Spec}(\mathbb{C} \times \mathbb{C}) \cong \operatorname{Spec}(\mathbb{C}) \sqcup \operatorname{Spec}(\mathbb{C}).$$

Therefore the fiber product has two points, whereas the set-theoretic product of two one-point spaces has only one point.

In other words, the two points of

$$\operatorname{Spec}(\mathbb{C}) \times_{\operatorname{Spec}(\mathbb{R})} \operatorname{Spec}(\mathbb{C})$$

both map to the same pair of points in

$$\operatorname{Spec}(\mathbb{C}) \times \operatorname{Spec}(\mathbb{C}).$$

Hence the natural map from the point set of the fiber product to the set-theoretic product is not injective.

Example 2. Let $C =$ category of differentiable manifolds, let z be the manifold with one point. Then for any manifold X,

$$\operatorname{Hom}_C(z, X) \cong X$$

as a point set.

Example 3. Let $C =$ category of groups, let $z = \mathbb{Z}$. Then for any group G,

$$\operatorname{Hom}_C(z, G) \cong G$$

as a point set.

Example 4. Let C = category of rings with 1 (and homomorphisms f such that $f(1) = 1$), let $z = \mathbb{Z}[X]$. Then for any ring R,

$$\mathrm{Hom}_C(z, R) \cong R$$

as a point set.

So if C is any category whose objects may not be point sets to begin with and z is an object, we can think of $\mathrm{Hom}_C(z, X)$ as the underlying set of points of the object X.

$$X \mapsto \mathrm{Hom}_C(z, X)$$

extends to a function $C \to$ Sets. But we can only call $\mathrm{Hom}_C(z, X)$ the set of points of X if this function is **faithful**, a morphism $f : X_1 \to X_2$ is determined by the map of sets $\tilde{f} : \mathrm{Hom}_C(z, X_1) \to \mathrm{Hom}_C(z, X_2)$.

Example 5. Let Hot be the category of CW complexes, where $\mathrm{Hom}(X, Y)$ is the set of homotopy-classes of continuous maps from X to Y. If $z =$ the one point complex, then

$$\mathrm{Hom}_{\mathrm{Hot}}(z, X) = \pi_0(X) \quad \text{(the set of components of } X\text{)}$$

and this does not give a faithful functor. The issue is that maps out of a point detect only connected components, so distinct homotopy classes of maps can induce the same map on π_0.

Example 6. Let C = category of preschemes. Take for z the final object of the category C: $z = \mathrm{Spec}(\mathbb{Z})$. Then $\mathrm{Hom}_C(\mathrm{Spec}(\mathbb{Z}), X)$ is really small and does not give a faithful functor.

The problem here is that no scheme plays the role of $\mathbb{Z}[X]$, so we must choose $\mathbb{Z}$, and then we get things like e.g. $\mathrm{Hom}(\mathrm{Spec}\,\mathbb{Z}, \mathrm{Spec}\,\mathbb{F}_{p^2}) = \varnothing$ (otherwise there would be a unital ring homomorphism $\mathbb{F}_{p^2} \to \mathbb{Z}$).

To remedy this, one considers all test objects at once; informally, attach to X the collection

$$\bigcup_z \mathrm{Hom}_C(z, X).$$

The correct formal version of this idea is the functor of points below. We have extra structure on the above set too:

1. the decomposition of $\bigcup_z \mathrm{Hom}_C(z, X)$ into subsets $S_z = \mathrm{Hom}_C(z, X)$
2. the natural maps from one S_z to another $S_{z'}$, one for each morphism $g : z \to z'$ in the category

Formally: attach to each X in C the functor h_X (contravariant from C to Sets) via

(*) $h_X(z) = \mathrm{Hom}_C(z, X)$, $\quad z$ an object in C

(**) $h_X(g) = \{\text{induced map from } \mathrm{Hom}_C(z, X) \text{ to } \mathrm{Hom}_C(z', X)\}$, $\quad g : z' \to z$ a morphism in C.
$(h_X(f : Y \to Z) = (-) \circ f : \mathrm{Hom}_C(Z, X) \to \mathrm{Hom}_C(Y, X))$

And h_X lives in the category $\mathrm{Funct}(C^{\mathrm{op}}, \mathrm{Sets})$. Also if $g : X_1 \to X_2$ is a morphism in C, then we get a morphism of functors $h_g : h_{X_1} \to h_{X_2}$. So we get one big functor

$$h : C \to \mathrm{Funct}(C^{\mathrm{op}}, \mathrm{Sets}).$$

Proposition 7 (Yoneda Lemma)**.** *h is fully faithful, i.e. if X_1, X_2 are objects of C, then under h,*

$$\mathrm{Hom}_C(X_1, X_2) \cong \mathrm{Hom}_{\mathrm{Funct}}(h_{X_1}, h_{X_2}).$$

Sketch of proof:

$$h_X(-) = \mathrm{Hom}_C(-, X).$$

Given a morphism $f : X_1 \to X_2$, composition with f gives a natural transformation

$$h_{X_1} \to h_{X_2}, \qquad (g : T \to X_1) \mapsto f \circ g.$$

Conversely, given a natural transformation $\eta : h_{X_1} \to h_{X_2}$, evaluate it at X_1 and at the identity:

$$f := \eta_{X_1}(\mathrm{id}_{X_1}) \in \mathrm{Hom}_C(X_1, X_2).$$

Naturality implies that for every $g : T \to X_1$,

$$\eta_T(g) = f \circ g.$$

Thus η is completely determined by f, and these two constructions are inverse. Hence

$$\mathrm{Hom}_C(X_1, X_2) \cong \mathrm{Hom}_{\mathrm{Funct}}(h_{X_1}, h_{X_2}).$$

Definition 8. If X and K are preschemes, a **K-valued point of** X is a morphism $f : K \to X$.
If $K = \mathrm{Spec}(R)$, we call this an **R-valued point of X.**
If X and K are preschemes over a third prescheme S (i.e. we are given morphisms $p_X : X \to S, p_K : K \to S$), then f is a **K-valued point of** X/S if $p_X \circ f = p_K$.
If $K = \mathrm{Spec}(R)$, we call this an **R-valued point of** X/S.
The set of all R-valued points of a prescheme X, or of X/S, is denoted $X(R)$.

So $X(K) =$ morphisms $K \to X$ of *S-schemes.*

If K, X, Y are preschemes over S then the set of K-valued points of $X \times_S Y/S$ is the set-theoretic product of the set of K-valued points of X/S and the set of K-valued points of Y/S (by def. of fiber product).

Example 9. What are the $\mathbb{C}$-valued points of

$$\mathrm{Spec}(\mathbb{C}) \times_{\mathrm{Spec}(\mathbb{R})} \mathrm{Spec}(\mathbb{C})?$$

Recall that

$$\mathrm{Spec}(\mathbb{C}) \times_{\mathrm{Spec}(\mathbb{R})} \mathrm{Spec}(\mathbb{C}) \cong \mathrm{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}).$$

There are two relevant $\mathbb{R}$-algebra maps:

1.
$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \to \mathbb{C}, \qquad a \otimes b \mapsto ab;$$

2.
$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \to \mathbb{C}, \qquad a \otimes b \mapsto a\sigma(b),$$

where σ is complex conjugation.

Their kernels are the two points of

$$\operatorname{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}) \cong \operatorname{Spec}(\mathbb{C}) \sqcup \operatorname{Spec}(\mathbb{C}).$$

These two points can be thought of as the "graphs" of id and σ.

Another way to see this is that

$$\operatorname{Spec}(\mathbb{C})(\mathbb{C}) = \operatorname{Hom}_{\mathbb{R}\text{-alg}}(\mathbb{C}, \mathbb{C}) = \{\mathrm{id}, \sigma\} = \operatorname{Gal}(\mathbb{C}/\mathbb{R}).$$

Thus, each copy of $\operatorname{Spec}(\mathbb{C})$ has two $\mathbb{C}$-valued points over $\mathbb{R}$, corresponding to id and σ.

Therefore the fiber product has

$$\operatorname{Spec}(\mathbb{C})(\mathbb{C}) \times \operatorname{Spec}(\mathbb{C})(\mathbb{C}) = \{\mathrm{id}, \sigma\} \times \{\mathrm{id}, \sigma\},$$

so it has four $\mathbb{C}$-valued points.

Remark 10. If $f_1, \ldots, f_m \in \mathbb{Z}[X_1, \ldots, X_n]$ and $X = \operatorname{Spec}(\mathbb{Z}[X_1, \ldots, X_n]/(f_1, \ldots, f_m))$, then an R-valued point of X is the same as an n-tuple $a_1, \ldots, a_n \in R$ such that $f_1(a_1, \ldots, a_n) = \cdots = f_m(a_1, \ldots, a_n) = 0$.

A morphism $g : \operatorname{Spec}(R) \to \operatorname{Spec}(\mathbb{Z}[X_1, \ldots, X_n]/(f_1, \ldots, f_m))$ is determined by the n-tuple $a_i = g^*(X_i), 1 \le i \le n$, and those n-tuples that occur are exactly those such that $h \mapsto h(a_1, \ldots, a_n)$ defines a homomorphism $g^* : \mathbb{Z}[X_1, \ldots, X_n]/(f_1, \ldots, f_m) \to R$, i.e. solutions of $f_1, \ldots, f_m$.

Remark 11. Let X be a prescheme and let $h_X^{(0)}$ be the covariant functor Rings $\to$ Sets defined by

$$h_X^{(0)}(R) = h_X(\operatorname{Spec}(R)) = \operatorname{Hom}(\operatorname{Spec}(R), X).$$

This deals with the idea of "representability."

Proposition 12. *For any two preschemes X_1, X_2,*

$$\operatorname{Hom}(X_1, X_2) \cong \operatorname{Hom}(h_{X_1}^{(0)}, h_{X_2}^{(0)}).$$

Hence $h^{(0)}$ is a fully faithful functor from the category of preschemes to $\operatorname{Func}(\mathrm{Rings}, \mathrm{Sets})$.

Remark 13 (Grothendieck's existence problem). Why not identify a prescheme X with its corresponding functor $h_X^{(0)}$ and try to define preschemes as suitable functors F : Rings $\to$ Sets? Problem: to find natural conditions on F to ensure it is isomorphic to a functor of type $h_X^{(0)}$.

For example, all $h_X^{(0)}$ have compatibility with faithfully flat descent: if $q : A \to B$ is a homomorphism of rings making B into a faithfully flat A-algebra, i.e. for all ideals $I \subseteq A$, $I \otimes_A B \cong I \cdot B$ and $q^{-1}(I \cdot B) = I$. Then if $p_1, p_2 : B \to B \otimes_A B$ are the homomorphisms $\beta \mapsto \beta \otimes 1$ and $\beta \mapsto 1 \otimes \beta$, the induced diagram of sets

$$F(A) \xrightarrow{F(q)} F(B) \underset{F(p_2)}{\overset{F(p_1)}{\rightrightarrows}} F(B \otimes_A B)$$

is exact, i.e. $F(q)$ is injective and $\mathrm{Im}(F(q)) = \{x \in F(B) \mid F(p_1)(x) = F(p_2)(x)\}$.

Say you want to look at all curves of genus g. Can you find a scheme X such that

$$X(T) = \mathrm{Hom}(T, X)$$

corresponds to families of genus-g curves parametrized by T? Such an X (when it exists) is called a **moduli space**.

More generally, start with some geometric objects you want to classify and define a functor

$$F : \mathrm{Sch}^{\mathrm{op}} \to \mathrm{Set},$$

where $F(T)$ is the set of families of those objects parametrized by T. We then ask: does there exist a scheme X such that

$$F \cong h_X = \mathrm{Hom}(-, X)?$$

That is, is F the functor of points of some scheme X? If so, we say that F is **representable** by X.

Thus the moduli problem becomes: define the desired functor F, and ask whether it comes from a scheme.

Example 14. For the case of a field:
$\{k$-valued points of $X\} \cong \{$ points $x \in X$, plus injections $\kappa(x) \hookrightarrow k\}$.

$\{k$-valued points of X over $k_0\} \cong \{x \in X$ and $\kappa(x) \hookrightarrow k$ that reduce to identity on $k_0\}$.

If $k = k_0$ is algebraically closed and X is of finite type over k, then
$\{k$-valued points of $X/k\} \cong \{x \in X$ such that $k \cong \kappa(x)\} = \{$closed points $x \in X\}$.

So if X_0 is a prevariety over k_0, we recover the underlying set of the old variety over k associated to X. Thus when k is an algebraically closed extension field of k_0, k-valued points of X_0/k_0 are often called **geometric points of** X_0.

Example 15. $X = \mathbb{A}^1_{\mathbb{R}} = \mathrm{Spec}\,\mathbb{R}[x]$.

The point $x = 0$ corresponds to the ideal (x), and

$$\mathbb{R}[x]/(x) \cong \mathbb{R}.$$

The point $x = 1$ corresponds to the ideal $(x - 1)$, and

$$\mathbb{R}[x]/(x - 1) \cong \mathbb{R}.$$

The ideal $(x^2 + 1)$ also corresponds to a point of $\operatorname{Spec} \mathbb{R}[x]$. Since $x^2 + 1$ is irreducible over $\mathbb{R}$, this is a closed point of degree 2, with residue field

$$\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}.$$

Over $\mathbb{C}$, there are two ways to realize this point, corresponding to the two $\mathbb{R}$-algebra homomorphisms

$$\varphi_1 : x \mapsto i \qquad \text{and} \qquad \varphi_2 : x \mapsto -i.$$

These give two $\mathbb{C}$-valued points lying over the same degree-2 point $(x^2 + 1)$ of $\mathbb{A}^1_{\mathbb{R}}$. The two are related by complex conjugation.

From the other perspective, to look at $X(\mathbb{R})$, we look at the morphisms

$$\operatorname{Spec} \mathbb{R} \to X = \mathbb{A}^1_{\mathbb{R}} = \operatorname{Spec} \mathbb{R}[x],$$

so we get homomorphisms of $\mathbb{R}$-algebras in the other direction:

$$\mathbb{R}[x] \to \mathbb{R}.$$

Here we can map x to any element of $\mathbb{R}$. Thus the $\mathbb{R}$-valued points are exactly the usual real points of the real line.

To look at $X(\mathbb{C})$, we instead look at morphisms

$$\operatorname{Spec} \mathbb{C} \to \operatorname{Spec} \mathbb{R}[x],$$

which correspond to homomorphisms of $\mathbb{R}$-algebras

$$\mathbb{R}[x] \to \mathbb{C}.$$

Now we may still send x to, for example, 0 or 1, but we can also send x to i or $-i$. The two choices

$$x \mapsto i \qquad \text{and} \qquad x \mapsto -i$$

give two $\mathbb{C}$-valued points, but both lie over the same point $(x^2 + 1) \in \operatorname{Spec} \mathbb{R}[x]$. Its residue field is $\mathbb{C}$, which has degree 2 over the base field $\mathbb{R}$. The two $\mathbb{C}$-valued points are related by complex conjugation, i.e. by the nontrivial element of $\operatorname{Gal}(\mathbb{C}/\mathbb{R})$.

Example 16. $X = \mathbb{A}^2_{\mathbb{R}} = \operatorname{Spec} \mathbb{R}[x, y]$
$Z = \mathbb{A}^1_{\mathbb{R}} = \operatorname{Spec} \mathbb{R}[t]$

$$X(Z) = \operatorname{Hom}_{\mathbb{R}\text{-alg}}(\mathbb{R}[x, y], \mathbb{R}[t])$$

We can send e.g. $x \mapsto 3, y \mapsto 4$ and get a point, or
$x \mapsto t, y \mapsto t$ and get a line, or
$x \mapsto t, y \mapsto t^2$ and get a parametrized parabola.
Thus

$$X(Z) = \{\text{polynomially parametrized maps } \mathbb{A}^1_{\mathbb{R}} \to \mathbb{A}^2_{\mathbb{R}}\}.$$

Example 17. $X = \mathbb{A}^1_{\mathbb{R}} = \operatorname{Spec} \mathbb{R}[x]$
$Z = \operatorname{Spec} \mathbb{R}(x)$
$X(Z) = \operatorname{Hom}_{\mathbb{R}\text{-alg}}(\mathbb{R}[X], \mathbb{R}(x))$
Distinguished element: $x \mapsto x$
Generic point before was $[(0)]$. The residue field of X at the generic point is

$$\mathbb{R}[x]_{(0)}/(0) = \mathbb{R}(x)/(0) \cong \mathbb{R}(x)$$

so the generic point is an $\mathbb{R}(x)$-valued point.

Example 18. If R is a local ring, if M is the maximal ideal of R, then if $f : \operatorname{Spec}(R) \to X$ is a morphism, then any open set on X containing $f([M])$ contains the whole image of f. So the image always lies in one affine piece of X.

This is because if M is maximal, then if $D(g) \supseteq M$, g is a unit so the preimage is everything.

Proposition 19. *Let $x \in X$. Then there is a bijection between the set of R-valued points of X such that $f([M]) = x$ and the set of local homomorphisms $q : \mathcal{O}_{X,x} \to R$.*

Definition 20. For X and K preschemes, if f, g are two K-valued points of X, then we say $f(x) \equiv g(x)$ if $f \cdot i_x = g \cdot i_x$ where $i_x : \operatorname{Spec}(\kappa(x)) \to K$ is the canonical morphism. Equivalently, this means that $f(x) = g(x)$, and that the two maps $f^*_x, g^*_x : \kappa(f(x)) \to \kappa(x)$ are equal.

If K is reduced, then $f = g \iff f(x) \equiv g(x)$ for all $x \in K$.

Proposition 21. *For all $f, g : K \to X$, $\{x \in K \mid f(x) \equiv g(x)\}$ is locally closed.*

Definition 22. A prescheme X is a **scheme** if for all preschemes K and all K-valued points f, g of X, $\{x \in K \mid f(x) \equiv g(x)\}$ is closed.

Proposition 23. *If X is a prescheme over a ring R, then the criterion for X to be a scheme is satisfied for all K, f, g if it is satisfied in the case:*

$$K = X \times_{\operatorname{Spec}(R)} X, \quad f = p_1, \quad g = p_2.$$

In modern language, Prop 21 says $\Delta : X \to X \times_R X$ is always an immersion, and Prop 23 says X/R is separated $\iff \Delta$ is a closed immersion with image exactly $\{p_1 \equiv p_2\}$.

Corollary 24. *If k is an algebraically closed field, then a prevariety over k is a variety in the sense of Ch. I $\iff$ it is a scheme.*

Proposition 25. *If X is a scheme and $U, V \subseteq X$ are open affine sets, then $U \cap V$ is affine and the canonical homomorphism*

$$\Gamma(U, \mathcal{O}_X) \otimes \Gamma(V, \mathcal{O}_X) \to \Gamma(U \cap V, \mathcal{O}_X)$$

is surjective.

Example 26. Let R be a ring. The **functor of points** of a scheme X is

$$R \longmapsto X(R), \qquad X(R) := \mathrm{Hom}_{\mathrm{Sch}}(\operatorname{Spec} R, X).$$

If $X = \operatorname{Spec} A$ is affine, then

$$X(R) \cong \mathrm{Hom}_{\mathrm{Ring}}(A, R).$$

Let's find the functor of points in some examples:

1. $\mathbb{A}^n_{\mathbb{Z}} = \operatorname{Spec}(\mathbb{Z}[x_1, \ldots, x_n])$.

   A morphism

   $$\operatorname{Spec} R \to \mathbb{A}^n_{\mathbb{Z}}$$

   corresponds to a ring homomorphism in the other direction

   $$\mathbb{Z}[x_1, \ldots, x_n] \to R.$$

   Such a homomorphism is completely determined by where we send $x_1, \ldots, x_n$, and each x_i can be sent to any element of R. Therefore

   $$\mathrm{Hom}(\mathbb{Z}[x_1, \ldots, x_n], R) \cong R^n,$$

   so

   $$\boxed{\mathbb{A}^n_{\mathbb{Z}}(R) = R^n.}$$

   In particular,

   $$\mathbb{G}_a = \mathbb{A}^1_{\mathbb{Z}}$$

   represents the additive group, since

   $$\mathbb{G}_a(R) = R$$

   with its additive group structure.

2. $\mathbb{G}_m = \operatorname{Spec} \mathbb{Z}[t, t^{-1}]$.

   A ring homomorphism

   $$\mathbb{Z}[t, t^{-1}] \to R$$

   must send t to an invertible element of R, since t has an inverse. Conversely, any $r \in R^\times$ determines such a homomorphism by $t \mapsto r$. Thus

   $$\mathrm{Hom}(\mathbb{Z}[t, t^{-1}], R) \cong R^\times,$$

   so

   $$\boxed{\mathbb{G}_m(R) = R^\times.}$$

   Thus $\mathbb{G}_m$ represents the multiplicative group.

3. $\mu_n = \operatorname{Spec}(\mathbb{Z}[t]/(t^n - 1))$.

   A ring homomorphism
   $$\mathbb{Z}[t]/(t^n - 1) \to R$$
   is determined by the image r of t, but the relation $t^n = 1$ forces
   $$r^n = 1.$$
   Therefore
   $$\boxed{\mu_n(R) = \{r \in R \mid r^n = 1\}.}$$
   Thus μ_n represents the group of nth roots of unity.

4. $X = \operatorname{Spec}(\mathbb{Z}[x]/(x^2))$.

   A ring homomorphism
   $$\mathbb{Z}[x]/(x^2) \to R$$
   is determined by the image r of x, and the relation $x^2 = 0$ forces $r^2 = 0$. Hence
   $$\boxed{X(R) = \{r \in R \mid r^2 = 0\}.}$$
   Thus the functor of points detects nilpotent elements of R.

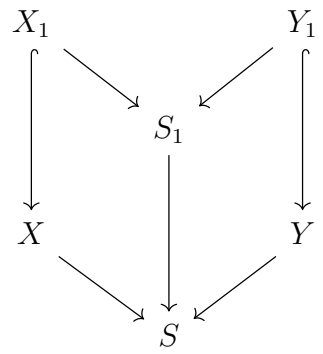
5. $$\mathrm{GL}_2 = \operatorname{Spec}(\mathbb{Z}[a, b, c, d, t]/((ad - bc)t - 1)).$$

   An R-point corresponds to elements
   $$a, b, c, d, t \in R$$
   satisfying
   $$(ad - bc)t = 1.$$
   Thus $ad - bc$ must be invertible in R, and t is then its inverse. Therefore an R-point is exactly a matrix
   $$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$
   with invertible determinant. Hence
   $$\boxed{\mathrm{GL}_2(R) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(R) \;\middle|\; ad - bc \in R^\times \right\} = \mathrm{GL}_2(R).}$$
   Thus the scheme GL_2 represents the functor sending a ring R to the group of invertible 2×2 matrices over R.

Problem 2.4. 1. **Check that a fiber product of schemes is a scheme.**

2. **Prove that $\mathbb{P}^n_{\mathbb{Z}}$ is a scheme**

3. **Given a diagram of schemes**



where X_1, Y_1 are closed subschemes of X and Y, show that $X_1 \times_{S_1} Y_1$ is a closed subscheme of $X \times_S Y$ (Hint: You must use the fact that S_1 is a scheme)

1) Let

$$X \longrightarrow S, \qquad Y \longrightarrow S$$

be morphisms of schemes, and let

$$P = X \times_S Y.$$

We already know that the fiber product exists as a prescheme. We must check that P is a scheme. Let K be any prescheme, and let

$$f, g : K \longrightarrow P$$

be two morphisms. Write

$$p_X : P \longrightarrow X, \qquad p_Y : P \longrightarrow Y$$

for the two projections. Consider the set

$$E = \{z \in K \mid f(z) \equiv g(z)\}.$$

By the universal property of the fiber product, two maps to P agree at z if and only if their compositions with both projections agree at z. Hence

$$E = \{z \in K \mid p_X f(z) \equiv p_X g(z)\} \cap \{z \in K \mid p_Y f(z) \equiv p_Y g(z)\}.$$

Since X and Y are schemes, both sets on the right are closed in K. Therefore E is closed.

Thus $P = X \times_S Y$ satisfies the defining condition for a scheme, and hence

$$\boxed{X \times_S Y \text{ is a scheme.}}$$

2) Let

$$P = \mathbb{P}^n_{\mathbb{Z}}.$$

By the criterion for a prescheme to be a scheme, it is enough to show that the diagonal is closed in

$$P \times_{\operatorname{Spec}\mathbb{Z}} P.$$

Use homogeneous coordinates

$$[X_0 : \cdots : X_n]$$

on the first factor and

$$[Y_0 : \cdots : Y_n]$$

on the second factor. Consider the equations

$$X_i Y_j - X_j Y_i = 0, \qquad 0 \le i, j \le n.$$

These equations define a closed subprescheme of

$$P \times_{\operatorname{Spec}\mathbb{Z}} P.$$

Indeed, on a standard affine open

$$U_r \times U_s, \qquad U_r = \{X_r \neq 0\}, \quad U_s = \{Y_s \neq 0\},$$

they become the ordinary polynomial equations

$$\frac{X_i}{X_r}\frac{Y_j}{Y_s} - \frac{X_j}{X_r}\frac{Y_i}{Y_s} = 0.$$

Thus their zero sets are closed on every standard affine piece and agree on the overlaps.

We claim that the closed subprescheme defined by these equations is the diagonal. To see this, let z be a point of

$$P \times_{\operatorname{Spec}\mathbb{Z}} P$$

and let $\kappa(z)$ be its residue field. Every point z of a prescheme comes with a morphism $\operatorname{Spec}\kappa(z) \to Z = P \times_{\operatorname{Spec}\mathbb{Z}} P$. So we can compose the morphisms with the projections $p_1, p_2 : Z \to \mathbb{P}^n_{\mathbb{Z}}$, and thus the two projections give two morphisms $\operatorname{Spec}\kappa(z) \to \mathbb{P}^n_{\mathbb{Z}}$, i.e. two $\kappa(z)$-valued points

$$[x_0 : \cdots : x_n], \qquad [y_0 : \cdots : y_n]$$

of projective space. They represent the same projective point if and only if the two vectors are proportional, which is equivalent to

$$x_i y_j - x_j y_i = 0 \qquad \text{for every } i, j.$$

Hence the locus on which the two projections agree is precisely

$$V(X_iY_j - X_jY_i \mid 0 \leq i, j \leq n),$$

which is closed.

Therefore the diagonal of $\mathbb{P}^n_{\mathbb{Z}}$ is closed, and so

$$\boxed{\mathbb{P}^n_{\mathbb{Z}} \text{ is a scheme.}}$$

3) We will do this in two steps.
Step 1: $X_1 \times_S Y_1$ is a closed subscheme of $X \times_S Y$.
Since

$$X_1 \hookrightarrow X$$

is a closed immersion, base-changing it by $Y \to S$ gives a closed immersion

$$X_1 \times_S Y \hookrightarrow X \times_S Y.$$

Similarly, since

$$Y_1 \hookrightarrow Y$$

is a closed immersion, base-changing it by $X_1 \to S$ gives a closed immersion

$$X_1 \times_S Y_1 \hookrightarrow X_1 \times_S Y.$$

Thus the composite

$$X_1 \times_S Y_1 \hookrightarrow X_1 \times_S Y \hookrightarrow X \times_S Y$$

is a closed immersion. Therefore

$$X_1 \times_S Y_1$$

is already a closed subscheme of $X \times_S Y$.

So it remains only to see that

$$X_1 \times_{S_1} Y_1$$

is a closed subscheme of

$$X_1 \times_S Y_1.$$

Step 2: Impose the condition that the two maps to S_1 agree.
Set

$$Z = X_1 \times_S Y_1.$$

A point of Z consists, roughly speaking, of a point of X_1 and a point of Y_1 which have the same image in S.

There are two natural maps

$$a, b : Z \longrightarrow S_1.$$

The first is obtained by projecting to X_1 and then using $X_1 \to S_1$:

$$a : Z \longrightarrow X_1 \longrightarrow S_1.$$

The second is obtained by projecting to Y_1 and then using $Y_1 \to S_1$:

$$b : Z \longrightarrow Y_1 \longrightarrow S_1.$$

Because $Z = X_1 \times_S Y_1$, the two maps a and b have the same image after composing with $S_1 \to S$. Hence together they define a map

$$(a, b) : Z \longrightarrow S_1 \times_S S_1.$$

Now,

$$X_1 \times_{S_1} Y_1$$

is exactly the part of Z on which the two maps a and b are equal. In other words, it is the inverse image of the diagonal

$$\Delta_{S_1/S} : S_1 \longrightarrow S_1 \times_S S_1.$$

Thus we have a cartesian square

$$\begin{array}{ccc} X_1 \times_{S_1} Y_1 & \longrightarrow & S_1 \\ \downarrow & & \downarrow{\scriptstyle \Delta_{S_1/S}} \\ X_1 \times_S Y_1 & \xrightarrow{(a,b)} & S_1 \times_S S_1. \end{array}$$

This is the point where we use the hypothesis that S_1 is a *scheme*. Since S_1 is a scheme, its diagonal is closed. Therefore

$$\Delta_{S_1/S} : S_1 \longrightarrow S_1 \times_S S_1$$

is a closed immersion.

Since closed immersions remain closed immersions after base change, the left vertical map in the cartesian square is also a closed immersion. Hence

$$X_1 \times_{S_1} Y_1 \hookrightarrow X_1 \times_S Y_1$$

is a closed immersion.

Finally, we already know from Step 1 that

$$X_1 \times_S Y_1 \hookrightarrow X \times_S Y$$

is a closed immersion. Composing the two closed immersions gives

$$X_1 \times_{S_1} Y_1 \hookrightarrow X \times_S Y.$$

Therefore

$$\boxed{X_1 \times_{S_1} Y_1 \text{ is a closed subscheme of } X \times_S Y.}$$